

Bd. 19 - Reelle Zahlen

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# Kapitel 1

## Einleitung

Band 18 konstruiert die rationalen Zahlen mit ihrer Addition, Multiplikation und totalen Ordnung; die maßgeblichen Grundlagen sind in Def. 18.5.1 zusammengefasst. In diesem Band werden daraus die reellen Zahlen als Dedekindsche Schnitte gewonnen. Diese Konstruktion verwendet nur Mengen, rationale Arithmetik und Ordnung; insbesondere setzt sie weder Folgen noch einen Grenzwertbegriff voraus.

Ein reeller Wert wird durch die Menge aller rationalen Zahlen beschrieben, die echt unter ihm liegen. Die Inklusion solcher Mengen liefert die Ordnung. Das Supremum einer nichtleeren, nach oben beschränkten Familie ist ihre Vereinigung. Damit ist die Vollständigkeit bereits auf der Ebene der Konstruktion sichtbar. Folgen und ihre Grenzwerte können deshalb erst in einem späteren Band ohne begrifflichen Zirkel eingeführt werden.

## Kapitel 2

# Dedekindsche Schnitte

### 2.1 Schnittprädikat und Trägermenge

**Definition 19.2.1.1 (Dedekindscher Schnitt der rationalen Zahlen).**

|                                      |                                                   |
|--------------------------------------|---------------------------------------------------|
| <i>Mengen:</i>                       | $A, p, q, r$                                      |
| <i>Relationsoperatoren:</i>          | $<$                                               |
| <i>Geordneter rationaler Körper:</i> | $\triangleright \mathbb{Q}$                       |
| <i>Bedingungen:</i>                  |                                                   |
| <i>Nichtleer und echt:</i>           | $A \neq \emptyset, A \neq \mathbb{Q}$             |
| <i>Nach unten abgeschlossen:</i>     | $p \in A, q \in \mathbb{Q}, q < p \vdash q \in A$ |
| <i>Kein größtes Element:</i>         | $p \in A \vdash \exists r \in A (p < r)$          |
| <i>Neues Symbol:</i>                 |                                                   |
| <i>Schnittprädikat:</i>              | $\text{Cut}_{\mathbb{Q}}(\cdot)$                  |

$$\begin{aligned}\text{Cut}_{\mathbb{Q}}(A) := & A \subseteq \mathbb{Q} \wedge A \neq \emptyset \wedge A \neq \mathbb{Q} \wedge \\ & \forall p \in A \forall q \in \mathbb{Q} (q < p \rightarrow q \in A) \wedge \\ & \forall p \in A \exists r \in A (p < r).\end{aligned}$$

Die vier Bedingungen sind unabhängig voneinander wichtig. Die Abwärts-abgeschlossenheit macht einen Schnitt zu einem rationalen Anfangsabschnitt; die letzte Bedingung entfernt die ansonsten mögliche Mehrdeutigkeit am Rand eines rationalen Anfangsabschnitts.

**Definition 19.2.1.2 (Die Menge der reellen Zahlen).**

|                       |                                  |
|-----------------------|----------------------------------|
| <i>Mengen:</i>        | $A, \mathbb{R}$                  |
| <i>Prädikate:</i>     | $\text{Cut}_{\mathbb{Q}}(\cdot)$ |
| <i>Neues Symbol:</i>  |                                  |
| <i>Reelle Zahlen:</i> | $\mathbb{R}$                     |

$$\mathbb{R} := \{A \in \mathcal{P}(\mathbb{Q}) \mid \text{Cut}_{\mathbb{Q}}(A)\}$$

**Theorem 19.2.1.1 (Elementcharakterisierung der reellen Zahlen).**

**Mengen:**  $A$

$$A \in \mathbb{R} \dashv\vdash \text{Cut}_{\mathbb{Q}}(A)$$

*Beweis.*

Hinrichtung:

$$\begin{array}{ll} 1 & 1 \ A \in \mathbb{R} \qquad A \\ & 2 \ \mathbb{R} = \{X \in \mathcal{P}(\mathbb{Q}) \mid \text{Cut}_{\mathbb{Q}}(X)\} \text{Def. 19.2.1.2} \\ 1 & 3 \ A \in \mathcal{P}(\mathbb{Q}) \wedge \text{Cut}_{\mathbb{Q}}(A) \qquad = E(1, 2) \\ 1 & 4 \ \text{Cut}_{\mathbb{Q}}(A) \qquad \wedge E 2(3) \end{array}$$

Rückrichtung:

$$\begin{array}{ll} 1 & 1 \ \text{Cut}_{\mathbb{Q}}(A) \qquad A \\ 1 & 2 \ A \subseteq \mathbb{Q} \qquad \text{Def. 19.2.1.1(1)} \\ 1 & 3 \ A \in \mathcal{P}(\mathbb{Q}) \qquad \text{Th. 3.16.2.1(2)} \\ 1 & 4 \ A \in \mathcal{P}(\mathbb{Q}) \wedge \text{Cut}_{\mathbb{Q}}(A) \wedge I(3, 1) \\ 1 & 5 \ A \in \mathbb{R} \qquad \text{Def. 19.2.1.2(4)} \end{array}$$

Zusammenfassung:

$$\begin{array}{l} 1 \ A \in \mathbb{R} \rightarrow \text{Cut}_{\mathbb{Q}}(A) \rightarrow I(1, 4) \\ 2 \ \text{Cut}_{\mathbb{Q}}(A) \rightarrow A \in \mathbb{R} \rightarrow I(1, 5) \\ 3 \ A \in \mathbb{R} \leftrightarrow \text{Cut}_{\mathbb{Q}}(A) \leftrightarrow I(1, 2) \end{array}$$

□

Diese Charakterisierung erlaubt es im Folgenden, eine reelle Zahl und den sie darstellenden Schnitt ohne zusätzliche Klassenklammern miteinander zu identifizieren.

## 2.2 Hauptschnitte und rationale Einbettung

**Definition 19.2.2.1 (Hauptschnitt einer rationalen Zahl).**

**Mengen:**  $p, q$

**Relationsoperatoren:**  $<$

**Neues Symbol:**

**Hauptschnitt:**  $\hat{q}$

$$q \in \mathbb{Q} \vdash \hat{q} := \{p \in \mathbb{Q} \mid p < q\}$$

**Theorem 19.2.2.1 (Hauptschnitte sind Dedekindsche Schnitte).**

**Mengen:**  $p, q, r$

$$q \in \mathbb{Q} \vdash \text{Cut}_{\mathbb{Q}}(\widehat{q})$$

*Beweis.*

**Nichtleerheit und Echtheit**

Th. 19.2.2.1(i)

$$q \in \mathbb{Q} \vdash \widehat{q} \neq \emptyset \wedge \widehat{q} \neq \mathbb{Q}$$

|      |                                                                      |                         |                                                        |
|------|----------------------------------------------------------------------|-------------------------|--------------------------------------------------------|
| 1    | $1 \ q \in \mathbb{Q}$                                               | $A$                     |                                                        |
|      | $2 \ 1 \in \mathbb{Z}_{>0}$                                          | Th. 17.4.3.1            |                                                        |
|      | $3 \ 0 < 1$                                                          | Def. 17.4.3.1(2)        |                                                        |
|      | $4 \ 0 \leq 1 \wedge 0 \neq 1$                                       | Def. 17.4.2.3(3)        |                                                        |
|      | $5 \ 0 \in \mathbb{Z}$                                               | Th. 17.2.3.7            |                                                        |
|      | $6 \ 1 \in \mathbb{Z}$                                               | Th. 17.2.3.8            |                                                        |
|      | $7 \ \iota_{\mathbb{Q}}(0) \leq \iota_{\mathbb{Q}}(1)$               | Th. 18.4.2.5(4, 5, 6)   |                                                        |
|      | $8 \ \iota_{\mathbb{Q}}(0) \neq \iota_{\mathbb{Q}}(1)$               | Th. 18.2.3.2(4, 5, 6)   |                                                        |
|      | $9 \ 0 = \iota_{\mathbb{Q}}(0) \wedge 1 = \iota_{\mathbb{Q}}(1)$     |                         | $\wedge I(\text{Def. 18.2.3.5}, \text{Def. 18.2.3.6})$ |
|      | $10 \ 0 < 1$                                                         |                         |                                                        |
|      |                                                                      |                         | Def. 18.4.1.4(7, 8, 9)                                 |
| 1,10 | $11 \ q - 1 < q$                                                     | Th. 18.4.2.1(1,10)      |                                                        |
| 1,10 | $12 \ \exists p \in \mathbb{Q} (q - 1 < p \wedge p < q)$             | Th. 18.6.1.2(11)        |                                                        |
| 1,10 | $13 \ p \in \mathbb{Q} \wedge p < q$                                 | $A$                     |                                                        |
| 1,10 | $14 \ p \in \widehat{q}$                                             | Def. 19.2.2.1(13)       |                                                        |
| 1,10 | $15 \ \widehat{q} \neq \emptyset$                                    | Th. 3.6.3.5(14)         |                                                        |
| 1    | $16 \ q \notin \widehat{q}$                                          |                         | Def. 19.2.2.1(Th. 18.4.1.4(1))                         |
| 1    | $17 \ \widehat{q} \neq \mathbb{Q}$                                   | Th. 3.4.2.2(1, 16)      |                                                        |
| 1    | $18 \ \widehat{q} \neq \emptyset \wedge \widehat{q} \neq \mathbb{Q}$ | $\wedge I(15, 17)$      |                                                        |
| 1    | $19 \ \widehat{q} \neq \emptyset \wedge \widehat{q} \neq \mathbb{Q}$ | $\exists E(12, 13, 18)$ |                                                        |

**Abwärtsabschluss**

Th. 19.2.2.1(ii)

$$q \in \mathbb{Q}, p \in \widehat{q}, s \in \mathbb{Q}, s < p \\ \vdash s \in \widehat{q}$$

- 1  $1 \ q \in \mathbb{Q} \ A$
- 2  $2 \ p \in \widehat{q} \ A$
- 3  $3 \ s \in \mathbb{Q} \ A$

$$\begin{array}{l}
4 \quad 4 \quad s < p \quad A \\
1,2 \quad 5 \quad p < q \quad \text{Def. 19.2.2.1(1,2)} \\
1,2,4 \quad 6 \quad s < q \quad \text{Th. 18.4.1.4(4,5)} \\
1,2,3,4 \quad 7 \quad s \in \widehat{q} \quad \text{Def. 19.2.2.1(3,6)}
\end{array}$$

**Kein größtes Element**

Th. 19.2.2.1(iii)

$$q \in \mathbb{Q}, \quad p \in \widehat{q} \vdash \exists r \in \widehat{q} (p < r)$$

$$\begin{array}{ll}
1 \quad 1 \quad q \in \mathbb{Q} & A \\
2 \quad 2 \quad p \in \widehat{q} & A \\
1,2 \quad 3 \quad p < q & \text{Def. 19.2.2.1(1,2)} \\
1,2 \quad 4 \quad \exists r \in \mathbb{Q} (p < r \wedge r < q) & \text{Th. 18.6.1.2(3)} \\
1,2 \quad 5 \quad r \in \mathbb{Q} \wedge p < r \wedge r < q & A \\
1,2 \quad 6 \quad r \in \widehat{q} & \text{Def. 19.2.2.1(5)} \\
1,2 \quad 7 \quad \exists r \in \widehat{q} (p < r) & \exists I(5,6) \\
1,2 \quad 8 \quad \exists r \in \widehat{q} (p < r) & \exists E(4,5,7)
\end{array}$$

Schluss:

$$\begin{array}{ll}
1 \quad 1 \quad q \in \mathbb{Q} & A \\
1 \quad 2 \quad \widehat{q} \subseteq \mathbb{Q} & \text{Def. 19.2.2.1(1)} \\
1 \quad 3 \quad \widehat{q} \neq \emptyset \wedge \widehat{q} \neq \mathbb{Q} & \text{Th. 19.2.2.1(i)(1)} \\
1 \quad 4 \quad \forall p \in \widehat{q} \forall s \in \mathbb{Q} (s < p \rightarrow s \in \widehat{q}) & \text{Th. 19.2.2.1(ii)(1)} \\
1 \quad 5 \quad \forall p \in \widehat{q} \exists r \in \widehat{q} (p < r) & \text{Th. 19.2.2.1(iii)(1)} \\
1 \quad 6 \quad \text{Cut}_{\mathbb{Q}}(\widehat{q}) & \text{Def. 19.2.1.1(2,3,4,5)}
\end{array}$$

□

**Definition 19.2.2.2 (Kanonische Einbettung der rationalen Zahlen).**

**Mengen:**  $p, \quad q$

**Funktionen:**  $\iota_{\mathbb{R}}$

**Neues Symbol:**

**Rationale Einbettung:**  $\iota_{\mathbb{R}}$

$$\begin{array}{l}
\iota_{\mathbb{R}}: \mathbb{Q} \rightarrow \mathbb{R}, \\
\forall q \in \mathbb{Q} \quad (\iota_{\mathbb{R}}(q) := \widehat{q})
\end{array}$$

$$\begin{array}{ll}
1 \quad 1 \quad q \in \mathbb{Q} & A \\
1 \quad 2 \quad \text{Cut}_{\mathbb{Q}}(\widehat{q}) & \text{Th. 19.2.2.1(1)} \\
1 \quad 3 \quad \widehat{q} \in \mathbb{R} & \text{Th. 19.2.1.1(2)} \\
4 \quad \forall q \in \mathbb{Q} \quad (\widehat{q} \in \mathbb{R}) & \forall I(\rightarrow I(1,3))
\end{array}$$

**Theorem 19.2.2.2** (Die rationale Einbettung ist eine Funktion).

*Funktionen:*  $\iota_{\mathbb{R}}$

$$\iota_{\mathbb{R}}: \mathbb{Q} \rightarrow \mathbb{R}$$

$$1 \quad \iota_{\mathbb{R}}: \mathbb{Q} \rightarrow \mathbb{R} \quad \text{Def. 19.2.2.2}$$

## Kapitel 3

# Ordnung und Vollständigkeit

### 3.1 Inklusionsordnung

**Definition 19.3.1.1** (Ordnung der reellen Zahlen).

*Mengen:*  $A, B$   
*Relationsoperatoren:*  $\leq_{\mathbb{R}}$   
*Neues Symbol:*  
*Reelle Ordnung:*  $\leq_{\mathbb{R}}$

$$A, B \in \mathbb{R} \vdash A \leq_{\mathbb{R}} B := A \subseteq B$$

**Definition 19.3.1.2** (Strikte Ordnung der reellen Zahlen).

*Mengen:*  $A, B$   
*Relationsoperatoren:*  $\leq_{\mathbb{R}}, <_{\mathbb{R}}$   
*Neues Symbol:*  
*Strikte reelle Ordnung:*  $<_{\mathbb{R}}$

$$A, B \in \mathbb{R} \vdash A <_{\mathbb{R}} B := A \subsetneq B$$

**Theorem 19.3.1.1** (Charakterisierung der reellen Vergleichsrelationen).

*Mengen:*  $A, B$   
*Relationsoperatoren:*  $\leq_{\mathbb{R}}, <_{\mathbb{R}}$

$$A, B \in \mathbb{R} \vdash (A \leq_{\mathbb{R}} B \leftrightarrow A \subseteq B) \wedge (A <_{\mathbb{R}} B \leftrightarrow A \subsetneq B)$$

|     |                                                       |                  |
|-----|-------------------------------------------------------|------------------|
| 1 1 | $A, B \in \mathbb{R}$                                 | $A$              |
| 1 2 | $A \leq_{\mathbb{R}} B \leftrightarrow A \subseteq B$ | Def. 19.3.1.1(1) |
| 1 3 | $A <_{\mathbb{R}} B \leftrightarrow A \subsetneq B$   | Def. 19.3.1.2(1) |

$$1 \quad 4 (A \leq_{\mathbb{R}} B \leftrightarrow A \subseteq B) \wedge (A <_{\mathbb{R}} B \leftrightarrow A \subsetneq B) \wedge I(2, 3)$$

**Theorem 19.3.1.2 (Vergleichbarkeit zweier Dedekindscher Schnitte).**

**Mengen:**  $A, B, a, b$

$$A, B \in \mathbb{R} \vdash A \subseteq B \vee B \subseteq A$$

*Beweis.*

**Ausschluss zweier Gegenzeugen**

Th. 19.3.1.2(i)

$$A, B \in \mathbb{R}, \quad a \in A \setminus B, \quad b \in B \setminus A \\ \vdash \perp$$

|          |    |                                                                |                                    |
|----------|----|----------------------------------------------------------------|------------------------------------|
| 1        | 1  | $A, B \in \mathbb{R}$                                          | $A$                                |
| 2        | 2  | $a \in A \setminus B$                                          | $A$                                |
| 3        | 3  | $b \in B \setminus A$                                          | $A$                                |
| 1        | 4  | $\text{Cut}_{\mathbb{Q}}(A) \wedge \text{Cut}_{\mathbb{Q}}(B)$ | Th. 19.2.1.1(1)                    |
| 2        | 5  | $a \in A \wedge a \notin B$                                    | Th. 3.12.1(2)                      |
| 3        | 6  | $b \in B \wedge b \notin A$                                    | Th. 3.12.1(3)                      |
| 1,2,3    | 7  | $a < b \vee a = b \vee b < a$                                  | Th. 18.4.1.4(1,5,6)                |
| 8        | 8  | $a < b$                                                        | $A$                                |
| 1,2,3,8  | 9  | $a \in B$                                                      | Def. 19.2.1.1(4,6,8)               |
| 1,2,3,8  | 10 | $\perp$                                                        | $\perp I(5, 9)$                    |
| 11       | 11 | $a = b$                                                        | $A$                                |
| 1,2,3,11 | 12 | $a \in B$                                                      | $= E(11, 6)$                       |
| 1,2,3,11 | 13 | $\perp$                                                        | $\perp I(5, 12)$                   |
| 14       | 14 | $b < a$                                                        | $A$                                |
| 1,2,3,14 | 15 | $b \in A$                                                      | Def. 19.2.1.1(4,5,14)              |
| 1,2,3,14 | 16 | $\perp$                                                        | $\perp I(6, 15)$                   |
| 1,2,3    | 17 | $\perp$                                                        | $\vee E(7, 8, 10, 11, 13, 14, 16)$ |

Schluss:

|     |   |                                              |                  |
|-----|---|----------------------------------------------|------------------|
| 1   | 1 | $A, B \in \mathbb{R}$                        | $A$              |
| 2   | 2 | $\neg(A \subseteq B \vee B \subseteq A)$     | $A$              |
| 2   | 3 | $A \not\subseteq B \wedge B \not\subseteq A$ | Th. 2.1.8.1(2)   |
| 1,2 | 4 | $\exists a \in A \setminus B$                | Th. 3.13.3.64(3) |
| 1,2 | 5 | $\exists b \in B \setminus A$                | Th. 3.13.3.64(3) |
| 1,2 | 6 | $a \in A \setminus B$                        | $A$              |
| 1,2 | 7 | $b \in B \setminus A$                        | $A$              |

|     |                                       |                        |
|-----|---------------------------------------|------------------------|
| 1,2 | $8 \perp$                             | Th. 19.3.1.2(i)(1,6,7) |
| 1,2 | $9 \perp$                             | $\exists E(5, 7, 8)$   |
| 1,2 | $10 \perp$                            | $\exists E(4, 6, 9)$   |
| 1   | $11 A \subseteq B \vee B \subseteq A$ | $\neg I(2, 10)$        |

□

**Theorem 19.3.1.3 (Die Inklusionsordnung der Schnitte ist total).**

**Mengen:**  $A, B$

**Relationsoperatoren:**  $\leq_{\mathbb{R}}$

$\text{TotOrd}(\mathbb{R}, \leq_{\mathbb{R}})$

*Beweis.*

**Partielle Ordnung**

Th. 19.3.1.3(i)

$\text{PartOrd}(\mathbb{R}, \leq_{\mathbb{R}})$

|       |                                                                                                                              |                      |                                                       |
|-------|------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------|
| 1     | $1 A \in \mathbb{R}$                                                                                                         | $A$                  |                                                       |
| 1     | $2 A \subseteq A$                                                                                                            | Th. 3.5.3.1          |                                                       |
| 1     | $3 A \leq_{\mathbb{R}} A$                                                                                                    | Def. 19.3.1.1(1, 2)  |                                                       |
|       | $4 \forall A \in \mathbb{R} (A \leq_{\mathbb{R}} A) \quad \forall I(\rightarrow I(1, 3))$                                    |                      |                                                       |
| 5     | $5 A, B, C \in \mathbb{R}$                                                                                                   | $A$                  |                                                       |
| 6     | $6 A \leq_{\mathbb{R}} B \wedge B \leq_{\mathbb{R}} C$                                                                       | $A$                  |                                                       |
| 5,6   | $7 A \subseteq B \wedge B \subseteq C$                                                                                       | Th. 19.3.1.1(5, 6)   |                                                       |
| 5,6   | $8 A \subseteq C$                                                                                                            | Th. 3.5.3.6(7)       |                                                       |
| 5,6   | $9 A \leq_{\mathbb{R}} C$                                                                                                    | Def. 19.3.1.1(5, 8)  |                                                       |
|       | $10 \forall A, B, C \in \mathbb{R} ((A \leq_{\mathbb{R}} B \wedge B \leq_{\mathbb{R}} C) \rightarrow A \leq_{\mathbb{R}} C)$ |                      | $\forall I(\rightarrow I(5, \rightarrow I(6, 9)))$    |
| 11    | $11 A, B \in \mathbb{R}$                                                                                                     | $A$                  |                                                       |
| 12    | $12 A \leq_{\mathbb{R}} B \wedge B \leq_{\mathbb{R}} A$                                                                      | $A$                  |                                                       |
| 11,12 | $13 A \subseteq B \wedge B \subseteq A$                                                                                      | Th. 19.3.1.1(11, 12) |                                                       |
| 11,12 | $14 A = B$                                                                                                                   | Th. 3.5.3.5(13)      |                                                       |
|       | $15 \forall A, B \in \mathbb{R} ((A \leq_{\mathbb{R}} B \wedge B \leq_{\mathbb{R}} A) \rightarrow A = B)$                    |                      | $\forall I(\rightarrow I(11, \rightarrow I(12, 14)))$ |
|       | $16 \text{PartOrd}(\mathbb{R}, \leq_{\mathbb{R}})$                                                                           |                      | Def. 12.2.1.1(4, 10, 15)                              |

Vergleichbarkeit und Schluss:

|   |                                                 |                 |
|---|-------------------------------------------------|-----------------|
| 1 | $\text{PartOrd}(\mathbb{R}, \leq_{\mathbb{R}})$ | Th. 19.3.1.3(i) |
| 2 | $2 A, B \in \mathbb{R}$                         | $A$             |

|     |                                                                                                                 |                    |
|-----|-----------------------------------------------------------------------------------------------------------------|--------------------|
| 2 3 | $A \subseteq B \vee B \subseteq A$                                                                              | Th. 19.3.1.2(2)    |
| 2 4 | $A \leq_{\mathbb{R}} B \vee B \leq_{\mathbb{R}} A$                                                              | Th. 19.3.1.1(2,3)  |
| 5   | $\forall A, B \in \mathbb{R} (A \leq_{\mathbb{R}} B \vee B \leq_{\mathbb{R}} A) \forall I(\rightarrow I(2, 4))$ |                    |
| 6   | $\text{TotOrd}(\mathbb{R}, \leq_{\mathbb{R}})$                                                                  | Def. 15.2.1.1(1,5) |

□

**Theorem 19.3.1.4 (Hauptschnitte bewahren und spiegeln die Ordnung).**

**Mengen:**  $p, q$   
**Funktionen:**  $\iota_{\mathbb{R}}$   
**Relationsoperatoren:**  $\leq_{\mathbb{R}}, <_{\mathbb{R}}$

$$p, q \in \mathbb{Q} \vdash (p \leq q \leftrightarrow \iota_{\mathbb{R}}(p) \leq_{\mathbb{R}} \iota_{\mathbb{R}}(q)) \wedge (p < q \leftrightarrow \iota_{\mathbb{R}}(p) <_{\mathbb{R}} \iota_{\mathbb{R}}(q))$$

*Beweis.*

**Nichtstrikte Hinrichtung**

Th. 19.3.1.4(i)

$$p, q \in \mathbb{Q}, p \leq q \vdash \hat{p} \subseteq \hat{q}$$

|       |   |                             |                                                  |
|-------|---|-----------------------------|--------------------------------------------------|
| 1     | 1 | $p, q \in \mathbb{Q}$       | $A$                                              |
| 2     | 2 | $p \leq q$                  | $A$                                              |
| 3     | 3 | $x \in \hat{p}$             | $A$                                              |
| 1,3   | 4 | $x < p$                     | Def. 19.2.2.1(1,3)                               |
| 1,2,3 | 5 | $x < q$                     | Th. 18.4.1.4(2,4)                                |
| 1,2,3 | 6 | $x \in \hat{q}$             | Def. 19.2.2.1(1,5)                               |
| 1,2   | 7 | $\hat{p} \subseteq \hat{q}$ | Def. 3.5.1.1( $\forall I(\rightarrow I(3, 6))$ ) |

**Nichtstrikte Rückrichtung**

Th. 19.3.1.4(ii)

$$p, q \in \mathbb{Q}, \hat{p} \subseteq \hat{q} \vdash p \leq q$$

|       |   |                                                 |                   |
|-------|---|-------------------------------------------------|-------------------|
| 1     | 1 | $p, q \in \mathbb{Q}$                           | $A$               |
| 2     | 2 | $\hat{p} \subseteq \hat{q}$                     | $A$               |
| 3     | 3 | $q < p$                                         | $A$               |
| 1,3   | 4 | $\exists r \in \mathbb{Q} (q < r \wedge r < p)$ | Th. 18.6.1.2(1,3) |
| 1,3   | 5 | $r \in \mathbb{Q} \wedge q < r \wedge r < p$    | $A$               |
| 1,3   | 6 | $r \in \hat{p}$                                 | Def. 19.2.2.1(5)  |
| 1,2,3 | 7 | $r \in \hat{q}$                                 | Th. 3.5.2.6(2,6)  |
| 1,3   | 8 | $r \notin \hat{q}$                              | Def. 19.2.2.1(5)  |
| 1,2,3 | 9 | $\perp$                                         | $\perp I(7, 8)$   |

|                      |                      |
|----------------------|----------------------|
| 1,2,3 10 $\perp$     | $\exists E(4, 5, 9)$ |
| 1,2 11 $\neg(q < p)$ | $\neg I(3, 10)$      |
| 1,2 12 $p \leq q$    | Th. 18.4.1.4(1,11)   |

### Strikte Hinrichtung

Th. 19.3.1.4(iii)

$$p, q \in \mathbb{Q}, p < q \vdash \hat{p} \subsetneq \hat{q}$$

|                                                       |                       |
|-------------------------------------------------------|-----------------------|
| 1 1 $p, q \in \mathbb{Q}$                             | $A$                   |
| 2 2 $p < q$                                           | $A$                   |
| 1,2 3 $p \leq q$                                      | Def. 18.4.1.4(1,2)    |
| 1,2 4 $\hat{p} \subseteq \hat{q}$                     | Th. 19.3.1.4(i)(1,3)  |
| 1,2 5 $\exists r \in \mathbb{Q} (p < r \wedge r < q)$ | Th. 18.6.1.2(1,2)     |
| 1,2 6 $r \in \mathbb{Q} \wedge p < r \wedge r < q$    | $A$                   |
| 1,2 7 $r \in \hat{q}$                                 | Def. 19.2.2.1(6)      |
| 1,2 8 $r \notin \hat{p}$                              | Def. 19.2.2.1(6)      |
| 1,2 9 $\hat{p} \neq \hat{q}$                          | Th. 3.4.2.2(7,8)      |
| 1,2 10 $\hat{p} \subsetneq \hat{q}$                   | Def. 3.5.1.2(4,9)     |
| 1,2 11 $\hat{p} \subsetneq \hat{q}$                   | $\exists E(5, 6, 10)$ |

### Strikte Rückrichtung

Th. 19.3.1.4(iv)

$$p, q \in \mathbb{Q}, \hat{p} \subsetneq \hat{q} \vdash p < q$$

|                                                             |                       |
|-------------------------------------------------------------|-----------------------|
| 1 1 $p, q \in \mathbb{Q}$                                   | $A$                   |
| 2 2 $\hat{p} \subsetneq \hat{q}$                            | $A$                   |
| 2 3 $\hat{p} \subseteq \hat{q} \wedge \hat{p} \neq \hat{q}$ | Def. 3.5.1.2(2)       |
| 1,2 4 $p \leq q$                                            | Th. 19.3.1.4(ii)(1,3) |
| 5 5 $p = q$                                                 | $A$                   |
| 1,5 6 $\hat{p} = \hat{q}$                                   | $= E(5)$              |
| 1,2,5 7 $\perp$                                             | $\perp I(3, 6)$       |
| 1,2 8 $p \neq q$                                            | $\neg I(5, 7)$        |
| 1,2 9 $p < q$                                               | Def. 18.4.1.4(1,4,8)  |

Einbettung und Zusammenfassung:

|                                                                                              |                                                                        |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 1 1 $p, q \in \mathbb{Q}$                                                                    | $A$                                                                    |
| 1 2 $p \leq q \leftrightarrow \hat{p} \subseteq \hat{q}$                                     | $\leftrightarrow I(\text{Th. 19.3.1.4}(i), \text{Th. 19.3.1.4}(ii))$   |
| 1 3 $p < q \leftrightarrow \hat{p} \subsetneq \hat{q}$                                       | $\leftrightarrow I(\text{Th. 19.3.1.4}(iii), \text{Th. 19.3.1.4}(iv))$ |
| 1 4 $\iota_{\mathbb{R}}(p) = \hat{p} \wedge \iota_{\mathbb{R}}(q) = \hat{q}$                 | Def. 19.2.2.2(1)                                                       |
| 1 5 $p \leq q \leftrightarrow \iota_{\mathbb{R}}(p) \leq_{\mathbb{R}} \iota_{\mathbb{R}}(q)$ | Th. 19.3.1.1(2,4)                                                      |
| 1 6 $p < q \leftrightarrow \iota_{\mathbb{R}}(p) <_{\mathbb{R}} \iota_{\mathbb{R}}(q)$       | Th. 19.3.1.1(3,4)                                                      |

$$\begin{array}{l} 1 \quad 7 \quad (p \leq q \leftrightarrow \iota_{\mathbb{R}}(p) \leq_{\mathbb{R}} \iota_{\mathbb{R}}(q)) \wedge \wedge I(5,6) \\ \quad (p < q \leftrightarrow \iota_{\mathbb{R}}(p) <_{\mathbb{R}} \iota_{\mathbb{R}}(q)) \end{array}$$

□

**Theorem 19.3.1.5 (Injektivität der rationalen Einbettung).**

**Funktionen:**  $\iota_{\mathbb{R}}$

$$\iota_{\mathbb{R}}: \mathbb{Q} \rightarrow \mathbb{R}$$

*Beweis.*

**Punktweise Injektivität**

Th. 19.3.1.5(i)

$$p, q \in \mathbb{Q}, \iota_{\mathbb{R}}(p) = \iota_{\mathbb{R}}(q) \vdash p = q$$

$$\begin{array}{ll} 1 \quad 1 \quad p, q \in \mathbb{Q} & A \\ 2 \quad 2 \quad \iota_{\mathbb{R}}(p) = \iota_{\mathbb{R}}(q) & A \\ 1,2 \quad 3 \quad \iota_{\mathbb{R}}(p) \leq_{\mathbb{R}} \iota_{\mathbb{R}}(q) \wedge \iota_{\mathbb{R}}(q) \leq_{\mathbb{R}} \iota_{\mathbb{R}}(p) & \text{Th. 19.3.1.1(1,2)} \\ 1,2 \quad 4 \quad p \leq q \wedge q \leq p & \text{Th. 19.3.1.4(1,3)} \\ 1,2 \quad 5 \quad p = q & \text{Th. 18.4.1.4(1,4)} \end{array}$$

Funktionsschluss:

$$\begin{array}{ll} 1 \quad \iota_{\mathbb{R}}: \mathbb{Q} \rightarrow \mathbb{R} & \text{Th. 19.2.2.2} \\ 2 \quad \forall p, q \in \mathbb{Q} (\iota_{\mathbb{R}}(p) = \iota_{\mathbb{R}}(q) \rightarrow p = q) & \text{Th. 19.3.1.5(i)} \\ 3 \quad \iota_{\mathbb{R}}: \mathbb{Q} \rightarrow \mathbb{R} & \text{Def. 6.2.1.1(1,2)} \end{array}$$

□

## 3.2 Kanonische Teilmengenkopien

Die rationale Einbettung transportiert die in  $\mathbb{Q}$  bereits geschachtelten natürlichen und ganzzahligen Kopien in den reellen Träger. Damit entsteht innerhalb von  $\mathbb{R}$  eine wörtliche Inklusionskette.

**Definition 19.3.2.1 (Kanonische rationale Kopie in den reellen Zahlen).**

**Mengen:**  $\mathbb{Q}_{\mathbb{R}}$   
**Funktion:**  $\iota_{\mathbb{R}}$   
**Neue Symbole:**  $\mathbb{Q}_{\mathbb{R}}$

$$\mathbb{Q}_{\mathbb{R}} := \iota_{\mathbb{R}}[\mathbb{Q}]$$

**Definition 19.3.2.2 (Kanonische ganzzahlige Kopie in den reellen Zahlen).**

*Mengen:*  $\mathbb{Z}_Q, \mathbb{Z}_R$   
*Funktion:*  $\iota_R$   
*Neue Symbole:*  $\mathbb{Z}_R$

$$\mathbb{Z}_R := \iota_R[\mathbb{Z}_Q]$$

**Definition 19.3.2.3 (Kanonische natürliche Kopie in den reellen Zahlen).**

*Mengen:*  $\mathbb{N}_Q, \mathbb{N}_R$   
*Funktion:*  $\iota_R$   
*Neue Symbole:*  $\mathbb{N}_R$

$$\mathbb{N}_R := \iota_R[\mathbb{N}_Q]$$

**Theorem 19.3.2.1 (Geschachtelter Zahlbereichsturm in den reellen Zahlen).**

*Mengen:*  $\mathbb{N}_Q, \mathbb{Z}_Q, \mathbb{N}_R, \mathbb{Z}_R, \mathbb{Q}_R$   
*Funktion:*  $\iota_R$

$$(\mathbb{N}_R \subseteq \mathbb{Z}_R \wedge \mathbb{Z}_R \subseteq \mathbb{Q}_R \wedge \mathbb{Q}_R \subseteq \mathbb{R}) \wedge (\mathbb{N}_R \subseteq \mathbb{R} \wedge \mathbb{Z}_R \subseteq \mathbb{R})$$

|                                                                                                                                                                                                                                                                          |                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 1 $\iota_R: \mathbb{Q} \rightarrow \mathbb{R}$                                                                                                                                                                                                                           | Th. 19.2.2.2              |
| 2 $(\mathbb{N}_Q \subseteq \mathbb{Z}_Q \wedge \mathbb{Z}_Q \subseteq \mathbb{Q}) \wedge \mathbb{N}_Q \subseteq \mathbb{Q}$                                                                                                                                              | Th. 18.2.3.3              |
| 3 $\mathbb{N}_Q \subseteq \mathbb{Z}_Q$                                                                                                                                                                                                                                  | $\wedge E1(\wedge E1(2))$ |
| 4 $\mathbb{Z}_Q \subseteq \mathbb{Q}$                                                                                                                                                                                                                                    | $\wedge E2(\wedge E1(2))$ |
| 5 $\mathbb{Q} \subseteq \mathbb{R}$                                                                                                                                                                                                                                      | Th. 3.5.3.1               |
| 6 $\iota_R[\mathbb{N}_Q] \subseteq \iota_R[\mathbb{Z}_Q]$                                                                                                                                                                                                                | Th. 5.2.4.10(1,3,4)       |
| 7 $\iota_R[\mathbb{Z}_Q] \subseteq \iota_R[\mathbb{Q}]$                                                                                                                                                                                                                  | Th. 5.2.4.10(1,4,5)       |
| 8 $\iota_R[\mathbb{Q}] \subseteq \mathbb{R}$                                                                                                                                                                                                                             | Th. 5.2.4.18(1)           |
| 9 $\mathbb{N}_R = \iota_R[\mathbb{N}_Q]$                                                                                                                                                                                                                                 | Def. 19.3.2.3             |
| 10 $\mathbb{Z}_R = \iota_R[\mathbb{Z}_Q]$                                                                                                                                                                                                                                | Def. 19.3.2.2             |
| 11 $\mathbb{Q}_R = \iota_R[\mathbb{Q}]$                                                                                                                                                                                                                                  | Def. 19.3.2.1             |
| 12 $\mathbb{N}_R \subseteq \mathbb{Z}_R$                                                                                                                                                                                                                                 | $= E(9, 10, 6)$           |
| 13 $\mathbb{Z}_R \subseteq \mathbb{Q}_R$                                                                                                                                                                                                                                 | $= E(10, 11, 7)$          |
| 14 $\mathbb{Q}_R \subseteq \mathbb{R}$                                                                                                                                                                                                                                   | $= E(11, 8)$              |
| 15 $\mathbb{Z}_R \subseteq \mathbb{R}$                                                                                                                                                                                                                                   | Th. 3.5.3.6(13,14)        |
| 16 $\mathbb{N}_R \subseteq \mathbb{R}$                                                                                                                                                                                                                                   | Th. 3.5.3.6(12,15)        |
| 17 $(\mathbb{N}_R \subseteq \mathbb{Z}_R \wedge \mathbb{Z}_R \subseteq \mathbb{Q}_R \wedge \mathbb{Q}_R \subseteq \mathbb{R}) \wedge \wedge I(\wedge I(12, 13, 14), \wedge I(16, 15))$<br>$(\mathbb{N}_R \subseteq \mathbb{R} \wedge \mathbb{Z}_R \subseteq \mathbb{R})$ |                           |

**Theorem 19.3.2.2 (Bijektive Identifikation der natürlichen Kopie).**

**Mengen:**  $\mathbb{N}_{\mathbb{Q}}, \mathbb{N}_{\mathbb{R}}$

**Funktion:**  $\iota_{\mathbb{R}}$

$$\iota_{\mathbb{R}} \upharpoonright_{\mathbb{N}_{\mathbb{Q}}} : \mathbb{N}_{\mathbb{Q}} \xrightarrow{\sim} \mathbb{N}_{\mathbb{R}}$$

|                                                                                                                                                                                                       |                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 1 $\iota_{\mathbb{R}} : \mathbb{Q} \hookrightarrow \mathbb{R}$                                                                                                                                        | Th. 19.3.1.5                     |
| 2 $\mathbb{N}_{\mathbb{Q}} \subseteq \mathbb{Q}$                                                                                                                                                      | $\wedge E2(Th. 18.2.3.3)$        |
| 3 $\iota_{\mathbb{R}} \upharpoonright_{\mathbb{N}_{\mathbb{Q}}} : \mathbb{N}_{\mathbb{Q}} \hookrightarrow \mathbb{R}$                                                                                 | Th. 6.3.1.4(1,2)                 |
| 4 $\iota_{\mathbb{R}} \upharpoonright_{\mathbb{N}_{\mathbb{Q}}} : \mathbb{N}_{\mathbb{Q}} \xrightarrow{\sim} (\iota_{\mathbb{R}} \upharpoonright_{\mathbb{N}_{\mathbb{Q}}})[\mathbb{N}_{\mathbb{Q}}]$ | Th. 8.3.2.3(3)                   |
| 5 $(\iota_{\mathbb{R}} \upharpoonright_{\mathbb{N}_{\mathbb{Q}}})[\mathbb{N}_{\mathbb{Q}}] = \iota_{\mathbb{R}}[\mathbb{N}_{\mathbb{Q}}]$                                                             | Th. 5.3.2.16(2, Def. 6.2.1.1(1)) |
| 6 $\mathbb{N}_{\mathbb{R}} = \iota_{\mathbb{R}}[\mathbb{N}_{\mathbb{Q}}]$                                                                                                                             | Def. 19.3.2.3                    |
| 7 $\iota_{\mathbb{R}} \upharpoonright_{\mathbb{N}_{\mathbb{Q}}} : \mathbb{N}_{\mathbb{Q}} \xrightarrow{\sim} \mathbb{N}_{\mathbb{R}}$                                                                 | $= E(5, 6, 4)$                   |

**Theorem 19.3.2.3 (Bijektive Identifikation der ganzzahligen Kopie).**

**Mengen:**  $\mathbb{Z}_{\mathbb{Q}}, \mathbb{Z}_{\mathbb{R}}$

**Funktion:**  $\iota_{\mathbb{R}}$

$$\iota_{\mathbb{R}} \upharpoonright_{\mathbb{Z}_{\mathbb{Q}}} : \mathbb{Z}_{\mathbb{Q}} \xrightarrow{\sim} \mathbb{Z}_{\mathbb{R}}$$

|                                                                                                                                                                                                       |                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 1 $\iota_{\mathbb{R}} : \mathbb{Q} \hookrightarrow \mathbb{R}$                                                                                                                                        | Th. 19.3.1.5                         |
| 2 $\mathbb{Z}_{\mathbb{Q}} \subseteq \mathbb{Q}$                                                                                                                                                      | $\wedge E2(\wedge E1(Th. 18.2.3.3))$ |
| 3 $\iota_{\mathbb{R}} \upharpoonright_{\mathbb{Z}_{\mathbb{Q}}} : \mathbb{Z}_{\mathbb{Q}} \hookrightarrow \mathbb{R}$                                                                                 | Th. 6.3.1.4(1,2)                     |
| 4 $\iota_{\mathbb{R}} \upharpoonright_{\mathbb{Z}_{\mathbb{Q}}} : \mathbb{Z}_{\mathbb{Q}} \xrightarrow{\sim} (\iota_{\mathbb{R}} \upharpoonright_{\mathbb{Z}_{\mathbb{Q}}})[\mathbb{Z}_{\mathbb{Q}}]$ | Th. 8.3.2.3(3)                       |
| 5 $(\iota_{\mathbb{R}} \upharpoonright_{\mathbb{Z}_{\mathbb{Q}}})[\mathbb{Z}_{\mathbb{Q}}] = \iota_{\mathbb{R}}[\mathbb{Z}_{\mathbb{Q}}]$                                                             | Th. 5.3.2.16(2, Def. 6.2.1.1(1))     |
| 6 $\mathbb{Z}_{\mathbb{R}} = \iota_{\mathbb{R}}[\mathbb{Z}_{\mathbb{Q}}]$                                                                                                                             | Def. 19.3.2.2                        |
| 7 $\iota_{\mathbb{R}} \upharpoonright_{\mathbb{Z}_{\mathbb{Q}}} : \mathbb{Z}_{\mathbb{Q}} \xrightarrow{\sim} \mathbb{Z}_{\mathbb{R}}$                                                                 | $= E(5, 6, 4)$                       |

**Theorem 19.3.2.4 (Bijektive Identifikation der rationalen Kopie).**

**Mengen:**  $\mathbb{Q}_{\mathbb{R}}$

**Funktion:**  $\iota_{\mathbb{R}}$

$$\iota_{\mathbb{R}} : \mathbb{Q} \xrightarrow{\sim} \mathbb{Q}_{\mathbb{R}}$$

|                                                                                       |                |
|---------------------------------------------------------------------------------------|----------------|
| 1 $\iota_{\mathbb{R}} : \mathbb{Q} \hookrightarrow \mathbb{R}$                        | Th. 19.3.1.5   |
| 2 $\iota_{\mathbb{R}} : \mathbb{Q} \xrightarrow{\sim} \iota_{\mathbb{R}}[\mathbb{Q}]$ | Th. 8.3.2.3(1) |
| 3 $\mathbb{Q}_{\mathbb{R}} = \iota_{\mathbb{R}}[\mathbb{Q}]$                          | Def. 19.3.2.1  |
| 4 $\iota_{\mathbb{R}} : \mathbb{Q} \xrightarrow{\sim} \mathbb{Q}_{\mathbb{R}}$        | $= E(3, 2)$    |

### 3.3 Suprema als Vereinigungen

**Theorem 19.3.3.1** (Vereinigung einer beschränkten Schnittfamilie).

*Mengen:*  $\mathcal{A}, A, U$

*Relationsoperatoren:*  $\leq_{\mathbb{R}}$

$$\emptyset \neq \mathcal{A} \subseteq \mathbb{R}, U \in \mathbb{R}, \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} U) \\ \vdash \text{Cut}_{\mathbb{Q}}\left(\bigcup \mathcal{A}\right).$$

*Beweis.*

**Träger, Nichtleerheit und Echtheit**

Th. 19.3.3.1(i)

$$\emptyset \neq \mathcal{A} \subseteq \mathbb{R}, U \in \mathbb{R}, \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} U) \\ \vdash \bigcup \mathcal{A} \subseteq \mathbb{Q} \wedge \bigcup \mathcal{A} \neq \emptyset \wedge \bigcup \mathcal{A} \neq \mathbb{Q}$$

|      |                                                       |                                                    |
|------|-------------------------------------------------------|----------------------------------------------------|
| 1    | 1 $\emptyset \neq \mathcal{A} \subseteq \mathbb{R}$   | $A$                                                |
| 2    | 2 $U \in \mathbb{R}$                                  | $A$                                                |
| 3    | 3 $\forall A \in \mathcal{A} (A \leq_{\mathbb{R}} U)$ | $A$                                                |
| 1    | 4 $\exists A_0 \in \mathcal{A}$                       | Th. 3.6.3.7(1)                                     |
| 1    | 5 $A_0 \in \mathcal{A}$                               | $A$                                                |
| 1,5  | 6 $A_0 \in \mathbb{R}$                                | Th. 3.5.2.6(1, 5)                                  |
| 1,5  | 7 $\text{Cut}_{\mathbb{Q}}(A_0)$                      | Th. 19.2.1.1(6)                                    |
| 1,5  | 8 $\exists a_0 \in A_0$                               | Def. 19.2.1.1(7)                                   |
| 1,5  | 9 $a_0 \in A_0$                                       | $A$                                                |
| 1,5  | 10 $a_0 \in \bigcup \mathcal{A}$                      | Th. 3.13.2.4(5, 9)                                 |
| 1,5  | 11 $\bigcup \mathcal{A} \neq \emptyset$               | Th. 3.6.3.5(10)                                    |
| 1    | 12 $\bigcup \mathcal{A} \neq \emptyset$               | $\exists E(8, 9, 11)$                              |
| 1    | 13 $\bigcup \mathcal{A} \neq \emptyset$               | $\exists E(4, 5, 12)$                              |
| 14   | 14 $x \in \bigcup \mathcal{A}$                        | $A$                                                |
| 14   | 15 $\exists A \in \mathcal{A} (x \in A)$              | Th. 3.13.2.1(14)                                   |
| 14   | 16 $A \in \mathcal{A} \wedge x \in A$                 | $A$                                                |
| 1,14 | 17 $A \in \mathbb{R}$                                 | Th. 3.5.2.6(1, 16)                                 |
| 1,14 | 18 $A \subseteq \mathbb{Q}$                           | Th. 19.2.1.1(17)                                   |
| 1,14 | 19 $x \in \mathbb{Q}$                                 | Th. 3.5.2.6(18, 16)                                |
| 1,14 | 20 $x \in \mathbb{Q}$                                 | $\exists E(15, 16, 19)$                            |
| 1    | 21 $\bigcup \mathcal{A} \subseteq \mathbb{Q}$         |                                                    |
|      |                                                       | Def. 3.5.1.1( $\forall I(\rightarrow I(14, 20))$ ) |
| 22   | 22 $x \in \bigcup \mathcal{A}$                        | $A$                                                |
| 22   | 23 $\exists A \in \mathcal{A} (x \in A)$              | Th. 3.13.2.1(22)                                   |
| 22   | 24 $A \in \mathcal{A} \wedge x \in A$                 | $A$                                                |
| 3,22 | 25 $A \leq_{\mathbb{R}} U$                            | $\forall E(3, 24)$                                 |

|                                                                                                                                                                         |                         |                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------|
| 3,22 26 $A \subseteq U$                                                                                                                                                 |                         | Th. 19.3.1.1(2, 24, 25)                            |
| 3,22 27 $x \in U$                                                                                                                                                       | Th. 3.5.2.6(26, 24)     |                                                    |
| 3,22 28 $x \in U$                                                                                                                                                       | $\exists E(23, 24, 27)$ |                                                    |
| 3 29 $\bigcup \mathcal{A} \subseteq U$                                                                                                                                  |                         | Def. 3.5.1.1( $\forall I(\rightarrow I(22, 28))$ ) |
| 2 30 $\text{Cut}_{\mathbb{Q}}(U)$                                                                                                                                       | Th. 19.2.1.1(2)         |                                                    |
| 2 31 $U \neq \mathbb{Q}$                                                                                                                                                | Def. 19.2.1.1(30)       |                                                    |
| 32 32 $\bigcup \mathcal{A} = \mathbb{Q}$                                                                                                                                | $A$                     |                                                    |
| 2,3,32 33 $\mathbb{Q} \subseteq U$                                                                                                                                      | $= E(32, 29)$           |                                                    |
| 2 34 $U \subseteq \mathbb{Q}$                                                                                                                                           | Def. 19.2.1.1(30)       |                                                    |
| 2,3,32 35 $U = \mathbb{Q}$                                                                                                                                              | Th. 3.5.3.5(34, 33)     |                                                    |
| 2,3,32 36 $\perp$                                                                                                                                                       | $\perp I(31, 35)$       |                                                    |
| 2,3 37 $\bigcup \mathcal{A} \neq \mathbb{Q}$                                                                                                                            | $\neg I(32, 36)$        |                                                    |
| 1,2,3 38 $\bigcup \mathcal{A} \subseteq \mathbb{Q} \wedge \bigcup \mathcal{A} \neq \emptyset \wedge \bigcup \mathcal{A} \neq \mathbb{Q} \wedge I(21, \wedge I(13, 37))$ |                         |                                                    |

#### Abwärtsabschluss

Th. 19.3.3.1(ii)

$$\mathcal{A} \subseteq \mathbb{R}, p \in \bigcup \mathcal{A}, q \in \mathbb{Q}, q < p$$

$$\vdash q \in \bigcup \mathcal{A}$$

|         |                                         |                        |
|---------|-----------------------------------------|------------------------|
| 1       | 1 $\mathcal{A} \subseteq \mathbb{R}$    | $A$                    |
| 2       | 2 $p \in \bigcup \mathcal{A}$           | $A$                    |
| 3       | 3 $q \in \mathbb{Q}$                    | $A$                    |
| 4       | 4 $q < p$                               | $A$                    |
| 2       | 5 $\exists A \in \mathcal{A} (p \in A)$ | Th. 3.13.2.1(2)        |
| 2       | 6 $A \in \mathcal{A} \wedge p \in A$    | $A$                    |
| 1,2     | 7 $A \in \mathbb{R}$                    | Th. 3.5.2.6(1,6)       |
| 1,2     | 8 $\text{Cut}_{\mathbb{Q}}(A)$          | Th. 19.2.1.1(7)        |
| 1,2,3,4 | 9 $q \in A$                             | Def. 19.2.1.1(8,6,3,4) |
| 1,2,3,4 | 10 $q \in \bigcup \mathcal{A}$          | Th. 3.13.2.4(6,9)      |
| 1,2,3,4 | 11 $q \in \bigcup \mathcal{A}$          | $\exists E(5, 6, 10)$  |

#### Kein größtes Element

Th. 19.3.3.1(iii)

$$\mathcal{A} \subseteq \mathbb{R}, p \in \bigcup \mathcal{A} \vdash \exists r \in \bigcup \mathcal{A} (p < r)$$

|     |                                         |                  |
|-----|-----------------------------------------|------------------|
| 1   | 1 $\mathcal{A} \subseteq \mathbb{R}$    | $A$              |
| 2   | 2 $p \in \bigcup \mathcal{A}$           | $A$              |
| 2   | 3 $\exists A \in \mathcal{A} (p \in A)$ | Th. 3.13.2.1(2)  |
| 2   | 4 $A \in \mathcal{A} \wedge p \in A$    | $A$              |
| 1,2 | 5 $A \in \mathbb{R}$                    | Th. 3.5.2.6(1,4) |

|     |    |                                                                 |                    |
|-----|----|-----------------------------------------------------------------|--------------------|
| 1,2 | 6  | $\text{Cut}_{\mathbb{Q}}(A)$                                    | Th. 19.2.1.1(5)    |
| 1,2 | 7  | $\exists r \in A (p < r)$                                       | Def. 19.2.1.1(6,4) |
| 1,2 | 8  | $r \in A \wedge p < r$                                          | $A$                |
| 1,2 | 9  | $r \in \bigcup \mathcal{A}$                                     | Th. 3.13.2.4(4,8)  |
| 1,2 | 10 | $\exists r \in \bigcup \mathcal{A} (p < r) \exists I(8, 9)$     |                    |
| 1,2 | 11 | $\exists r \in \bigcup \mathcal{A} (p < r) \exists E(7, 8, 10)$ |                    |
| 1,2 | 12 | $\exists r \in \bigcup \mathcal{A} (p < r) \exists E(3, 4, 11)$ |                    |

Schluss:

|       |   |                                                                                                                                 |                        |
|-------|---|---------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1     | 1 | $\emptyset \neq \mathcal{A} \subseteq \mathbb{R}$                                                                               | $A$                    |
| 2     | 2 | $U \in \mathbb{R}$                                                                                                              | $A$                    |
| 3     | 3 | $\forall A \in \mathcal{A} (A \leq_{\mathbb{R}} U)$                                                                             | $A$                    |
| 1,2,3 | 4 | $\bigcup \mathcal{A} \subseteq \mathbb{Q} \wedge \bigcup \mathcal{A} \neq \emptyset \wedge \bigcup \mathcal{A} \neq \mathbb{Q}$ | Th. 19.3.3.1(i)(1,2,3) |
| 1     | 5 | $\forall p \in \bigcup \mathcal{A} \forall q \in \mathbb{Q} (q < p \rightarrow q \in \bigcup \mathcal{A})$                      | Th. 19.3.3.1(ii)(1)    |
| 1     | 6 | $\forall p \in \bigcup \mathcal{A} \exists r \in \bigcup \mathcal{A} (p < r)$                                                   | Th. 19.3.3.1(iii)(1)   |
| 1,2,3 | 7 | $\text{Cut}_{\mathbb{Q}}(\bigcup \mathcal{A})$                                                                                  | Def. 19.2.1.1(4,5,6)   |

□

**Theorem 19.3.3.2 (Vollständigkeit der reellen Ordnung).**

**Mengen:**  $\mathcal{A}, A, S, U, V$

**Relationsoperatoren:**  $\leq_{\mathbb{R}}$

$$\emptyset \neq \mathcal{A} \subseteq \mathbb{R}, \exists U \in \mathbb{R} \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} U) \\ \vdash \exists S \in \mathbb{R} \left( \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} S) \wedge \right. \\ \left. \forall V \in \mathbb{R} (\forall A \in \mathcal{A} (A \leq_{\mathbb{R}} V) \rightarrow S \leq_{\mathbb{R}} V) \right).$$

*Beweis.*

**Vereinigung ist obere Schranke**

Th. 19.3.3.2(i)

$$\emptyset \neq \mathcal{A} \subseteq \mathbb{R}, S = \bigcup \mathcal{A} \vdash \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} S)$$

|       |   |                                                   |                    |
|-------|---|---------------------------------------------------|--------------------|
| 1     | 1 | $\emptyset \neq \mathcal{A} \subseteq \mathbb{R}$ | $A$                |
| 2     | 2 | $S = \bigcup \mathcal{A}$                         | $A$                |
| 3     | 3 | $A \in \mathcal{A}$                               | $A$                |
| 1,3   | 4 | $A \in \mathbb{R}$                                | Th. 3.5.2.6(1,3)   |
| 3     | 5 | $A \subseteq \bigcup \mathcal{A}$                 | Th. 3.13.2.4(3)    |
| 2,3   | 6 | $A \subseteq S$                                   | $= E(2, 5)$        |
| 1,2,3 | 7 | $A \leq_{\mathbb{R}} S$                           | Def. 19.3.1.1(4,6) |

$$1,2 \ 8 \ \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} S) \forall I(\rightarrow I(3,7))$$

**Vereinigung liegt unter jeder oberen Schranke**

Th. 19.3.3.2(ii)

$$\mathcal{A} \subseteq \mathbb{R}, \ V \in \mathbb{R}, \ \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} V) \\ \vdash \bigcup \mathcal{A} \leq_{\mathbb{R}} V$$

$$\begin{array}{ll} 1 \ 1 \ \mathcal{A} \subseteq \mathbb{R} & A \\ 2 \ 2 \ V \in \mathbb{R} & A \\ 3 \ 3 \ \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} V) A & \\ 4 \ 4 \ x \in \bigcup \mathcal{A} & A \\ 4 \ 5 \ \exists A \in \mathcal{A} (x \in A) & \text{Th. 3.13.2.1(4)} \\ 4 \ 6 \ A \in \mathcal{A} \wedge x \in A & A \\ 1,4 \ 7 \ A \in \mathbb{R} & \text{Th. 3.5.2.6(1,6)} \\ 3,4 \ 8 \ A \leq_{\mathbb{R}} V & \forall E(3,6) \\ 1,2,3,4 \ 9 \ A \subseteq V & \text{Th. 19.3.1.1(7,2,8)} \\ 1,2,3,4 \ 10 \ x \in V & \text{Th. 3.5.2.6(9,6)} \\ 1,2,3,4 \ 11 \ x \in V & \exists E(5,6,10) \\ 1,2,3 \ 12 \ \bigcup \mathcal{A} \subseteq V & \text{Def. 3.5.1.1}(\forall I(\rightarrow I(4,11))) \\ 1,2,3 \ 13 \ \bigcup \mathcal{A} \leq_{\mathbb{R}} V & \text{Def. 19.3.1.1(2,12)} \end{array}$$

Existenz der kleinsten oberen Schranke:

$$\begin{array}{ll} 1 \ 1 \ \emptyset \neq \mathcal{A} \subseteq \mathbb{R} & A \\ 2 \ 2 \ \exists U \in \mathbb{R} \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} U) & A \\ 2 \ 3 \ U \in \mathbb{R} \wedge \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} U) & A \\ 1,2 \ 4 \ \text{Cut}_{\mathbb{Q}}(\bigcup \mathcal{A}) & \text{Th. 19.3.3.1(1,3)} \\ 1,2 \ 5 \ \bigcup \mathcal{A} \in \mathbb{R} & \text{Th. 19.2.1.1(4)} \\ 1 \ 6 \ \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} \bigcup \mathcal{A}) & \text{Th. 19.3.3.2(i)(1)} \\ 1 \ 7 \ \forall V \in \mathbb{R} (\forall A \in \mathcal{A} (A \leq_{\mathbb{R}} V) \rightarrow \bigcup \mathcal{A} \leq_{\mathbb{R}} V) & \text{Th. 19.3.3.2(ii)(1)} \\ 1,2 \ 8 \ \exists S \in \mathbb{R} (\forall A \in \mathcal{A} (A \leq_{\mathbb{R}} S) \wedge \forall V \in \mathbb{R} (\forall A \in \mathcal{A} (A \leq_{\mathbb{R}} V) \rightarrow S \leq_{\mathbb{R}} V)) & \\ & \exists I(5, \wedge I(6,7)) \\ 1,2 \ 9 \ \exists S \in \mathbb{R} (\forall A \in \mathcal{A} (A \leq_{\mathbb{R}} S) \wedge \forall V \in \mathbb{R} (\forall A \in \mathcal{A} (A \leq_{\mathbb{R}} V) \rightarrow S \leq_{\mathbb{R}} V)) & \\ & \exists E(2,3,8) \end{array}$$

□

Erst die vorstehende Existenz- und Kleinste-Obergrenzen-Aussage erfüllt die Definitionsprämisse des allgemeinen Supremumsbegriffs aus Band 13. Daher ist der folgende Supremumsterm wohldefiniert.

**Theorem 19.3.3.3 (Supremum einer beschränkten Schnittfamilie).**

**Mengen:**  $\mathcal{A}, A, U$   
**Relationsoperatoren:**  $\leq_{\mathbb{R}}$

$$\emptyset \neq \mathcal{A} \subseteq \mathbb{R}, U \in \mathbb{R}, \forall A \in \mathcal{A} (A \leq_{\mathbb{R}} U)$$

$$\vdash \sup_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) = \bigcup \mathcal{A}.$$

|                                            |    |                                                                                                                                    |                            |
|--------------------------------------------|----|------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1                                          | 1  | $\emptyset \neq \mathcal{A} \subseteq \mathbb{R}$                                                                                  | $A$                        |
| 2                                          | 2  | $U \in \mathbb{R}$                                                                                                                 | $A$                        |
| 3                                          | 3  | $\forall A \in \mathcal{A} (A \leq_{\mathbb{R}} U)$                                                                                | $A$                        |
| 1,2,3                                      | 4  | $\bigcup \mathcal{A} \in \mathbb{R}$                                                                                               |                            |
| <i>Th. 19.2.1.1(Th. 19.3.3.1(1, 2, 3))</i> |    |                                                                                                                                    |                            |
| 1                                          | 5  | $\forall A \in \mathcal{A} (A \leq_{\mathbb{R}} \bigcup \mathcal{A})$                                                              | <i>Th. 19.3.3.2(i)(1)</i>  |
| 1                                          | 6  | $\forall V \in \mathbb{R} (\forall A \in \mathcal{A} (A \leq_{\mathbb{R}} V) \rightarrow \bigcup \mathcal{A} \leq_{\mathbb{R}} V)$ | <i>Th. 19.3.3.2(ii)(1)</i> |
| 1,2,3                                      | 7  | $\bigcup \mathcal{A} \in \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A})$                                                   | <i>Th. 13.2.1.3(4, 5)</i>  |
| 1,2,3                                      | 8  | $\forall V \in \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) (\bigcup \mathcal{A} \leq_{\mathbb{R}} V)$                   | <i>Th. 13.2.1.2(6)</i>     |
| 1,2,3                                      | 9  | $\bigcup \mathcal{A} = \sup_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A})$                                                          |                            |
| <i>Th. 13.2.1.15(1, 7, 8)</i>              |    |                                                                                                                                    |                            |
| 1,2,3                                      | 10 | $\sup_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) = \bigcup \mathcal{A}$                                                          | <i>Th. 2.9.1.1(9)</i>      |

**Theorem 19.3.3.4 (Infimumsvollständigkeit der reellen Ordnung).**

**Mengen:**  $\mathcal{A}, A, L, V$

**Relationsoperatoren:**  $\leq_{\mathbb{R}}$

$$\emptyset \neq \mathcal{A} \subseteq \mathbb{R}, \exists L \in \mathbb{R} \forall A \in \mathcal{A} (L \leq_{\mathbb{R}} A)$$

$$\vdash \exists I \in \mathbb{R} (\forall A \in \mathcal{A} (I \leq_{\mathbb{R}} A) \wedge \forall V \in \mathbb{R} (\forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A) \rightarrow V \leq_{\mathbb{R}} I)).$$

*Beweis.*

**Schrankenmenge ist nichtleer und beschränkt** *Th. 19.3.3.4(i)*

$$\emptyset \neq \mathcal{A} \subseteq \mathbb{R}, L \in \mathbb{R}, \forall A \in \mathcal{A} (L \leq_{\mathbb{R}} A)$$

$$\vdash \emptyset \neq \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) \subseteq \mathbb{R} \wedge$$

$$\exists A_0 \in \mathbb{R} \forall V \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) (V \leq_{\mathbb{R}} A_0)$$

|       |   |                                                                               |                              |
|-------|---|-------------------------------------------------------------------------------|------------------------------|
| 1     | 1 | $\emptyset \neq \mathcal{A} \subseteq \mathbb{R}$                             | $A$                          |
| 2     | 2 | $L \in \mathbb{R}$                                                            | $A$                          |
| 3     | 3 | $\forall A \in \mathcal{A} (L \leq_{\mathbb{R}} A)$                           | $A$                          |
| 1,2,3 | 4 | $L \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A})$                | <i>Th. 13.2.1.7(1, 2, 3)</i> |
| 1,2,3 | 5 | $\text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) \neq \emptyset$       | <i>Th. 3.6.3.5(4)</i>        |
| 1     | 6 | $\text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) \subseteq \mathbb{R}$ | <i>Th. 13.2.1.8(1)</i>       |
| 1     | 7 | $\exists A_0 \in \mathcal{A}$                                                 | <i>Th. 3.6.3.7(1)</i>        |
| 1     | 8 | $A_0 \in \mathcal{A}$                                                         | $A$                          |

|       |    |                                                                                                                                                                                                                           |                                    |
|-------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1     | 9  | $A_0 \in \mathbb{R}$                                                                                                                                                                                                      | <i>Th.</i> 3.5.2.6(1, 8)           |
| 10    | 10 | $V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A})$                                                                                                                                                              | $A$                                |
| 1,10  | 11 | $V \leq_{\mathbb{R}} A_0$                                                                                                                                                                                                 |                                    |
|       |    |                                                                                                                                                                                                                           | <i>Th.</i> 13.2.1.6(1, 10, 8)      |
| 1     | 12 | $\forall V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) (V \leq_{\mathbb{R}} A_0)$                                                                                                                            | $\forall I(\rightarrow I(10, 11))$ |
| 1     | 13 | $\exists A_0 \in \mathbb{R} \forall V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) (V \leq_{\mathbb{R}} A_0) \exists I(9, 12)$                                                                                |                                    |
| 1     | 14 | $\exists A_0 \in \mathbb{R} \forall V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) (V \leq_{\mathbb{R}} A_0) \exists E(7, 8, 13)$                                                                             |                                    |
| 1,2,3 | 15 | $\emptyset \neq \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) \subseteq \mathbb{R} \wedge \exists A_0 \in \mathbb{R} \forall V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) (V \leq_{\mathbb{R}} A_0)$ |                                    |
|       |    |                                                                                                                                                                                                                           | $\wedge I(\wedge I(5, 6), 14)$     |

Supremum der unteren Schranken:

|           |    |                                                                                                                                                                                                                                                                                                         |                                                       |
|-----------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 1         | 1  | $\emptyset \neq \mathcal{A} \subseteq \mathbb{R}$                                                                                                                                                                                                                                                       | $A$                                                   |
| 2         | 2  | $\exists L \in \mathbb{R} \forall A \in \mathcal{A} (L \leq_{\mathbb{R}} A)$                                                                                                                                                                                                                            | $A$                                                   |
| 2         | 3  | $L \in \mathbb{R} \wedge \forall A \in \mathcal{A} (L \leq_{\mathbb{R}} A)$                                                                                                                                                                                                                             | $A$                                                   |
| 1,2       | 4  | $\emptyset \neq \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) \subseteq \mathbb{R} \wedge$<br>$\exists A_0 \in \mathbb{R} \forall V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) (V \leq_{\mathbb{R}} A_0)$                                                                          |                                                       |
|           |    |                                                                                                                                                                                                                                                                                                         | <i>Th.</i> 19.3.3.4(i)(1, 3)                          |
| 1,2       | 5  | $\exists I \in \mathbb{R} ($<br>$\forall V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) (V \leq_{\mathbb{R}} I) \wedge$<br>$\forall W \in \mathbb{R} ($<br>$\forall V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) (V \leq_{\mathbb{R}} W) \rightarrow I \leq_{\mathbb{R}} W))$ | <i>Th.</i> 19.3.3.2(4)                                |
| 1,2       | 6  | $I \in \mathbb{R} \wedge$<br>$\forall V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) (V \leq_{\mathbb{R}} I) \wedge$<br>$\forall W \in \mathbb{R} ($<br>$\forall V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) (V \leq_{\mathbb{R}} W) \rightarrow I \leq_{\mathbb{R}} W)$     | $A$                                                   |
| 7         | 7  | $A \in \mathcal{A}$                                                                                                                                                                                                                                                                                     | $A$                                                   |
| 1,7       | 8  | $A \in \mathbb{R}$                                                                                                                                                                                                                                                                                      | <i>Th.</i> 3.5.2.6(1, 7)                              |
| 1,7       | 9  | $\forall V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A}) (V \leq_{\mathbb{R}} A)$                                                                                                                                                                                                            | <i>Th.</i> 13.2.1.6(1, 7)                             |
| 1,2,7     | 10 | $I \leq_{\mathbb{R}} A$                                                                                                                                                                                                                                                                                 | $\forall E(6, 8, 9)$                                  |
| 1,2       | 11 | $\forall A \in \mathcal{A} (I \leq_{\mathbb{R}} A)$                                                                                                                                                                                                                                                     | $\forall I(\rightarrow I(7, 10))$                     |
| 12        | 12 | $V \in \mathbb{R}$                                                                                                                                                                                                                                                                                      | $A$                                                   |
| 13        | 13 | $\forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A)$                                                                                                                                                                                                                                                     | $A$                                                   |
| 1,12,13   | 14 | $V \in \text{LB}_{\mathbb{R}, \leq \mathbb{R}}(\mathcal{A})$                                                                                                                                                                                                                                            |                                                       |
|           |    |                                                                                                                                                                                                                                                                                                         | <i>Th.</i> 13.2.1.7(1, 12, 13)                        |
| 1,2,12,13 | 15 | $V \leq_{\mathbb{R}} I$                                                                                                                                                                                                                                                                                 | $\forall E(6, 14)$                                    |
| 1,2       | 16 | $\forall V \in \mathbb{R} (\forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A) \rightarrow V \leq_{\mathbb{R}} I)$                                                                                                                                                                                        |                                                       |
|           |    |                                                                                                                                                                                                                                                                                                         | $\forall I(\rightarrow I(12, \rightarrow I(13, 15)))$ |
| 1,2       | 17 | $\exists I \in \mathbb{R} ($<br>$\forall A \in \mathcal{A} (I \leq_{\mathbb{R}} A) \wedge$<br>$\forall V \in \mathbb{R} (\forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A)$<br>$\rightarrow V \leq_{\mathbb{R}} I))$                                                                                    | $\exists I(6, \wedge I(11, 16))$                      |
| 1,2       | 18 | $\exists I \in \mathbb{R} ($<br>$\forall A \in \mathcal{A} (I \leq_{\mathbb{R}} A) \wedge$<br>$\forall V \in \mathbb{R} (\forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A)$<br>$\rightarrow V \leq_{\mathbb{R}} I))$                                                                                    | $\exists E(5, 6, 17)$                                 |

$$\begin{array}{ll}
1,2 & 19 \exists I \in \mathbb{R} ( \quad \quad \quad \exists E(2, 3, 18) \\
& \quad \quad \quad \forall A \in \mathcal{A} (I \leq_{\mathbb{R}} A) \wedge \\
& \quad \quad \quad \forall V \in \mathbb{R} (\forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A) \\
& \quad \quad \quad \rightarrow V \leq_{\mathbb{R}} I))
\end{array}$$

□

Auch hier wird der allgemeine Infimumsterm erst nach diesem Existenzsatz eingeführt.

**Theorem 19.3.3.5 (Charakterisierung des reellen Infimums).**

**Mengen:**  $\mathcal{A}, A, L, V$   
**Relationsoperatoren:**  $\leq_{\mathbb{R}}$

$$\begin{array}{l}
\emptyset \neq \mathcal{A} \subseteq \mathbb{R}, \exists L \in \mathbb{R} \forall A \in \mathcal{A} (L \leq_{\mathbb{R}} A) \\
\vdash \inf_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A}) \in \mathbb{R} \wedge \\
\forall A \in \mathcal{A} (\inf_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A}) \leq_{\mathbb{R}} A) \wedge \\
\forall V \in \mathbb{R} (\forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A) \rightarrow V \leq_{\mathbb{R}} \inf_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A})).
\end{array}$$

*Beweis.*

**Existenz eines größten Elements der Schrankenmenge** Th. 19.3.3.5(i)

$$\begin{array}{l}
\emptyset \neq \mathcal{A} \subseteq \mathbb{R}, \exists L \in \mathbb{R} \forall A \in \mathcal{A} (L \leq_{\mathbb{R}} A) \\
\vdash \exists I \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A}) \forall V \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A}) (V \leq_{\mathbb{R}} I)
\end{array}$$

$$\begin{array}{ll}
1 & 1 \emptyset \neq \mathcal{A} \subseteq \mathbb{R} \quad A \\
2 & 2 \exists L \in \mathbb{R} \forall A \in \mathcal{A} (L \leq_{\mathbb{R}} A) \quad A \\
1,2 & 3 \exists I \in \mathbb{R} (\forall A \in \mathcal{A} (I \leq_{\mathbb{R}} A) \wedge \forall V \in \mathbb{R} (\forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A) \rightarrow V \leq_{\mathbb{R}} I)) \quad Th. 19.3.3.4(1,2) \\
1,2 & 4 I \in \mathbb{R} \wedge \forall A \in \mathcal{A} (I \leq_{\mathbb{R}} A) \wedge \forall V \in \mathbb{R} (\forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A) \rightarrow V \leq_{\mathbb{R}} I) \quad A \\
1,2 & 5 I \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A}) \quad Th. 13.2.1.7(1,4) \\
6 & 6 V \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A}) \quad A \\
1,6 & 7 V \in \mathbb{R} \wedge \forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A) \quad Th. 13.2.1.5(1,6) \\
1,2,6 & 8 V \leq_{\mathbb{R}} I \quad \forall E(4,7) \\
1,2 & 9 \forall V \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A}) (V \leq_{\mathbb{R}} I) \quad \forall I(\rightarrow I(6,8)) \\
1,2 & 10 \exists I \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A}) \forall V \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A}) (V \leq_{\mathbb{R}} I) \exists I(5,9) \\
1,2 & 11 \exists I \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A}) \forall V \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}} (\mathcal{A}) (V \leq_{\mathbb{R}} I) \exists E(3,4,10)
\end{array}$$

Eigenschaften des definierten Infimums:

$$1 \quad 1 \emptyset \neq \mathcal{A} \subseteq \mathbb{R} \quad A$$

$$\begin{array}{ll}
2 & 2 \exists L \in \mathbb{R} \forall A \in \mathcal{A} (L \leq_{\mathbb{R}} A) \quad A \\
1,2 & 3 \exists I \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) \forall V \in \text{LB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) (V \leq_{\mathbb{R}} I) \quad Th. 19.3.3.5(i)(1, 2) \\
1,2 & 4 \inf_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) \in \mathbb{R} \quad Th. 13.2.1.18(1, 3) \\
1,2 & 5 \forall A \in \mathcal{A} (\inf_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) \leq_{\mathbb{R}} A) \quad Th. 13.2.1.19(1, 3) \\
1,2 & 6 \forall V \in \mathbb{R} (\forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A) \rightarrow V \leq_{\mathbb{R}} \inf_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A})) \quad Th. 13.2.1.20(1, 3) \\
1,2 & 7 \inf_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) \in \mathbb{R} \wedge \wedge I(4, \wedge I(5, 6)) \\
& \quad \forall A \in \mathcal{A} (\inf_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}) \leq_{\mathbb{R}} A) \wedge \\
& \quad \forall V \in \mathbb{R} (\forall A \in \mathcal{A} (V \leq_{\mathbb{R}} A) \\
& \quad \quad \rightarrow V \leq_{\mathbb{R}} \inf_{\mathbb{R}, \leq_{\mathbb{R}}}(\mathcal{A}))
\end{array}$$

□

## Kapitel 4

# Arithmetik der Schnitte

### 4.1 Addition und additive Inverse

**Definition 19.4.1.1** (Addition reeller Schnitte).

*Mengen:*  $A, B, a, b, q$   
*Funktionsoperatoren:*  $\oplus$   
*Neues Symbol:*  
*Reelle Addition:*  $\oplus$

$$A, B \in \mathbb{R} \vdash A \oplus B := \{q \in \mathbb{Q} \mid \exists a \in A \exists b \in B (q < a + b)\}$$

**Definition 19.4.1.2** (Additive Null der reellen Zahlen).

*Mengen:*  $0_{\mathbb{R}}$   
*Neues Symbol:*  
*Reelle Null:*  $0_{\mathbb{R}}$

$$0_{\mathbb{R}} := \hat{0}$$

**Definition 19.4.1.3** (Additives Inverses eines reellen Schnittes).

*Mengen:*  $A, q, r$   
*Funktionsoperatoren:*  $\ominus \cdot$   
*Neues Symbol:*  
*Reelle Negation:*  $\ominus \cdot$

$$A \in \mathbb{R} \vdash \ominus A := \{q \in \mathbb{Q} \mid \exists r \in \mathbb{Q} (r \notin A \wedge q < -r)\}$$

**Theorem 19.4.1.1** (Abgeschlossenheit der additiven Schnittoperationen).

*Mengen:*  $A, B$   
*Funktionsoperatoren:*  $\oplus, \ominus \cdot$

$$A, B \in \mathbb{R} \vdash A \oplus B \in \mathbb{R} \wedge \ominus A \in \mathbb{R} \wedge 0_{\mathbb{R}} \in \mathbb{R}$$

*Beweis.*

**Summe zweier Schnitte**

Th. 19.4.1.1(i)

$$A, B \in \mathbb{R} \vdash \text{Cut}_{\mathbb{Q}}(A \oplus B)$$

|       |    |                                                                                             |                                                       |
|-------|----|---------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 1     | 1  | $A, B \in \mathbb{R}$                                                                       | $A$                                                   |
| 1     | 2  | $\text{Cut}_{\mathbb{Q}}(A) \wedge \text{Cut}_{\mathbb{Q}}(B)$                              | Th. 19.2.1.1(1)                                       |
| 1     | 3  | $\exists a_0 \in A \wedge \exists b_0 \in B$                                                | Def. 19.2.1.1(2)                                      |
| 1     | 4  | $a_0 \in A \wedge b_0 \in B$                                                                | $A$                                                   |
| 1     | 5  | $a_0 + b_0 - 1 \in \mathbb{Q} \wedge a_0 + b_0 - 1 < a_0 + b_0$                             | Def. 18.5.1(2, 4)                                     |
| 1     | 6  | $a_0 + b_0 - 1 \in A \oplus B$                                                              | Def. 19.4.1.1(4, 5)                                   |
| 1     | 7  | $A \oplus B \neq \emptyset$                                                                 | Th. 3.6.3.5(6)                                        |
| 1     | 8  | $A \oplus B \neq \emptyset$                                                                 | $\exists E(3, 4, 7)$                                  |
| 1     | 9  | $\exists \alpha \in \mathbb{Q} \setminus A \wedge \exists \beta \in \mathbb{Q} \setminus B$ | Def. 19.2.1.1(2)                                      |
| 1     | 10 | $\alpha \in \mathbb{Q} \setminus A \wedge \beta \in \mathbb{Q} \setminus B$                 | $A$                                                   |
| 1     | 11 | $\forall a \in A (a \leq \alpha) \wedge \forall b \in B (b \leq \beta)$                     |                                                       |
|       |    |                                                                                             | Def. 19.2.1.1(2, 10)                                  |
| 1     | 12 | $\alpha + \beta \notin A \oplus B$                                                          |                                                       |
|       |    |                                                                                             | Th. 18.4.2.1(Def. 19.4.1.1(10, 11))                   |
| 1     | 13 | $A \oplus B \neq \mathbb{Q}$                                                                |                                                       |
|       |    |                                                                                             | Th. 3.4.2.2(Th. 18.3.2.4(10), 12)                     |
| 14    | 14 | $q \in A \oplus B$                                                                          | $A$                                                   |
| 15    | 15 | $s \in \mathbb{Q} \wedge s < q$                                                             | $A$                                                   |
| 14    | 16 | $\exists a \in A \exists b \in B (q < a + b)$                                               | Def. 19.4.1.1(14)                                     |
| 14    | 17 | $a \in A \wedge b \in B \wedge q < a + b$                                                   | $A$                                                   |
| 14,15 | 18 | $s < a + b$                                                                                 | Th. 18.4.1.4(15, 17)                                  |
| 14,15 | 19 | $s \in A \oplus B$                                                                          |                                                       |
|       |    |                                                                                             | Def. 19.4.1.1(15, 17, 18)                             |
| 14,15 | 20 | $s \in A \oplus B$                                                                          | $\exists E(16, 17, 19)$                               |
| 1     | 21 | $\forall q \in A \oplus B \forall s \in \mathbb{Q} (s < q \rightarrow s \in A \oplus B)$    |                                                       |
|       |    |                                                                                             | $\forall I(\rightarrow I(14, \rightarrow I(15, 20)))$ |
| 22    | 22 | $q \in A \oplus B$                                                                          | $A$                                                   |
| 22    | 23 | $\exists a \in A \exists b \in B (q < a + b)$                                               | Def. 19.4.1.1(22)                                     |
| 22    | 24 | $a \in A \wedge b \in B \wedge q < a + b$                                                   | $A$                                                   |
| 22    | 25 | $\exists r \in \mathbb{Q} (q < r \wedge r < a + b)$                                         | Th. 18.6.1.2(24)                                      |
| 22    | 26 | $r \in \mathbb{Q} \wedge q < r \wedge r < a + b$                                            | $A$                                                   |
| 22    | 27 | $r \in A \oplus B$                                                                          |                                                       |
|       |    |                                                                                             | Def. 19.4.1.1(24, 26)                                 |
| 22    | 28 | $\exists r \in A \oplus B (q < r)$                                                          | $\exists I(26, 27)$                                   |
| 22    | 29 | $\exists r \in A \oplus B (q < r)$                                                          | $\exists E(25, 26, 28)$                               |

|       |                                                             |                                    |
|-------|-------------------------------------------------------------|------------------------------------|
| 22 30 | $\exists r \in A \oplus B (q < r)$                          | $\exists E(23, 24, 29)$            |
| 1 31  | $\forall q \in A \oplus B \exists r \in A \oplus B (q < r)$ | $\forall I(\rightarrow I(22, 30))$ |
| 1 32  | $A \oplus B \subseteq \mathbb{Q}$                           | <i>Def.</i> 19.4.1.1(1)            |
| 1 33  | $\text{Cut}_{\mathbb{Q}}(A \oplus B)$                       |                                    |

*Def.* 19.2.1.1(32, 8, 13, 21, 31)

## Negation eines Schnittes

Th. 19.4.1.1(ii)

$$A \in \mathbb{R} \vdash \text{Cut}_{\mathbb{Q}}(\ominus A)$$

|      |                                          |                         |
|------|------------------------------------------|-------------------------|
| 1 1  | $A \in \mathbb{R}$                       | $A$                     |
| 1 2  | $\text{Cut}_{\mathbb{Q}}(A)$             | Th. 19.2.1.1(1)         |
| 1 3  | $\exists r_0 \in \mathbb{Q} \setminus A$ | Def. 19.2.1.1(2)        |
| 1 4  | $r_0 \in \mathbb{Q} \wedge r_0 \notin A$ | $A$                     |
| 1 5  | $-r_0 - 1 < -r_0$                        | Def. 18.5.1(4)          |
| 1 6  | $-r_0 - 1 \in \ominus A$                 | Def. 19.4.1.3(4,5)      |
| 1 7  | $\ominus A \neq \emptyset$               | Th. 3.6.3.5(6)          |
| 1 8  | $\ominus A \neq \emptyset$               | $\exists E(3, 4, 7)$    |
| 1 9  | $\exists a_0 \in A$                      | <i>Def.</i> 19.2.1.1(2) |
| 1 10 | $a_0 \in A$                              | $A$                     |
| 1 11 | $-a_0 \notin \ominus A$                  | Def. 19.4.1.3(2,10)     |
| 1 12 | $\ominus A \neq \mathbb{Q}$              |                         |

*Th.* 3.4.2.2(*Th.* 18.3.1.4(10), 11)

|          |                                                                                        |                          |
|----------|----------------------------------------------------------------------------------------|--------------------------|
| 1 13     | $\ominus A \neq \mathbb{Q}$                                                            | $\exists E(9, 10, 12)$   |
| 14 14    | $q \in \ominus A$                                                                      | $A$                      |
| 15 15    | $s \in \mathbb{Q} \wedge s < q$                                                        | $A$                      |
| 14 16    | $\exists r \in \mathbb{Q} (r \notin A \wedge q < -r)$                                  | <i>Def.</i> 19.4.1.3(14) |
| 14 17    | $r \in \mathbb{Q} \wedge r \notin A \wedge q < -rA$                                    |                          |
| 14,15 18 | $s < -r$                                                                               | Th. 18.4.1.4(15,17)      |
| 14,15 19 | $s \in \ominus A$                                                                      | Def. 19.4.1.3(15,17,18)  |
| 14,15 20 | $s \in \ominus A$                                                                      | $\exists E(16, 17, 19)$  |
| 1 21     | $\forall q \in \ominus A \forall s \in \mathbb{Q} (s < q \rightarrow s \in \ominus A)$ |                          |

$\forall I(\rightarrow I(14, \rightarrow I(15, 20)))$

|       |                                                           |                                    |
|-------|-----------------------------------------------------------|------------------------------------|
| 22 22 | $q \in \ominus A$                                         | $A$                                |
| 22 23 | $\exists r \in \mathbb{Q} (r \notin A \wedge q < -r)$     | <i>Def.</i> 19.4.1.3(22)           |
| 22 24 | $r \in \mathbb{Q} \wedge r \notin A \wedge q < -rA$       |                                    |
| 22 25 | $\exists s \in \mathbb{Q} (q < s \wedge s < -r)$          | <i>Th.</i> 18.6.1.2(24)            |
| 22 26 | $s \in \mathbb{Q} \wedge q < s \wedge s < -rA$            |                                    |
| 22 27 | $s \in \ominus A$                                         | Def. 19.4.1.3(24,26)               |
| 22 28 | $\exists s \in \ominus A (q < s)$                         | $\exists I(26, 27)$                |
| 22 29 | $\exists s \in \ominus A (q < s)$                         | $\exists E(25, 26, 28)$            |
| 22 30 | $\exists s \in \ominus A (q < s)$                         | $\exists E(23, 24, 29)$            |
| 1 31  | $\forall q \in \ominus A \exists s \in \ominus A (q < s)$ | $\forall I(\rightarrow I(22, 30))$ |

|                                           |                              |
|-------------------------------------------|------------------------------|
| 1 32 $\ominus A \subseteq \mathbb{Q}$     | Def. 19.4.1.3(1)             |
| 1 33 $\text{Cut}_{\mathbb{Q}}(\ominus A)$ | Def. 19.2.1.1(32,8,13,21,31) |

Nullschnitt und Zusammenfassung:

|                                                                                                                                  |                                   |
|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1 1 $A, B \in \mathbb{R}$                                                                                                        | $A$                               |
| 1 2 $A \oplus B \in \mathbb{R}$                                                                                                  | Th. 19.2.1.1(Th. 19.4.1.1(i)(1))  |
| 1 3 $\ominus A \in \mathbb{R}$                                                                                                   | Th. 19.2.1.1(Th. 19.4.1.1(ii)(1)) |
| 4 $0 \in \mathbb{Q}$                                                                                                             | Th. 18.2.3.7                      |
| 5 $\text{Cut}_{\mathbb{Q}}(\hat{0})$                                                                                             | Th. 19.2.2.1(4)                   |
| 6 $0_{\mathbb{R}} \in \mathbb{R}$                                                                                                | Def. 19.4.1.2(Th. 19.2.1.1(5))    |
| 1 7 $A \oplus B \in \mathbb{R} \wedge \ominus A \in \mathbb{R} \wedge 0_{\mathbb{R}} \in \mathbb{R} \wedge I(2, \wedge I(3, 6))$ |                                   |

□

**Theorem 19.4.1.2 (Rationale Randapproximation eines Schnittes).**

**Mengen:**  $A, a, r, h$

$$A \in \mathbb{R}, h \in \mathbb{Q}, 0 < h \vdash \\ \exists a \in A \exists r \in \mathbb{Q} (r \notin A \wedge a < r \wedge r - a < h).$$

*Beweis.*

**Ein Gitterpunkt liegt außerhalb des Schnittes** Th. 19.4.1.2(i)

$$A \in \mathbb{R}, h \in \mathbb{Q}, 0 < h, p \in A, u \in \mathbb{Q} \setminus A \\ \vdash \exists m \in \mathbb{N} \left( u < p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m)) \frac{h}{2} \right)$$

|                                                                                                                            |                                   |
|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1 1 $A \in \mathbb{R}$                                                                                                     | $A$                               |
| 2 2 $h \in \mathbb{Q}$                                                                                                     | $A$                               |
| 3 3 $0 < h$                                                                                                                | $A$                               |
| 4 4 $p \in A$                                                                                                              | $A$                               |
| 5 5 $u \in \mathbb{Q} \setminus A$                                                                                         | $A$                               |
| 1,4 6 $p \in \mathbb{Q}$                                                                                                   |                                   |
|                                                                                                                            | Def. 19.2.1.1(Th. 19.2.1.1(1), 4) |
| 2,3 7 $\frac{h}{2} \in \mathbb{Q} \wedge 0 < \frac{h}{2}$                                                                  | Def. 18.5.1(2, 3)                 |
| 1,2,3,4,5 8 $(u - p) \left( \frac{h}{2} \right)^{-1} \in \mathbb{Q}$                                                       |                                   |
|                                                                                                                            | Def. 18.5.1(2, 5, 6, 7)           |
| 1,2,3,4,5 9 $\exists m \in \mathbb{N} (u - p) \left( \frac{h}{2} \right)^{-1} < \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m))$ | Th. 18.6.1.3(8)                   |
| 1,2,3,4,5 10 $m \in \mathbb{N} \wedge (u - p)(h/2)^{-1} < \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m))A$                      |                                   |

|           |    |                                                                                                         |                            |
|-----------|----|---------------------------------------------------------------------------------------------------------|----------------------------|
| 1,2,3,4,5 | 11 | $u < p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m)) \frac{h}{2}$                                         | <i>Th.</i> 18.4.2.3(7, 10) |
| 1,2,3,4,5 | 12 | $\exists m \in \mathbb{N} \left( u < p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m)) \frac{h}{2} \right)$ | $\exists I(10, 11)$        |
| 1,2,3,4,5 | 13 | $\exists m \in \mathbb{N} \left( u < p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m)) \frac{h}{2} \right)$ | $\exists E(9, 10, 12)$     |

### Erster äußerer Gitterpunkt

Th. 19.4.1.2(ii)

$$\begin{aligned}
& A \in \mathbb{R}, \quad h \in \mathbb{Q}, \quad 0 < h, \quad p \in A, \quad u \in \mathbb{Q} \setminus A \\
& \vdash \exists m_0 \in \mathbb{N} \setminus \{0\} \left( p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m_0)) \frac{h}{2} \notin A \wedge \right. \\
& \quad \left. p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(\text{pred}(m_0))) \frac{h}{2} \in A \right)
\end{aligned}$$

|           |   |                                                                                                         |     |
|-----------|---|---------------------------------------------------------------------------------------------------------|-----|
| 1         | 1 | $A \in \mathbb{R}$                                                                                      | $A$ |
| 2         | 2 | $h \in \mathbb{Q}$                                                                                      | $A$ |
| 3         | 3 | $0 < h$                                                                                                 | $A$ |
| 4         | 4 | $p \in A$                                                                                               | $A$ |
| 5         | 5 | $u \in \mathbb{Q} \setminus A$                                                                          | $A$ |
| 1,2,3,4,5 | 6 | $\exists m \in \mathbb{N} \left( u < p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m)) \frac{h}{2} \right)$ |     |

*Th.* 19.4.1.2(i)(1, 2, 3, 4, 5)

|           |   |                                                                                                                      |  |
|-----------|---|----------------------------------------------------------------------------------------------------------------------|--|
| 1,2,3,4,5 | 7 | $M := \left\{ n \in \mathbb{N} \mid p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)) \frac{h}{2} \notin A \right\} = I$ |  |
| 1,2,3,4,5 | 8 | $M \neq \emptyset$                                                                                                   |  |

*Def.* 19.2.1.1(5, 6, 7)

|           |    |                                                  |                            |
|-----------|----|--------------------------------------------------|----------------------------|
| 1,2,3,4,5 | 9  | $\exists m_0 \in M \forall n \in M (m_0 \leq n)$ | <i>Th.</i> 10.4.5.26(7, 8) |
| 1,2,3,4,5 | 10 | $m_0 \in M \wedge \forall n \in M (m_0 \leq n)$  | $A$                        |
| 1,2,3,4,5 | 11 | $m_0 \neq 0$                                     |                            |

*Th.* 10.4.4.2(4, 7, 10)

|           |    |                                   |  |
|-----------|----|-----------------------------------|--|
| 1,2,3,4,5 | 12 | $\text{pred}(m_0) \in \mathbb{N}$ |  |
|-----------|----|-----------------------------------|--|

*Th.* 10.4.2.17(10, 11)

|           |    |                          |  |
|-----------|----|--------------------------|--|
| 1,2,3,4,5 | 13 | $\text{pred}(m_0) < m_0$ |  |
|-----------|----|--------------------------|--|

*Th.* 10.4.2.21(10, 11, 12)

|           |    |                                                                                  |  |
|-----------|----|----------------------------------------------------------------------------------|--|
| 1,2,3,4,5 | 14 | $p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(\text{pred}(m_0))) \frac{h}{2} \in A$ |  |
|-----------|----|----------------------------------------------------------------------------------|--|

*Th.* 10.4.5.26(7, 10, 13)

|           |    |                                                                        |                |
|-----------|----|------------------------------------------------------------------------|----------------|
| 1,2,3,4,5 | 15 | $p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m_0)) \frac{h}{2} \notin A$ | $\wedge E(10)$ |
|-----------|----|------------------------------------------------------------------------|----------------|

|           |    |                                      |                           |
|-----------|----|--------------------------------------|---------------------------|
| 1,2,3,4,5 | 16 | $m_0 \in \mathbb{N} \setminus \{0\}$ | <i>Th.</i> 3.12.1(10, 11) |
|-----------|----|--------------------------------------|---------------------------|

|           |    |                                                                                                                                                                                                                        |                                   |
|-----------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1,2,3,4,5 | 17 | $\exists m_0 \in \mathbb{N} \setminus \{0\} \left( p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m_0)) \frac{h}{2} \notin A \wedge p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(\text{pred}(m_0))) \frac{h}{2} \in A \right)$ | $\exists I(16, \wedge I(15, 14))$ |
|-----------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|

|           |    |                                                                                                                                                                                                                        |                        |
|-----------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1,2,3,4,5 | 18 | $\exists m_0 \in \mathbb{N} \setminus \{0\} \left( p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m_0)) \frac{h}{2} \notin A \wedge p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(\text{pred}(m_0))) \frac{h}{2} \in A \right)$ | $\exists E(9, 10, 17)$ |
|-----------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|

Schluss:

|   |   |                    |     |
|---|---|--------------------|-----|
| 1 | 1 | $A \in \mathbb{R}$ | $A$ |
| 2 | 2 | $h \in \mathbb{Q}$ | $A$ |
| 3 | 3 | $0 < h$            | $A$ |

|       |    |                                                                                                                                                                                                                        |  |                          |                                                                |
|-------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------|----------------------------------------------------------------|
| 1     | 4  | $\exists p \in A$                                                                                                                                                                                                      |  |                          |                                                                |
|       |    |                                                                                                                                                                                                                        |  |                          | <i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1(1))                  |
| 1     | 5  | $\exists u \in \mathbb{Q} \setminus A$                                                                                                                                                                                 |  |                          |                                                                |
|       |    |                                                                                                                                                                                                                        |  |                          | <i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1(1))                  |
| 1     | 6  | $p \in A$                                                                                                                                                                                                              |  | $A$                      |                                                                |
| 1     | 7  | $u \in \mathbb{Q} \setminus A$                                                                                                                                                                                         |  | $A$                      |                                                                |
| 1,2,3 | 8  | $\exists m_0 \in \mathbb{N} \setminus \{0\} \left( p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m_0)) \frac{h}{2} \notin A \wedge p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(\text{pred}(m_0))) \frac{h}{2} \in A \right)$ |  |                          |                                                                |
|       |    |                                                                                                                                                                                                                        |  |                          | <i>Th.</i> 19.4.1.2(ii)(1, 2, 3, 6, 7)                         |
| 1,2,3 | 9  | $m_0 \in \mathbb{N} \setminus \{0\} \wedge p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m_0)) \frac{h}{2} \notin A \wedge p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(\text{pred}(m_0))) \frac{h}{2} \in A$                 |  |                          |                                                                |
|       |    |                                                                                                                                                                                                                        |  |                          | $A$                                                            |
| 1,2,3 | 10 | $\iota_{\mathbb{Z}}(m_0) = \iota_{\mathbb{Z}}(\text{pred}(m_0)) + 1$                                                                                                                                                   |  | <i>Th.</i> 10.4.2.21(9)  |                                                                |
| 1,2,3 | 11 | $\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m_0)) = \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(\text{pred}(m_0))) + 1$                                                                                                           |  | <i>Th.</i> 18.3.4.10(10) |                                                                |
| 1,2,3 | 12 | $a := p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(\text{pred}(m_0))) \frac{h}{2} \in A$                                                                                                                                  |  | $= E(9)$                 |                                                                |
| 1,2,3 | 13 | $r := p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m_0)) \frac{h}{2} \notin A$                                                                                                                                            |  | $= E(9)$                 |                                                                |
| 1,2,3 | 14 | $r - a = h/2$                                                                                                                                                                                                          |  |                          |                                                                |
|       |    |                                                                                                                                                                                                                        |  |                          | <i>Def.</i> 18.5.1(11, 12, 13)                                 |
| 1,2,3 | 15 | $0 < r - a < h$                                                                                                                                                                                                        |  |                          |                                                                |
|       |    |                                                                                                                                                                                                                        |  |                          | <i>Def.</i> 18.5.1(2, 3, 14)                                   |
| 1,2,3 | 16 | $a < r$                                                                                                                                                                                                                |  | <i>Th.</i> 18.4.2.1(15)  |                                                                |
| 1,2,3 | 17 | $\exists a \in A \exists r \in \mathbb{Q} (r \notin A \wedge a < r \wedge r - a < h)$                                                                                                                                  |  |                          |                                                                |
|       |    |                                                                                                                                                                                                                        |  |                          | $\exists I(12, \exists I(13, \wedge I(13, \wedge I(16, 15))))$ |
| 1,2,3 | 18 | $\exists a \in A \exists r \in \mathbb{Q} (r \notin A \wedge a < r \wedge r - a < h) \exists E(8, 9, 17)$                                                                                                              |  |                          |                                                                |
| 1,2,3 | 19 | $\exists a \in A \exists r \in \mathbb{Q} (r \notin A \wedge a < r \wedge r - a < h) \exists E(5, 7, 18)$                                                                                                              |  |                          |                                                                |
| 1,2,3 | 20 | $\exists a \in A \exists r \in \mathbb{Q} (r \notin A \wedge a < r \wedge r - a < h) \exists E(4, 6, 19)$                                                                                                              |  |                          |                                                                |

□

### Theorem 19.4.1.3 (Gesetze der reellen Addition).

**Mengen:**  $A, B, C$

**Funktionsoperatoren:**  $\oplus, \ominus$ .

$$A, B, C \in \mathbb{R} \vdash (A \oplus B) \oplus C = A \oplus (B \oplus C),$$

$$A \oplus B = B \oplus A, \quad A \oplus 0_{\mathbb{R}} = A, \quad A \oplus \ominus A = 0_{\mathbb{R}}.$$

*Beweis.*

**Assoziativität**

*Th.* 19.4.1.3(i)

$$A, B, C \in \mathbb{R} \vdash (A \oplus B) \oplus C = A \oplus (B \oplus C)$$

|   |   |                                                              |                        |
|---|---|--------------------------------------------------------------|------------------------|
| 1 | 1 | $A, B, C \in \mathbb{R}$                                     | $A$                    |
| 1 | 2 | $A \oplus B \in \mathbb{R} \wedge B \oplus C \in \mathbb{R}$ | <i>Th.</i> 19.4.1.1(1) |

|     |                                                                          |                         |                                                   |
|-----|--------------------------------------------------------------------------|-------------------------|---------------------------------------------------|
| 3   | $3 \ q \in (A \oplus B) \oplus C$                                        | $A$                     |                                                   |
| 1,3 | $4 \ \exists x \in A \oplus B \exists c \in C (q < x + c)$               | $Def. \ 19.4.1.1(2, 3)$ |                                                   |
| 1,3 | $5 \ \exists a \in A \exists b \in B \exists c \in C (q < (a + b) + c)$  | $Def. \ 19.4.1.1(4)$    |                                                   |
| 1,3 | $6 \ \exists a \in A \exists y \in B \oplus C (q < a + y)$               | $Th. \ 18.3.4.1(5)$     |                                                   |
| 1,3 | $7 \ q \in A \oplus (B \oplus C)$                                        | $Def. \ 19.4.1.1(2, 6)$ |                                                   |
| 1   | $8 \ (A \oplus B) \oplus C \subseteq A \oplus (B \oplus C)$              |                         | $Def. \ 3.5.1.1(\forall I(\rightarrow I(3, 7)))$  |
| 9   | $9 \ q \in A \oplus (B \oplus C)$                                        | $A$                     |                                                   |
| 1,9 | $10 \ \exists a \in A \exists b \in B \exists c \in C (q < a + (b + c))$ | $Def. \ 19.4.1.1(2, 9)$ |                                                   |
| 1,9 | $11 \ \exists x \in A \oplus B \exists c \in C (q < x + c)$              | $Th. \ 18.3.4.1(10)$    |                                                   |
| 1,9 | $12 \ q \in (A \oplus B) \oplus C$                                       |                         | $Def. \ 19.4.1.1(2, 11)$                          |
| 1   | $13 \ A \oplus (B \oplus C) \subseteq (A \oplus B) \oplus C$             |                         | $Def. \ 3.5.1.1(\forall I(\rightarrow I(9, 12)))$ |
| 1   | $14 \ (A \oplus B) \oplus C = A \oplus (B \oplus C)$                     | $Th. \ 3.5.3.5(8, 13)$  |                                                   |

### Kommutativität

Th. 19.4.1.3(ii)

$$A, B \in \mathbb{R} \vdash A \oplus B = B \oplus A$$

|     |                                                   |                                                  |  |
|-----|---------------------------------------------------|--------------------------------------------------|--|
| 1   | $1 \ A, B \in \mathbb{R}$                         | $A$                                              |  |
| 2   | $2 \ q \in A \oplus B$                            | $A$                                              |  |
| 1,2 | $3 \ \exists a \in A \exists b \in B (q < a + b)$ | $Def. \ 19.4.1.1(1, 2)$                          |  |
| 1,2 | $4 \ \exists b \in B \exists a \in A (q < b + a)$ | $Th. \ 18.3.4.2(3)$                              |  |
| 1,2 | $5 \ q \in B \oplus A$                            | $Def. \ 19.4.1.1(1, 4)$                          |  |
| 1   | $6 \ A \oplus B \subseteq B \oplus A$             | $Def. \ 3.5.1.1(\forall I(\rightarrow I(2, 5)))$ |  |
| 1   | $7 \ B \oplus A \subseteq A \oplus B$             | $= E(6)$                                         |  |
| 1   | $8 \ A \oplus B = B \oplus A$                     | $Th. \ 3.5.3.5(6, 7)$                            |  |

### Nullgesetz

Th. 19.4.1.3(iii)

$$A \in \mathbb{R} \vdash A \oplus 0_{\mathbb{R}} = A$$

|     |                                                 |                                                  |  |
|-----|-------------------------------------------------|--------------------------------------------------|--|
| 1   | $1 \ A \in \mathbb{R}$                          | $A$                                              |  |
| 2   | $2 \ q \in A \oplus 0_{\mathbb{R}}$             | $A$                                              |  |
| 1,2 | $3 \ \exists a \in A \exists r < 0 (q < a + r)$ | $Def. \ 19.4.1.1(Def. \ 19.4.1.2, 2)$            |  |
| 1,2 | $4 \ q < a$                                     | $Th. \ 18.4.2.1(3)$                              |  |
| 1,2 | $5 \ q \in A$                                   | $Def. \ 19.2.1.1(Th. \ 19.2.1.1(1), 3, 4)$       |  |
| 1   | $6 \ A \oplus 0_{\mathbb{R}} \subseteq A$       | $Def. \ 3.5.1.1(\forall I(\rightarrow I(2, 5)))$ |  |
| 7   | $7 \ q \in A$                                   | $A$                                              |  |
| 1,7 | $8 \ \exists a \in A (q < a)$                   | $Def. \ 19.2.1.1(Th. \ 19.2.1.1(1), 7)$          |  |
| 1,7 | $9 \ a \in A \wedge q < a$                      | $A$                                              |  |
| 1,7 | $10 \ q - a < 0$                                | $Th. \ 18.4.2.1(9)$                              |  |

|        |                                                     |                                                   |
|--------|-----------------------------------------------------|---------------------------------------------------|
| 1,7 11 | $\exists r \in \mathbb{Q} (q - a < r \wedge r < 0)$ | Th. 18.6.1.2(10)                                  |
| 1,7 12 | $r \in \mathbb{Q} \wedge q - a < r \wedge r < 0$    | $A$                                               |
| 1,7 13 | $r \in 0_{\mathbb{R}}$                              | Def. 19.4.1.2(Def. 19.2.2.1(12))                  |
| 1,7 14 | $q < a + r$                                         | Th. 18.4.2.1(12)                                  |
| 1,7 15 | $q \in A \oplus 0_{\mathbb{R}}$                     | Def. 19.4.1.1(9,12,13,14)                         |
| 1,7 16 | $q \in A \oplus 0_{\mathbb{R}}$                     | $\exists E(11, 12, 15)$                           |
| 1,7 17 | $q \in A \oplus 0_{\mathbb{R}}$                     | $\exists E(8, 9, 16)$                             |
| 1 18   | $A \subseteq A \oplus 0_{\mathbb{R}}$               | Def. 3.5.1.1( $\forall I(\rightarrow I(7, 17))$ ) |
| 1 19   | $A \oplus 0_{\mathbb{R}} = A$                       | Th. 3.5.3.5(6,18)                                 |

## Additives Inverses

Th. 19.4.1.3(iv)

$$A \in \mathbb{R} \vdash A \oplus \ominus A = 0_{\mathbb{R}}$$

|        |                                                                                       |                                                  |
|--------|---------------------------------------------------------------------------------------|--------------------------------------------------|
| 1 1    | $A \in \mathbb{R}$                                                                    | $A$                                              |
| 2 2    | $q \in A \oplus \ominus A$                                                            | $A$                                              |
| 1,2 3  | $\exists a \in A \exists b \in \ominus A (q < a + b)$                                 | Def. 19.4.1.1(1, 2)                              |
| 1,2 4  | $\exists a \in A \exists r \notin A (q < a - r)$                                      | Def. 19.4.1.3(3)                                 |
| 1,2 5  | $q < 0$                                                                               | Def. 19.2.1.1(Th. 19.2.1.1(1), 4)                |
| 1,2 6  | $q \in 0_{\mathbb{R}}$                                                                | Def. 19.4.1.2(Def. 19.2.2.1(5))                  |
| 1 7    | $A \oplus \ominus A \subseteq 0_{\mathbb{R}}$                                         | Def. 3.5.1.1( $\forall I(\rightarrow I(2, 6))$ ) |
| 8 8    | $q \in 0_{\mathbb{R}}$                                                                | $A$                                              |
| 8 9    | $q < 0$                                                                               | Def. 19.4.1.2(Def. 19.2.2.1(8))                  |
| 8 10   | $\exists h \in \mathbb{Q} (0 < h \wedge h < -q)$                                      | Th. 18.6.1.2(9)                                  |
| 8 11   | $h \in \mathbb{Q} \wedge 0 < h \wedge h < -q$                                         | $A$                                              |
| 1,8 12 | $\exists a \in A \exists r \in \mathbb{Q} (r \notin A \wedge a < r \wedge r - a < h)$ | Th. 19.4.1.2(1, 11)                              |
| 1,8 13 | $a \in A \wedge r \in \mathbb{Q} \wedge r \notin A \wedge a < r \wedge r - a < h$     | $A$                                              |
| 1,8 14 | $q < a - r$                                                                           | Th. 18.4.2.1(9, 11, 13)                          |
| 1,8 15 | $\exists b \in \mathbb{Q} (q - a < b \wedge b < -r)$                                  | Th. 18.6.1.2(14)                                 |
| 1,8 16 | $b \in \mathbb{Q} \wedge q - a < b \wedge b < -r$                                     | $A$                                              |
| 1,8 17 | $b \in \ominus A$                                                                     | Def. 19.4.1.3(13, 16)                            |
| 1,8 18 | $q < a + b$                                                                           | Th. 18.4.2.1(16)                                 |
| 1,8 19 | $q \in A \oplus \ominus A$                                                            | Def. 19.4.1.1(13, 17, 18)                        |
| 1,8 20 | $q \in A \oplus \ominus A$                                                            | $\exists E(15, 16, 19)$                          |
| 1,8 21 | $q \in A \oplus \ominus A$                                                            | $\exists E(12, 13, 20)$                          |
| 1,8 22 | $q \in A \oplus \ominus A$                                                            | $\exists E(10, 11, 21)$                          |

$$1 \quad 23 \quad 0_{\mathbb{R}} \subseteq A \oplus \ominus A$$

Def. 3.5.1.1( $\forall I(\rightarrow I(8, 22))$ )

$$1 \quad 24 \quad A \oplus \ominus A = 0_{\mathbb{R}}$$

Th. 3.5.3.5(7, 23)

Zusammenfassung:

$$1 \quad 1 \quad A, B, C \in \mathbb{R}$$

$$A$$

$$1 \quad 2 \quad (A \oplus B) \oplus C = A \oplus (B \oplus C) \text{ Th. 19.4.1.3(i)(1)}$$

$$1 \quad 3 \quad A \oplus B = B \oplus A \quad \text{Th. 19.4.1.3(ii)(1)}$$

$$1 \quad 4 \quad A \oplus 0_{\mathbb{R}} = A \quad \text{Th. 19.4.1.3(iii)(1)}$$

$$1 \quad 5 \quad A \oplus \ominus A = 0_{\mathbb{R}} \quad \text{Th. 19.4.1.3(iv)(1)}$$

$$1 \quad 6 \quad (A \oplus B) \oplus C = A \oplus (B \oplus C) \wedge A \oplus B = B \oplus A \wedge A \oplus 0_{\mathbb{R}} = A \wedge A \oplus \ominus A = 0_{\mathbb{R}}$$

$$\wedge I(2, \wedge I(3, \wedge I(4, 5)))$$

□

**Theorem 19.4.1.4 (Negation kehrt die Schnittordnung um).**

**Mengen:**  $A, B$

**Relationsoperatoren:**  $\leq_{\mathbb{R}}, <_{\mathbb{R}}$

**Funktionsoperatoren:**  $\ominus \cdot$

$$A, B \in \mathbb{R} \vdash (A \leq_{\mathbb{R}} B \leftrightarrow \ominus B \leq_{\mathbb{R}} \ominus A), \quad (A <_{\mathbb{R}} B \leftrightarrow \ominus B <_{\mathbb{R}} \ominus A), \\ \ominus \ominus A = A, \quad \ominus 0_{\mathbb{R}} = 0_{\mathbb{R}}.$$

*Beweis.*

**Eindeutigkeit des additiven Inversen**

Th. 19.4.1.4(i)

$$X, Y \in \mathbb{R}, \quad X \oplus Y = 0_{\mathbb{R}} \vdash Y = \ominus X$$

$$1 \quad 1 \quad X, Y \in \mathbb{R} \quad A$$

$$2 \quad 2 \quad X \oplus Y = 0_{\mathbb{R}} \quad A$$

$$1 \quad 3 \quad Y = Y \oplus 0_{\mathbb{R}} \quad \text{Th. 19.4.1.3(1)}$$

$$1 \quad 4 \quad Y \oplus 0_{\mathbb{R}} = Y \oplus (X \oplus \ominus X) \text{ Th. 19.4.1.3(1)}$$

$$1 \quad 5 \quad Y \oplus (X \oplus \ominus X) = (Y \oplus X) \oplus \ominus X \text{ Th. 19.4.1.3(i)(1)}$$

$$1 \quad 6 \quad (Y \oplus X) \oplus \ominus X = (X \oplus Y) \oplus \ominus X \text{ Th. 19.4.1.3(ii)(1)}$$

$$1,2 \quad 7 \quad (X \oplus Y) \oplus \ominus X = 0_{\mathbb{R}} \oplus \ominus X = E(2)$$

$$1 \quad 8 \quad 0_{\mathbb{R}} \oplus \ominus X = \ominus X \quad \text{Th. 19.4.1.3(ii)(Th. 19.4.1.3(iii)(1))}$$

$$1,2 \quad 9 \quad Y = \ominus X \quad \text{Tr. (3, 4, 5, 6, 7, 8)}$$

**Involutivität und Null**

Th. 19.4.1.4(ii)

$$A \in \mathbb{R} \vdash \ominus \ominus A = A \wedge \ominus 0_{\mathbb{R}} = 0_{\mathbb{R}}$$

|   |                                                                                             |                        |
|---|---------------------------------------------------------------------------------------------|------------------------|
| 1 | $1 \ A \in \mathbb{R}$                                                                      | $A$                    |
| 1 | $2 \ \ominus A \in \mathbb{R}$                                                              | Th. 19.4.1.1(1)        |
| 1 | $3 \ \ominus A \oplus A = A \oplus \ominus A$                                               | Th. 19.4.1.3(ii)(1)    |
| 1 | $4 \ A \oplus \ominus A = 0_{\mathbb{R}}$                                                   | Th. 19.4.1.3(iv)(1)    |
| 1 | $5 \ A = \ominus \ominus A$                                                                 | Th. 19.4.1.4(i)(2,3,4) |
| 1 | $6 \ \ominus \ominus A = A$                                                                 | Th. 2.9.1.1(5)         |
|   | $7 \ 0_{\mathbb{R}} \in \mathbb{R}$                                                         | Th. 19.4.1.1           |
|   | $8 \ 0_{\mathbb{R}} \oplus 0_{\mathbb{R}} = 0_{\mathbb{R}}$                                 | Th. 19.4.1.3(iii)(7)   |
|   | $9 \ 0_{\mathbb{R}} = \ominus 0_{\mathbb{R}}$                                               | Th. 19.4.1.4(i)(7,8)   |
|   | $10 \ \ominus 0_{\mathbb{R}} = 0_{\mathbb{R}}$                                              | Th. 2.9.1.1(9)         |
| 1 | $11 \ \ominus \ominus A = A \wedge \ominus 0_{\mathbb{R}} = 0_{\mathbb{R}} \wedge I(6, 10)$ |                        |

### Antitonie

Th. 19.4.1.4(iii)

$$A, B \in \mathbb{R}, \ A \subseteq B \vdash \ominus B \subseteq \ominus A$$

|       |                                                           |                                                  |
|-------|-----------------------------------------------------------|--------------------------------------------------|
| 1     | $1 \ A, B \in \mathbb{R}$                                 | $A$                                              |
| 2     | $2 \ A \subseteq B$                                       | $A$                                              |
| 3     | $3 \ q \in \ominus B$                                     | $A$                                              |
| 1,3   | $4 \ \exists r \in \mathbb{Q} (r \notin B \wedge q < -r)$ | Def. 19.4.1.3(1,3)                               |
| 1,3   | $5 \ r \in \mathbb{Q} \wedge r \notin B \wedge q < -r$    | $A$                                              |
| 2,3   | $6 \ r \notin A$                                          |                                                  |
|       |                                                           | Th. 2.2.2.1(Def. 3.5.1.1(2), 5)                  |
| 1,2,3 | $7 \ q \in \ominus A$                                     |                                                  |
|       |                                                           | Def. 19.4.1.3(1, 5, 6)                           |
| 1,2,3 | $8 \ q \in \ominus A$                                     | $\exists E(4, 5, 7)$                             |
| 1,2   | $9 \ \ominus B \subseteq \ominus A$                       |                                                  |
|       |                                                           | Def. 3.5.1.1( $\forall I(\rightarrow I(3, 8))$ ) |

### Ordnungsäquivalenzen:

|     |                                                                                           |                         |
|-----|-------------------------------------------------------------------------------------------|-------------------------|
| 1   | $1 \ A, B \in \mathbb{R}$                                                                 | $A$                     |
| 2   | $2 \ A \subseteq B$                                                                       | $A$                     |
| 1,2 | $3 \ \ominus B \subseteq \ominus A$                                                       |                         |
|     |                                                                                           | Th. 19.4.1.4(iii)(1, 2) |
| 4   | $4 \ \ominus B \subseteq \ominus A$                                                       | $A$                     |
| 1,4 | $5 \ \ominus \ominus A \subseteq \ominus \ominus B$                                       |                         |
|     |                                                                                           | Th. 19.4.1.4(iii)(1, 4) |
| 1,4 | $6 \ A \subseteq B$                                                                       |                         |
|     |                                                                                           | Th. 19.4.1.4(ii)(1, 5)  |
| 1   | $7 \ A \subseteq B \leftrightarrow \ominus B \subseteq \ominus A \leftrightarrow I(3, 6)$ |                         |
| 1   | $8 \ A \leq_{\mathbb{R}} B \leftrightarrow \ominus B \leq_{\mathbb{R}} \ominus A$         | Th. 19.3.1.1(1,7)       |
| 9   | $9 \ A \subsetneq B$                                                                      | $A$                     |

|                                                                                                                                                                                                                                                                |                             |                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------|
| 1,9 10 $A \subseteq B \wedge A \neq B$                                                                                                                                                                                                                         | <i>Def.</i> 3.5.1.2(9)      |                                               |
| 1,9 11 $\ominus B \subseteq \ominus A$                                                                                                                                                                                                                         |                             | <i>Th.</i> 19.4.1.4(iii)(1, 10)               |
| 12 12 $\ominus B = \ominus A$                                                                                                                                                                                                                                  | $A$                         |                                               |
| 1,12 13 $A = B$                                                                                                                                                                                                                                                |                             | <i>Th.</i> 19.4.1.4(ii)(1, 12)                |
| 1,9,12 14 $\perp$                                                                                                                                                                                                                                              | $\perp I(10, 13)$           |                                               |
| 1,9 15 $\ominus B \neq \ominus A$                                                                                                                                                                                                                              | $\neg I(12, 14)$            |                                               |
| 1,9 16 $\ominus B \subsetneq \ominus A$                                                                                                                                                                                                                        |                             | <i>Def.</i> 3.5.1.2(11, 15)                   |
| 17 17 $\ominus B \subsetneq \ominus A$                                                                                                                                                                                                                         | $A$                         |                                               |
| 1,17 18 $\ominus B \subseteq \ominus A$                                                                                                                                                                                                                        | <i>Th.</i> 3.5.2.1(17)      |                                               |
| 1,17 19 $A \subseteq B$                                                                                                                                                                                                                                        | $\leftrightarrow E2(7, 18)$ |                                               |
| 20 20 $A = B$                                                                                                                                                                                                                                                  | $A$                         |                                               |
| 1,20 21 $\ominus B = \ominus A$                                                                                                                                                                                                                                | $= E(20)$                   |                                               |
| 1,17,20 22 $\perp$                                                                                                                                                                                                                                             | <i>Th.</i> 3.5.2.2(17, 21)  |                                               |
| 1,17 23 $A \neq B$                                                                                                                                                                                                                                             | $\neg I(20, 22)$            |                                               |
| 1,17 24 $A \subsetneq B$                                                                                                                                                                                                                                       |                             | <i>Def.</i> 3.5.1.2(19, 23)                   |
| 1 25 $A \subsetneq B \leftrightarrow \ominus B \subsetneq \ominus A \leftrightarrow I(16, 24)$                                                                                                                                                                 |                             |                                               |
| 1 26 $A <_{\mathbb{R}} B \leftrightarrow \ominus B <_{\mathbb{R}} \ominus A$                                                                                                                                                                                   | <i>Th.</i> 19.3.1.1(1,25)   |                                               |
| 1 27 $\ominus \ominus A = A$                                                                                                                                                                                                                                   | <i>Th.</i> 19.4.1.4(ii)(1)  |                                               |
| 28 $\ominus 0_{\mathbb{R}} = 0_{\mathbb{R}}$                                                                                                                                                                                                                   | <i>Th.</i> 19.4.1.4(ii)     |                                               |
| 1 29 $(A \leq_{\mathbb{R}} B \leftrightarrow \ominus B \leq_{\mathbb{R}} \ominus A) \wedge$<br>$(A <_{\mathbb{R}} B \leftrightarrow \ominus B <_{\mathbb{R}} \ominus A) \wedge$<br>$\ominus \ominus A = A \wedge$<br>$\ominus 0_{\mathbb{R}} = 0_{\mathbb{R}}$ |                             | $\wedge I(8, \wedge I(26, \wedge I(27, 28)))$ |

□

## 4.2 Multiplikation und multiplikative Inverse

Zunächst wird das Produkt nichtnegativer Schnitte definiert. Die rationale negative Halbachse  $0_{\mathbb{R}}$  wird ausdrücklich beigelegt. Dadurch ist das Produkt mit  $0_{\mathbb{R}}$  wieder  $0_{\mathbb{R}}$ ; der rationale Nullpunkt gehört genau dann zum positiven Produkt, wenn er zu beiden Faktoren gehört.

**Definition 19.4.2.1 (Positives Produkt reeller Schnitte).**

|                                  |                     |
|----------------------------------|---------------------|
| <b>Mengen:</b>                   | $A, B, a, b, q$     |
| <b>Relationsoperatoren:</b>      | $\leq_{\mathbb{R}}$ |
| <b>Funktionsoperatoren:</b>      | $\odot_+$           |
| <b>Neues Symbol:</b>             |                     |
| <b>Positives Schnittprodukt:</b> | $\odot_+$           |

$$A, B \in \mathbb{R}, \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} A, \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} B \\ \vdash A \odot_+ B := 0_{\mathbb{R}} \cup \{q \in \mathbb{Q} \mid \exists a \in A \exists b \in B (0 < a \wedge 0 < b \wedge q < ab)\}.$$

**Definition 19.4.2.2 (Multiplikation beliebiger reeller Schnitte).**

*Mengen:*  $A, B$   
*Relationsoperatoren:*  $\leq_{\mathbb{R}}, <_{\mathbb{R}}$   
*Funktionsoperatoren:*  $\odot, \odot_+, \ominus \cdot$   
*Neues Symbol:*  
*Reelle Multiplikation:*  $\odot$

$$A, B \in \mathbb{R} \vdash A \odot B := \begin{cases} A \odot_+ B, & 0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B, \\ \ominus((\ominus A) \odot_+ B), & A <_{\mathbb{R}} 0_{\mathbb{R}} \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B, \\ \ominus(A \odot_+ (\ominus B)), & 0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge B <_{\mathbb{R}} 0_{\mathbb{R}}, \\ (\ominus A) \odot_+ (\ominus B), & A <_{\mathbb{R}} 0_{\mathbb{R}} \wedge B <_{\mathbb{R}} 0_{\mathbb{R}}. \end{cases}$$

**Definition 19.4.2.3 (Multiplikative Eins der reellen Zahlen).**

*Mengen:*  $1_{\mathbb{R}}$   
*Neues Symbol:*  
*Reelle Eins:*  $1_{\mathbb{R}}$

$$1_{\mathbb{R}} := \hat{1}$$

**Definition 19.4.2.4 (Positives Reziprokes eines Schnittes).**

*Mengen:*  $A, q, r$   
*Relationsoperatoren:*  $<_{\mathbb{R}}$   
*Funktionsoperatoren:*  $\text{inv}_+(\cdot)$   
*Neues Symbol:*  
*Positives Reziprokes:*  $\text{inv}_+(\cdot)$

$$A \in \mathbb{R}, \quad 0_{\mathbb{R}} <_{\mathbb{R}} A \\ \vdash \text{inv}_+(A) := 0_{\mathbb{R}} \cup \{q \in \mathbb{Q} \mid \exists r \in \mathbb{Q} (0 < r \wedge r \notin A \wedge q < r^{-1})\}.$$

**Definition 19.4.2.5 (Reziprokes einer von Null verschiedenen reellen Zahl).**

*Mengen:*  $A$   
*Relationsoperatoren:*  $<_{\mathbb{R}}$   
*Funktionsoperatoren:*  $\text{inv}_{\mathbb{R}}(\cdot), \text{inv}_+(\cdot), \ominus \cdot$   
*Neues Symbol:*  
*Reelles Reziprokes:*  $\text{inv}_{\mathbb{R}}(\cdot)$

$$A \in \mathbb{R}, A \neq 0_{\mathbb{R}} \vdash \text{inv}_{\mathbb{R}}(A) := \begin{cases} \text{inv}_+(A), & 0_{\mathbb{R}} <_{\mathbb{R}} A, \\ \ominus(\text{inv}_+(\ominus A)), & A <_{\mathbb{R}} 0_{\mathbb{R}}. \end{cases}$$

**Theorem 19.4.2.1 (Abgeschlossenheit der multiplikativen Schnittoperationen).**

**Mengen:**  $A, B$   
**Funktionsoperatoren:**  $\odot_+, \odot, \text{inv}_+(\cdot), \text{inv}_{\mathbb{R}}(\cdot)$

$$\begin{aligned} A, B \in \mathbb{R}, 0_{\mathbb{R}} \leq_{\mathbb{R}} A, 0_{\mathbb{R}} \leq_{\mathbb{R}} B &\vdash A \odot_+ B \in \mathbb{R}, \\ A, B \in \mathbb{R} &\vdash A \odot B \in \mathbb{R} \wedge 1_{\mathbb{R}} \in \mathbb{R}, \\ A \in \mathbb{R}, 0_{\mathbb{R}} <_{\mathbb{R}} A &\vdash \text{inv}_+(A) \in \mathbb{R}, \\ A \in \mathbb{R}, A \neq 0_{\mathbb{R}} &\vdash \text{inv}_{\mathbb{R}}(A) \in \mathbb{R}. \end{aligned}$$

*Beweis.*

**Schnittpunkt unter rationalem Außenpunkt** Th. 19.4.2.1(i)

$$A \in \mathbb{R}, a \in A, r \in \mathbb{Q}, r \notin A \vdash a < r$$

$$\begin{array}{ll} 1 & 1 \ A \in \mathbb{R} \quad A \\ 2 & 2 \ a \in A \quad A \\ 3 & 3 \ r \in \mathbb{Q} \quad A \\ 4 & 4 \ r \notin A \quad A \\ 1 & 5 \ \text{Cut}_{\mathbb{Q}}(A) \text{Th. 19.2.1.1(1)} \\ 1,2 & 6 \ a \in \mathbb{Q} \quad \text{Def. 19.2.1.1(5,2)} \\ 7 & 7 \ r < a \quad A \\ 1,2,3,7 & 8 \ r \in A \quad \text{Def. 19.2.1.1(5,2,3,7)} \\ 1,2,3,4,7 & 9 \perp \quad \perp I(4,8) \\ 1,2,3,4 & 10 \neg(r < a) \neg I(7,9) \\ 1,2,3,4 & 11 \ a \leq r \quad \text{Th. 18.4.1.4(6,3,10)} \\ 2,4 & 12 \ a \neq r \quad \text{Th. 3.3.1(2,4)} \\ 1,2,3,4 & 13 \ a < r \quad \text{Def. 18.4.1.4(6,3,11,12)} \end{array}$$

**Positiver Punkt eines positiven Schnittes** Th. 19.4.2.1(ii)

$$A \in \mathbb{R}, 0_{\mathbb{R}} <_{\mathbb{R}} A \vdash \exists a \in A (0 < a)$$

$$\begin{array}{ll} 1 & 1 \ A \in \mathbb{R} \quad A \\ 2 & 2 \ 0_{\mathbb{R}} <_{\mathbb{R}} A \quad A \\ 1 & 3 \ \text{Cut}_{\mathbb{Q}}(A) \quad \text{Th. 19.2.1.1(1)} \\ 1,2 & 4 \ 0_{\mathbb{R}} \subsetneq A \quad \text{Th. 19.3.1.1(1,2)} \\ 1,2 & 5 \ 0_{\mathbb{R}} \subseteq A \wedge 0_{\mathbb{R}} \neq A \text{Def. 3.5.1.2(4)} \end{array}$$

|                                                                    |                                                                 |
|--------------------------------------------------------------------|-----------------------------------------------------------------|
| 1,2 6 $\exists x \in A \setminus 0_{\mathbb{R}}$                   | <i>Th.</i> 3.5.2.3( $\wedge E1(5), \wedge E2(5)$ )              |
| 1,2 7 $x \in A \setminus 0_{\mathbb{R}} \quad A$                   |                                                                 |
| 1,2 8 $x \in A \wedge x \notin 0_{\mathbb{R}} \quad Th. 3.12.1(7)$ |                                                                 |
| 1,2 9 $x \in \mathbb{Q}$                                           | <i>Def.</i> 19.2.1.1(3, $\wedge E1(8)$ )                        |
| 1,2 10 $\neg(x < 0)$                                               | <i>Def.</i> 19.4.1.2( <i>Def.</i> 19.2.2.1(9, $\wedge E2(8)$ )) |
| 1,2 11 $0 \leq x$                                                  | <i>Th.</i> 18.4.1.4( <i>Th.</i> 18.2.3.7, 9, 10)                |
| 1,2 12 $\exists a \in A (x < a)$                                   | <i>Def.</i> 19.2.1.1(3, $\wedge E1(8)$ )                        |
| 1,2 13 $a \in A \wedge x < a \quad A$                              |                                                                 |
| 1,2 14 $a \in \mathbb{Q}$                                          | <i>Def.</i> 19.2.1.1(3, $\wedge E1(13)$ )                       |
| 1,2 15 $0 < a$                                                     | <i>Th.</i> 18.4.1.4(11, $\wedge E2(13)$ )                       |
| 1,2 16 $\exists a \in A (0 < a)$                                   | $\exists I(\wedge I(\wedge E1(13), 15))$                        |
| 1,2 17 $\exists a \in A (0 < a) \exists E(12, 13, 16)$             |                                                                 |
| 1,2 18 $\exists a \in A (0 < a) \exists E(6, 7, 17)$               |                                                                 |

**Positive Außengrenzen begrenzen das Schnittprodukt** *Th.* 19.4.2.1(iii)

$$\begin{aligned}
& A, B \in \mathbb{R}, \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} A, \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} B, \quad \alpha, \beta \in \mathbb{Q}, \\
& 0 < \alpha, \quad \alpha \notin A, \quad 0 < \beta, \quad \beta \notin B \\
& \vdash \alpha\beta \in \mathbb{Q} \wedge \alpha\beta \notin A \odot_+ B
\end{aligned}$$

|                                                                                                                                |                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1 1 $A, B \in \mathbb{R}$                                                                                                      | $A$                                                                 |
| 2 2 $0_{\mathbb{R}} \leq_{\mathbb{R}} A$                                                                                       | $A$                                                                 |
| 3 3 $0_{\mathbb{R}} \leq_{\mathbb{R}} B$                                                                                       | $A$                                                                 |
| 4 4 $\alpha, \beta \in \mathbb{Q}$                                                                                             | $A$                                                                 |
| 5 5 $0 < \alpha$                                                                                                               | $A$                                                                 |
| 6 6 $\alpha \notin A$                                                                                                          | $A$                                                                 |
| 7 7 $0 < \beta$                                                                                                                | $A$                                                                 |
| 8 8 $\beta \notin B$                                                                                                           | $A$                                                                 |
| 4 9 $\alpha\beta \in \mathbb{Q}$                                                                                               | <i>Th.</i> 18.3.3.4(4)                                              |
| 4,5,7 10 $0 < \alpha\beta$                                                                                                     | <i>Th.</i> 18.4.2.3(4, 5, 7), <i>Th.</i> 18.3.4.8( $\wedge E1(4)$ ) |
| 11 11 $\alpha\beta \in A \odot_+ B$                                                                                            | $A$                                                                 |
| 1,2,3,11 12 $\alpha\beta \in 0_{\mathbb{R}} \vee \exists a \in A \exists b \in B (0 < a \wedge 0 < b \wedge \alpha\beta < ab)$ | <i>Def.</i> 19.4.2.1(1, 2, 3, 11)                                   |
| 13 13 $\alpha\beta \in 0_{\mathbb{R}}$                                                                                         | $A$                                                                 |

|                                                                                       |                                                                                                 |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 9,13 14 $\alpha\beta < 0$                                                             |                                                                                                 |
|                                                                                       | <i>Def.</i> 19.4.1.2( <i>Def.</i> 19.2.2.1(9, 13))                                              |
| 4,5,7,13 15 $\perp$                                                                   |                                                                                                 |
|                                                                                       | <i>Th.</i> 18.4.1.4(9, 10, 14)                                                                  |
| 16 16 $\exists a \in A \exists b \in B (0 < a \wedge 0 < b \wedge \alpha\beta < ab)A$ |                                                                                                 |
| 16 17 $a \in A \wedge b \in B \wedge 0 < a \wedge 0 < b \wedge \alpha\beta < abA$     |                                                                                                 |
| 1,16 18 $a \in \mathbb{Q}$                                                            | <i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1( $\wedge E1(1)$ ), 17)                                |
| 1,16 19 $b \in \mathbb{Q}$                                                            | <i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1( $\wedge E2(1)$ ), 17)                                |
| 1,4,6,16 20 $a < \alpha$                                                              | <i>Th.</i> 19.4.2.1( <i>i</i> )( $\wedge E1(1)$ , 17, $\wedge E1(4)$ , 6)                       |
| 1,4,8,16 21 $b < \beta$                                                               | <i>Th.</i> 19.4.2.1( <i>i</i> )( $\wedge E2(1)$ , 17, $\wedge E2(4)$ , 8)                       |
| 1,4,7,16 22 $ab < \alpha\beta$                                                        | <i>Th.</i> 18.4.2.3(18, 19, 20, $\wedge E2(17)$ ), <i>Th.</i> 18.3.4.6(18, 19, $\wedge E1(4)$ ) |
| 4,5,16 23 $\alpha\beta < \alpha\beta$                                                 | <i>Th.</i> 18.4.2.3(4, 19, 21, 5)                                                               |
| 1,4,5,7,16 24 $ab < \alpha\beta$                                                      | <i>Th.</i> 18.4.1.4(22, 23)                                                                     |
| 1,4,5,7,16 25 $\perp$                                                                 | <i>Th.</i> 18.4.1.4(17, 24)                                                                     |
| 1,4-8,16 26 $\perp$                                                                   | $\exists E(16, 17, 25)$                                                                         |
| 1-8,11 27 $\perp$                                                                     | $\vee E(12, 13, 15, 16, 26)$                                                                    |
| 1-8 28 $\alpha\beta \notin A \odot_+ B$                                               | $\neg I(11, 27)$                                                                                |
| 1-8 29 $\alpha\beta \in \mathbb{Q} \wedge \alpha\beta \notin A \odot_+ B$             | $\wedge I(9, 28)$                                                                               |

### Positives Produkt ist ein Schnitt

*Th.* 19.4.2.1(iv)

$$A, B \in \mathbb{R}, 0_{\mathbb{R}} \leq_{\mathbb{R}} A, 0_{\mathbb{R}} \leq_{\mathbb{R}} B \\ \vdash \text{Cut}_{\mathbb{Q}}(A \odot_+ B)$$

|                                                                         |                               |
|-------------------------------------------------------------------------|-------------------------------|
| 1 1 $A, B \in \mathbb{R}$                                               | $A$                           |
| 2 2 $0_{\mathbb{R}} \leq_{\mathbb{R}} A$                                | $A$                           |
| 3 3 $0_{\mathbb{R}} \leq_{\mathbb{R}} B$                                | $A$                           |
| <i>Nichtleerheit</i>                                                    |                               |
| 4 $0_{\mathbb{R}} \in \mathbb{R}$                                       | <i>Th.</i> 19.4.1.1           |
| 5 $\text{Cut}_{\mathbb{Q}}(0_{\mathbb{R}})$                             | <i>Th.</i> 19.2.1.1(4)        |
| 6 $0_{\mathbb{R}} \neq \emptyset \wedge 0_{\mathbb{R}} \neq \mathbb{Q}$ | <i>Def.</i> 19.2.1.1(5)       |
| 1,2,3 7 $0_{\mathbb{R}} \subseteq A \odot_+ B$                          | <i>Def.</i> 19.4.2.1(1, 2, 3) |
| 8 $\exists z \in 0_{\mathbb{R}}$                                        | <i>Def.</i> 19.2.1.1(5)       |
| 9 $z \in 0_{\mathbb{R}}$                                                | $A$                           |
| 1,2,3 10 $z \in A \odot_+ B$                                            | <i>Th.</i> 3.5.2.6(7, 9)      |

|                                                                                                                                     |                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1,2,3 11 $A \odot_+ B \neq \emptyset$                                                                                               | <i>Th.</i> 3.6.3.5(10)                                                                |
| 1,2,3 12 $A \odot_+ B \neq \emptyset$                                                                                               | $\exists E(8, 9, 11)$                                                                 |
| <i>Echtheit</i>                                                                                                                     |                                                                                       |
| 1,2,3 13 $A = 0_{\mathbb{R}} \vee B = 0_{\mathbb{R}} \vee (0_{\mathbb{R}} <_{\mathbb{R}} A \wedge 0_{\mathbb{R}} <_{\mathbb{R}} B)$ | <i>Th.</i> 19.3.1.3(1, 2, 3)                                                          |
| 14 14 $A = 0_{\mathbb{R}}$                                                                                                          | $A$                                                                                   |
| 1,2,3,14 15 $A \odot_+ B = 0_{\mathbb{R}}$                                                                                          | <i>Def.</i> 19.4.2.1(1, 2, 3, 14), <i>Def.</i> 19.4.1.2, <i>Def.</i> 19.2.2.1         |
| 1,2,3,14 16 $A \odot_+ B \neq \mathbb{Q}$                                                                                           | $= E(15, \wedge E2(6))$                                                               |
| 17 17 $B = 0_{\mathbb{R}}$                                                                                                          | $A$                                                                                   |
| 1,2,3,17 18 $A \odot_+ B = 0_{\mathbb{R}}$                                                                                          | <i>Def.</i> 19.4.2.1(1, 2, 3, 17), <i>Def.</i> 19.4.1.2, <i>Def.</i> 19.2.2.1         |
| 1,2,3,17 19 $A \odot_+ B \neq \mathbb{Q}$                                                                                           | $= E(18, \wedge E2(6))$                                                               |
| 20 20 $0_{\mathbb{R}} <_{\mathbb{R}} A \wedge 0_{\mathbb{R}} <_{\mathbb{R}} B$                                                      | $A$                                                                                   |
| 1,2,3,20 21 $\exists a \in A (0 < a)$                                                                                               | <i>Th.</i> 19.4.2.1(ii)( $\wedge E1(1), \wedge E1(20)$ )                              |
| 1,2,3,20 22 $a \in A \wedge 0 < a$                                                                                                  | $A$                                                                                   |
| 1 23 $A \subseteq \mathbb{Q} \wedge A \neq \mathbb{Q}$                                                                              | <i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1( $\wedge E1(1)$ )))                         |
| 1,2,3,20 24 $\exists \alpha \in \mathbb{Q} \setminus A$                                                                             | <i>Th.</i> 3.5.2.3( $\wedge E1(23), \wedge E2(23)$ )                                  |
| 1,2,3,20 25 $\alpha \in \mathbb{Q} \setminus A$                                                                                     | $A$                                                                                   |
| 1,2,3,20 26 $\alpha \in \mathbb{Q} \wedge \alpha \notin A$                                                                          | <i>Th.</i> 3.12.1(25)                                                                 |
| 1,2,3,20 27 $a < \alpha$                                                                                                            | <i>Th.</i> 19.4.2.1(i)( $\wedge E1(1), \wedge E1(22), \wedge E1(26), \wedge E2(26)$ ) |
| 1,2,3,20 28 $0 < \alpha$                                                                                                            | <i>Th.</i> 18.4.1.4(22, 27)                                                           |
| 1,2,3,20 29 $\exists b \in B (0 < b)$                                                                                               | <i>Th.</i> 19.4.2.1(ii)( $\wedge E2(1), \wedge E2(20)$ )                              |
| 1,2,3,20 30 $b \in B \wedge 0 < b$                                                                                                  | $A$                                                                                   |
| 1 31 $B \subseteq \mathbb{Q} \wedge B \neq \mathbb{Q}$                                                                              | <i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1( $\wedge E2(1)$ )))                         |
| 1,2,3,20 32 $\exists \beta \in \mathbb{Q} \setminus B$                                                                              | <i>Th.</i> 3.5.2.3( $\wedge E1(31), \wedge E2(31)$ )                                  |
| 1,2,3,20 33 $\beta \in \mathbb{Q} \setminus B$                                                                                      | $A$                                                                                   |
| 1,2,3,20 34 $\beta \in \mathbb{Q} \wedge \beta \notin B$                                                                            | <i>Th.</i> 3.12.1(33)                                                                 |
| 1,2,3,20 35 $b < \beta$                                                                                                             | <i>Th.</i> 19.4.2.1(i)( $\wedge E2(1), \wedge E1(30), \wedge E1(34), \wedge E2(34)$ ) |
| 1,2,3,20 36 $0 < \beta$                                                                                                             | <i>Th.</i> 18.4.1.4(30, 35)                                                           |
| 1,2,3,20 37 $\alpha\beta \in \mathbb{Q} \wedge \alpha\beta \notin A \odot_+ B$                                                      | <i>Th.</i> 19.4.2.1(iii)(1, 2, 3, 26, 28, 34, 36)                                     |
| 1,2,3,20 38 $\mathbb{Q} \neq A \odot_+ B$                                                                                           | <i>Th.</i> 3.4.2.2( $\wedge E1(37), \wedge E2(37)$ )                                  |
| 1,2,3,20 39 $A \odot_+ B \neq \mathbb{Q}$                                                                                           | <i>Th.</i> 2.9.3.1(38)                                                                |
| 1,2,3,20 40 $A \odot_+ B \neq \mathbb{Q}$                                                                                           | $\exists E(32, 33, 39)$                                                               |
| 1,2,3,20 41 $A \odot_+ B \neq \mathbb{Q}$                                                                                           | $\exists E(29, 30, 40)$                                                               |

|                                 |                                                                                                |                                                                                               |
|---------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 1,2,3,20 42                     | $A \odot_+ B \neq \mathbb{Q}$                                                                  | $\exists E(24, 25, 41)$                                                                       |
| 1,2,3,20 43                     | $A \odot_+ B \neq \mathbb{Q}$                                                                  | $\exists E(21, 22, 42)$                                                                       |
| 1,2,3 44                        | $A \odot_+ B \neq \mathbb{Q}$                                                                  | $\vee E(13, 14, 16, 17, 19, 20, 43)$                                                          |
| <i>Abwärtsabgeschlossenheit</i> |                                                                                                |                                                                                               |
| 45 45                           | $q \in A \odot_+ B$                                                                            | $A$                                                                                           |
| 46 46                           | $s \in \mathbb{Q} \wedge s < q$                                                                | $A$                                                                                           |
| 1,2,3,45 47                     | $q \in 0_{\mathbb{R}} \vee \exists a \in A \exists b \in B (0 < a \wedge 0 < b \wedge q < ab)$ | $Def. 19.4.2.1(1, 2, 3, 45)$                                                                  |
| 48 48                           | $q \in 0_{\mathbb{R}}$                                                                         | $A$                                                                                           |
| 46,48 49                        | $s \in 0_{\mathbb{R}}$                                                                         | $Def. 19.4.1.2(Th. 19.2.2.1(ii)(Th. 18.2.3.7, Def. 19.4.1.2(48), 46))$                        |
| 1,2,3,46,48 50                  | $s \in A \odot_+ B$                                                                            | $Th. 3.5.2.6(7, 49)$                                                                          |
| 51 51                           | $\exists a \in A \exists b \in B (0 < a \wedge 0 < b \wedge q < ab)$                           | $A$                                                                                           |
| 51 52                           | $a \in A \wedge b \in B \wedge 0 < a \wedge 0 < b \wedge q < ab$                               | $A$                                                                                           |
| 46,51 53                        | $s < ab$                                                                                       | $Th. 18.4.1.4(46, 52)$                                                                        |
| 1,2,3,46,51 54                  | $s \in A \odot_+ B$                                                                            | $Def. 19.4.2.1(1, 2, 3, 46, 52, 53)$                                                          |
| 1,2,3,46,51 55                  | $s \in A \odot_+ B$                                                                            | $\exists E(51, 52, 54)$                                                                       |
| 1,2,3,45,46 56                  | $s \in A \odot_+ B$                                                                            | $\vee E(47, 48, 50, 51, 55)$                                                                  |
| 1,2,3 57                        | $\forall q \in A \odot_+ B \forall s \in \mathbb{Q} (s < q \rightarrow s \in A \odot_+ B)$     | $\forall I(\rightarrow I(45, \rightarrow I(46, 56)))$                                         |
| <i>Kein größtes Element</i>     |                                                                                                |                                                                                               |
| 58 58                           | $q \in A \odot_+ B$                                                                            | $A$                                                                                           |
| 1,2,3,58 59                     | $q \in 0_{\mathbb{R}} \vee \exists a \in A \exists b \in B (0 < a \wedge 0 < b \wedge q < ab)$ | $Def. 19.4.2.1(1, 2, 3, 58)$                                                                  |
| 60 60                           | $q \in 0_{\mathbb{R}}$                                                                         | $A$                                                                                           |
| 60 61                           | $\exists r \in 0_{\mathbb{R}} (q < r)$                                                         | $Def. 19.4.1.2(Th. 19.2.2.1(iii)(Th. 18.2.3.7, Def. 19.4.1.2(60)))$                           |
| 62 62                           | $r \in 0_{\mathbb{R}} \wedge q < r$                                                            | $A$                                                                                           |
| 1,2,3,60 63                     | $r \in A \odot_+ B$                                                                            | $Th. 3.5.2.6(7, 62)$                                                                          |
| 1,2,3,60 64                     | $\exists r \in A \odot_+ B (q < r)$                                                            | $\exists I(\wedge I(63, \wedge E2(62)))$                                                      |
| 1,2,3,60 65                     | $\exists r \in A \odot_+ B (q < r)$                                                            | $\exists E(61, 62, 64)$                                                                       |
| 66 66                           | $\exists a \in A \exists b \in B (0 < a \wedge 0 < b \wedge q < ab)$                           | $A$                                                                                           |
| 66 67                           | $a \in A \wedge b \in B \wedge 0 < a \wedge 0 < b \wedge q < ab$                               | $A$                                                                                           |
| 1,2,3,58 68                     | $q \in \mathbb{Q}$                                                                             | $Th. 3.5.2.6(7, 58)$                                                                          |
| 1,66 69                         | $a, b \in \mathbb{Q}$                                                                          | $\wedge I^*(Wdh_2(Def. 19.2.1.1 \circ Th. 19.2.1.1; \wedge E1(1), 67 \mid \wedge E2(1), 67))$ |
| 1,66 70                         | $ab \in \mathbb{Q}$                                                                            | $Th. 18.3.3.4(69)$                                                                            |
| 1,2,3,58,66 71                  | $\exists r \in \mathbb{Q} (q < r \wedge r < ab)$                                               | $Th. 18.6.1.2(68, 70, 67)$                                                                    |
| 66 72                           | $r \in \mathbb{Q} \wedge q < r \wedge r < ab$                                                  | $A$                                                                                           |

|                                                 |    |                                                               |                                                     |
|-------------------------------------------------|----|---------------------------------------------------------------|-----------------------------------------------------|
| 1,2,3,66                                        | 73 | $r \in A \odot_+ B$                                           |                                                     |
|                                                 |    |                                                               | <i>Def.</i> 19.4.2.1(1, 2, 3, 67, 72)               |
| 1,2,3,66                                        | 74 | $\exists r \in A \odot_+ B (q < r)$                           |                                                     |
|                                                 |    |                                                               | $\exists I(\wedge I(73, \wedge E1(\wedge E2(72))))$ |
| 1,2,3,66                                        | 75 | $\exists r \in A \odot_+ B (q < r)$                           | $\exists E(71, 72, 74)$                             |
| 1,2,3,66                                        | 76 | $\exists r \in A \odot_+ B (q < r)$                           | $\exists E(66, 67, 75)$                             |
| 1,2,3,58                                        | 77 | $\exists r \in A \odot_+ B (q < r)$                           |                                                     |
|                                                 |    |                                                               | $\vee E(59, 60, 65, 66, 76)$                        |
| 1,2,3                                           | 78 | $\forall q \in A \odot_+ B \exists r \in A \odot_+ B (q < r)$ | $\forall I(\rightarrow I(58, 77))$                  |
| <i>Zusammensetzung der Schnitteigenschaften</i> |    |                                                               |                                                     |
| 1,2,3                                           | 79 | $A \odot_+ B \subseteq \mathbb{Q}$                            |                                                     |
|                                                 |    |                                                               | <i>Def.</i> 19.4.2.1(1, 2, 3)                       |
| 1,2,3                                           | 80 | $\text{Cut}_{\mathbb{Q}}(A \odot_+ B)$                        |                                                     |
|                                                 |    |                                                               | <i>Def.</i> 19.2.1.1(79, 12, 44, 57, 78)            |

**Positives Reziprokes ist ein Schnitt** *Th.* 19.4.2.1(v)

$$A \in \mathbb{R}, 0_{\mathbb{R}} <_{\mathbb{R}} A \vdash \text{Cut}_{\mathbb{Q}}(\text{inv}_+(A))$$

|                      |    |                                                                       |                                                                                 |
|----------------------|----|-----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1                    | 1  | $A \in \mathbb{R}$                                                    | $A$                                                                             |
| 2                    | 2  | $0_{\mathbb{R}} <_{\mathbb{R}} A$                                     | $A$                                                                             |
| <i>Nichtleerheit</i> |    |                                                                       |                                                                                 |
|                      | 3  | $0_{\mathbb{R}} \in \mathbb{R}$                                       | <i>Th.</i> 19.4.1.1                                                             |
|                      | 4  | $\text{Cut}_{\mathbb{Q}}(0_{\mathbb{R}})$                             | <i>Th.</i> 19.2.1.1(3)                                                          |
|                      | 5  | $0_{\mathbb{R}} \neq \emptyset \wedge 0_{\mathbb{R}} \neq \mathbb{Q}$ | <i>Def.</i> 19.2.1.1(4)                                                         |
| 1,2                  | 6  | $0_{\mathbb{R}} \subseteq \text{inv}_+(A)$                            | <i>Def.</i> 19.4.2.4(1, 2)                                                      |
|                      | 7  | $\exists z \in 0_{\mathbb{R}}$                                        | <i>Def.</i> 19.2.1.1(4)                                                         |
|                      | 8  | $z \in 0_{\mathbb{R}}$                                                | $A$                                                                             |
| 1,2                  | 9  | $z \in \text{inv}_+(A)$                                               | <i>Th.</i> 3.5.2.6(6, 8)                                                        |
| 1,2                  | 10 | $\text{inv}_+(A) \neq \emptyset$                                      | <i>Th.</i> 3.6.3.5(9)                                                           |
| 1,2                  | 11 | $\text{inv}_+(A) \neq \emptyset$                                      | $\exists E(7, 8, 10)$                                                           |
| <i>Echtheit</i>      |    |                                                                       |                                                                                 |
| 1,2                  | 12 | $\exists a \in A (0 < a)$                                             |                                                                                 |
|                      |    |                                                                       | <i>Th.</i> 19.4.2.1(ii)(1, 2)                                                   |
| 1,2                  | 13 | $a \in A \wedge 0 < a$                                                | $A$                                                                             |
| 1                    | 14 | $A \subseteq \mathbb{Q}$                                              |                                                                                 |
|                      |    |                                                                       | <i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1(1))                                   |
| 1,2                  | 15 | $a \in \mathbb{Q}$                                                    | <i>Th.</i> 3.5.2.6(14, 13)                                                      |
| 1,2                  | 16 | $a \neq 0$                                                            |                                                                                 |
|                      |    |                                                                       | <i>Th.</i> 2.9.3.1( <i>Def.</i> 18.4.1.4( $\wedge E2(13)$ )))                   |
| 1,2                  | 17 | $a^{-1} \in \mathbb{Q} \wedge 0 < a^{-1}$                             |                                                                                 |
|                      |    |                                                                       | $\wedge I(\text{Ax. } 18.5.14(15, 16), \text{Ax. } 18.5.22(15, \wedge E2(13)))$ |
| 18                   | 18 | $a^{-1} \in \text{inv}_+(A)$                                          | $A$                                                                             |

|                                 |    |                                                                                                            |                             |                                                                                                    |
|---------------------------------|----|------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------|
| 1,2,18                          | 19 | $a^{-1} \in 0_{\mathbb{R}} \vee \exists r \in \mathbb{Q} (0 < r \wedge r \notin A \wedge a^{-1} < r^{-1})$ |                             | <i>Def.</i> 19.4.2.4(1, 2, 18)                                                                     |
| 20                              | 20 | $a^{-1} \in 0_{\mathbb{R}}$                                                                                | $A$                         |                                                                                                    |
| 20                              | 21 | $a^{-1} < 0$                                                                                               |                             |                                                                                                    |
|                                 |    |                                                                                                            |                             | <i>Def.</i> 19.4.1.2( <i>Def.</i> 19.2.2.1(20))                                                    |
| 1,2,20                          | 22 | $\perp$                                                                                                    | <i>Th.</i> 18.4.1.4(17, 21) |                                                                                                    |
| 23                              | 23 | $\exists r \in \mathbb{Q} (0 < r \wedge r \notin A \wedge a^{-1} < r^{-1})A$                               |                             |                                                                                                    |
| 1,2,23                          | 24 | $r \in \mathbb{Q} \wedge 0 < r \wedge r \notin A \wedge a^{-1} < r^{-1} A$                                 |                             |                                                                                                    |
| 1,2,23                          | 25 | $r < a$                                                                                                    |                             | <i>Th.</i> 18.6.1.1(15, $\wedge E$ 2(13), 24)                                                      |
| 1,2,23                          | 26 | $r \in A$                                                                                                  |                             | <i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1(1), 13, 24, 25)                                          |
| 1,2,23                          | 27 | $\perp$                                                                                                    |                             | $\perp I(\wedge E1(\wedge E2(\wedge E2(24))), 26)$                                                 |
| 1,2,23                          | 28 | $\perp$                                                                                                    | $\exists E(23, 24, 27)$     |                                                                                                    |
| 1,2,18                          | 29 | $\perp$                                                                                                    |                             | $\vee E(19, 20, 22, 23, 28)$                                                                       |
| 1,2                             | 30 | $a^{-1} \notin \text{inv}_+(A)$                                                                            | $\neg I(18, 29)$            |                                                                                                    |
| 1,2                             | 31 | $\mathbb{Q} \neq \text{inv}_+(A)$                                                                          |                             | <i>Th.</i> 3.4.2.2( $\wedge E$ 1(17), 30)                                                          |
| 1,2                             | 32 | $\text{inv}_+(A) \neq \mathbb{Q}$                                                                          | <i>Th.</i> 2.9.3.1(31)      |                                                                                                    |
| 1,2                             | 33 | $\text{inv}_+(A) \neq \mathbb{Q}$                                                                          | $\exists E(12, 13, 32)$     |                                                                                                    |
| <i>Abwärtsabgeschlossenheit</i> |    |                                                                                                            |                             |                                                                                                    |
| 34                              | 34 | $q \in \text{inv}_+(A)$                                                                                    | $A$                         |                                                                                                    |
| 35                              | 35 | $s \in \mathbb{Q} \wedge s < q$                                                                            | $A$                         |                                                                                                    |
| 1,2,34                          | 36 | $q \in 0_{\mathbb{R}} \vee \exists r \in \mathbb{Q} (0 < r \wedge r \notin A \wedge q < r^{-1})$           |                             | <i>Def.</i> 19.4.2.4(1, 2, 34)                                                                     |
| 37                              | 37 | $q \in 0_{\mathbb{R}}$                                                                                     | $A$                         |                                                                                                    |
| 35,37                           | 38 | $s \in 0_{\mathbb{R}}$                                                                                     |                             | <i>Def.</i> 19.4.1.2( <i>Th.</i> 19.2.2.1(ii)( <i>Th.</i> 18.2.3.7, <i>Def.</i> 19.4.1.2(37), 35)) |
| 1,2,35,37                       | 39 | $s \in \text{inv}_+(A)$                                                                                    | <i>Th.</i> 3.5.2.6(6, 38)   |                                                                                                    |
| 40                              | 40 | $\exists r \in \mathbb{Q} (0 < r \wedge r \notin A \wedge q < r^{-1})$                                     | $A$                         |                                                                                                    |
| 40                              | 41 | $r \in \mathbb{Q} \wedge 0 < r \wedge r \notin A \wedge q < r^{-1}$                                        | $A$                         |                                                                                                    |
| 35,40                           | 42 | $s < r^{-1}$                                                                                               | <i>Th.</i> 18.4.1.4(35, 41) |                                                                                                    |
| 1,2,35,40                       | 43 | $s \in \text{inv}_+(A)$                                                                                    |                             | <i>Def.</i> 19.4.2.4(1, 2, 35, 41, 42)                                                             |
| 1,2,35,40                       | 44 | $s \in \text{inv}_+(A)$                                                                                    | $\exists E(40, 41, 43)$     |                                                                                                    |
| 1,2,34,35                       | 45 | $s \in \text{inv}_+(A)$                                                                                    |                             | $\vee E(36, 37, 39, 40, 44)$                                                                       |
| 1,2                             | 46 | $\forall q \in \text{inv}_+(A) \forall s \in \mathbb{Q} (s < q \rightarrow s \in \text{inv}_+(A))$         |                             | $\forall I(\rightarrow I(34, \rightarrow I(35, 45)))$                                              |
| <i>Kein größtes Element</i>     |    |                                                                                                            |                             |                                                                                                    |
| 47                              | 47 | $q \in \text{inv}_+(A)$                                                                                    | $A$                         |                                                                                                    |

|                                                 |    |                                                                                                        |                               |                                                                                                 |
|-------------------------------------------------|----|--------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------|
| 1,2,47                                          | 48 | $q \in 0_{\mathbb{R}} \vee \exists r \in \mathbb{Q} (0 < r \wedge r \notin A \wedge q < r^{-1})$       |                               | <i>Def.</i> 19.4.2.4(1, 2, 47)                                                                  |
| 49                                              | 49 | $q \in 0_{\mathbb{R}}$                                                                                 | $A$                           |                                                                                                 |
| 49                                              | 50 | $\exists s \in 0_{\mathbb{R}} (q < s)$                                                                 |                               | <i>Def.</i> 19.4.1.2( <i>Th.</i> 19.2.2.1(iii)( <i>Th.</i> 18.2.3.7, <i>Def.</i> 19.4.1.2(49))) |
| 49                                              | 51 | $s \in 0_{\mathbb{R}} \wedge q < s$                                                                    | $A$                           |                                                                                                 |
| 1,2,49                                          | 52 | $s \in \text{inv}_+(A)$                                                                                | $\text{Th.}$ 3.5.2.6(6, 51)   |                                                                                                 |
| 1,2,49                                          | 53 | $\exists s \in \text{inv}_+(A) (q < s)$                                                                |                               | $\exists I(\wedge I(52, \wedge E2(51)))$                                                        |
| 1,2,49                                          | 54 | $\exists s \in \text{inv}_+(A) (q < s)$                                                                | $\exists E(50, 51, 53)$       |                                                                                                 |
| 55                                              | 55 | $\exists r \in \mathbb{Q} (0 < r \wedge r \notin A \wedge q < r^{-1})$                                 | $A$                           |                                                                                                 |
| 55                                              | 56 | $r \in \mathbb{Q} \wedge 0 < r \wedge r \notin A \wedge q < r^{-1}$                                    | $A$                           |                                                                                                 |
| 1,2,47                                          | 57 | $q \in \mathbb{Q}$                                                                                     |                               | <i>Def.</i> 19.4.2.4(1, 2, 47)                                                                  |
| 55                                              | 58 | $r \neq 0$                                                                                             |                               | <i>Th.</i> 2.9.3.1( <i>Def.</i> 18.4.1.4(56))                                                   |
| 55                                              | 59 | $r^{-1} \in \mathbb{Q}$                                                                                | $\text{Th.}$ 18.3.5.7(56, 58) |                                                                                                 |
| 1,2,47,55                                       | 60 | $\exists s \in \mathbb{Q} (q < s \wedge s < r^{-1})$                                                   |                               | <i>Th.</i> 18.6.1.2(57, 59, 56)                                                                 |
| 55                                              | 61 | $s \in \mathbb{Q} \wedge q < s \wedge s < r^{-1}$                                                      | $A$                           |                                                                                                 |
| 1,2,55                                          | 62 | $s \in \text{inv}_+(A)$                                                                                |                               | <i>Def.</i> 19.4.2.4(1, 2, 56, 61)                                                              |
| 1,2,55                                          | 63 | $\exists s \in \text{inv}_+(A) (q < s)$                                                                |                               | $\exists I(\wedge I(62, \wedge E1(\wedge E2(61))))$                                             |
| 1,2,55                                          | 64 | $\exists s \in \text{inv}_+(A) (q < s)$                                                                | $\exists E(60, 61, 63)$       |                                                                                                 |
| 1,2,55                                          | 65 | $\exists s \in \text{inv}_+(A) (q < s)$                                                                | $\exists E(55, 56, 64)$       |                                                                                                 |
| 1,2,47                                          | 66 | $\exists s \in \text{inv}_+(A) (q < s)$                                                                |                               | $\vee E(48, 49, 54, 55, 65)$                                                                    |
| 1,2                                             | 67 | $\forall q \in \text{inv}_+(A) \exists s \in \text{inv}_+(A) (q < s) \forall I(\rightarrow I(47, 66))$ |                               |                                                                                                 |
| <i>Zusammensetzung der Schnitteigenschaften</i> |    |                                                                                                        |                               |                                                                                                 |
| 1,2                                             | 68 | $\text{inv}_+(A) \subseteq \mathbb{Q}$                                                                 | <i>Def.</i> 19.4.2.4(1, 2)    |                                                                                                 |
| 1,2                                             | 69 | $\text{Cut}_{\mathbb{Q}}(\text{inv}_+(A))$                                                             |                               | <i>Def.</i> 19.2.1.1(68, 11, 33, 46, 67)                                                        |

Fallzerlegung und Zusammenfassung:

|     |   |                                                                                                                                                                                                                                                               |     |                                                     |
|-----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------------------------------------------------|
| 1   | 1 | $A, B \in \mathbb{R}$                                                                                                                                                                                                                                         | $A$ |                                                     |
| 2   | 2 | $0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B$                                                                                                                                                                                | $A$ |                                                     |
| 1,2 | 3 | $A \odot_+ B \in \mathbb{R}$                                                                                                                                                                                                                                  |     | <i>Th.</i> 19.2.1.1( <i>Th.</i> 19.4.2.1(iv)(1, 2)) |
| 1   | 4 | $(A <_{\mathbb{R}} 0_{\mathbb{R}} \rightarrow (0_{\mathbb{R}} <_{\mathbb{R}} \ominus A \wedge \ominus A \in \mathbb{R})) \wedge$<br>$(B <_{\mathbb{R}} 0_{\mathbb{R}} \rightarrow (0_{\mathbb{R}} <_{\mathbb{R}} \ominus B \wedge \ominus B \in \mathbb{R}))$ |     | <i>Th.</i> 19.4.1.4(1), <i>Th.</i> 19.4.1.1(1)      |

|     |    |                                                                                                                               |                                                                                                               |
|-----|----|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1   | 5  | $A \odot B \in \mathbb{R}$                                                                                                    | <i>Def.</i> 19.4.2.2( <i>Th.</i> 19.3.1.3(1), 3, 4, <i>Th.</i> 19.4.2.1( <i>iv</i> ), <i>Th.</i> 19.4.1.1(1)) |
|     | 6  | $1 \in \mathbb{Q}$                                                                                                            | <i>Th.</i> 18.2.3.7                                                                                           |
|     | 7  | $1_{\mathbb{R}} \in \mathbb{R}$                                                                                               | <i>Def.</i> 19.4.2.3( <i>Th.</i> 19.2.1.1( <i>Th.</i> 19.2.2.1(6)))                                           |
| 8   | 8  | $A \in \mathbb{R} \wedge 0_{\mathbb{R}} <_{\mathbb{R}} A$                                                                     | $A$                                                                                                           |
| 8   | 9  | $\text{inv}_+(A) \in \mathbb{R}$                                                                                              | <i>Th.</i> 19.2.1.1( <i>Th.</i> 19.4.2.1( <i>v</i> )(8))                                                      |
| 10  | 10 | $A \in \mathbb{R} \wedge A \neq 0_{\mathbb{R}}$                                                                               | $A$                                                                                                           |
| 10  | 11 | $0_{\mathbb{R}} <_{\mathbb{R}} A \vee A <_{\mathbb{R}} 0_{\mathbb{R}}$                                                        | <i>Th.</i> 19.3.1.3(10)                                                                                       |
| 10  | 12 | $\text{inv}_{\mathbb{R}}(A) \in \mathbb{R}$                                                                                   | <i>Def.</i> 19.4.2.5(10, 11, 9, <i>Th.</i> 19.4.1.4(10), <i>Th.</i> 19.4.1.1(10))                             |
| 1,2 | 13 | $A \odot_+ B \in \mathbb{R} \wedge A \odot B \in \mathbb{R} \wedge 1_{\mathbb{R}} \in \mathbb{R} \wedge I(3, \wedge I(5, 7))$ |                                                                                                               |
|     | 8  | $14 \text{ inv}_+(A) \in \mathbb{R}$                                                                                          | $= E(9)$                                                                                                      |
| 10  | 15 | $\text{inv}_{\mathbb{R}}(A) \in \mathbb{R}$                                                                                   | $= E(12)$                                                                                                     |

□

**Theorem 19.4.2.2 (Multiplikative Randapproximation eines positiven Schnittes).**

**Mengen:**  $A, a, r, q$

**Relationsoperatoren:**  $<_{\mathbb{R}}$

$$A \in \mathbb{R}, 0_{\mathbb{R}} <_{\mathbb{R}} A, q \in \mathbb{Q}, 0 < q < 1 \vdash \\ \exists a \in A \exists r \in \mathbb{Q} (0 < a < r \wedge r \notin A \wedge q < ar^{-1}).$$

*Beweis.*

**Wahl einer positiven Maschenweite**

*Th.* 19.4.2.2(i)

$$p, q \in \mathbb{Q}, 0 < p, 0 < q < 1 \\ \vdash \exists h \in \mathbb{Q} (0 < h \wedge qh < (1 - q)p)$$

|       |   |                                                              |                                |
|-------|---|--------------------------------------------------------------|--------------------------------|
| 1     | 1 | $p, q \in \mathbb{Q}$                                        | $A$                            |
| 2     | 2 | $0 < p$                                                      | $A$                            |
| 3     | 3 | $0 < q < 1$                                                  | $A$                            |
| 1,2,3 | 4 | $0 < (1 - q)pq^{-1}$                                         | <i>Th.</i> 18.4.2.4(1,2,3)     |
| 1,2,3 | 5 | $\exists h \in \mathbb{Q} (0 < h \wedge h < (1 - q)pq^{-1})$ | <i>Th.</i> 18.6.1.2(4)         |
| 1,2,3 | 6 | $h \in \mathbb{Q} \wedge 0 < h \wedge h < (1 - q)pq^{-1}$    | $A$                            |
| 1,2,3 | 7 | $qh < (1 - q)p$                                              | <i>Th.</i> 18.4.2.3(3,6)       |
| 1,2,3 | 8 | $\exists h \in \mathbb{Q} (0 < h \wedge qh < (1 - q)p)$      | $\exists I(6, \wedge I(6, 7))$ |
| 1,2,3 | 9 | $\exists h \in \mathbb{Q} (0 < h \wedge qh < (1 - q)p)$      | $\exists E(5, 6, 8)$           |

Gitterrand und Quotientenvergleich:

|         |    |                                                                                                                                                                                                                      |                                                                |
|---------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 1       | 1  | $A \in \mathbb{R}$                                                                                                                                                                                                   | $A$                                                            |
| 2       | 2  | $0_{\mathbb{R}} <_{\mathbb{R}} A$                                                                                                                                                                                    | $A$                                                            |
| 3       | 3  | $q \in \mathbb{Q}$                                                                                                                                                                                                   | $A$                                                            |
| 4       | 4  | $0 < q < 1$                                                                                                                                                                                                          | $A$                                                            |
| 1,2     | 5  | $\exists p \in A (0 < p)$                                                                                                                                                                                            | <i>Th.</i> 19.3.1.1(1, 2)                                      |
| 1,2     | 6  | $p \in A \wedge 0 < p$                                                                                                                                                                                               | $A$                                                            |
| 1       | 7  | $\exists u \in \mathbb{Q} \setminus A$                                                                                                                                                                               |                                                                |
|         |    |                                                                                                                                                                                                                      | <i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1(1))                  |
| 1       | 8  | $u \in \mathbb{Q} \setminus A$                                                                                                                                                                                       | $A$                                                            |
| 1,2,3,4 | 9  | $\exists h \in \mathbb{Q} (0 < h \wedge qh < (1 - q)p)$                                                                                                                                                              |                                                                |
|         |    |                                                                                                                                                                                                                      | <i>Th.</i> 19.4.2.2(i)(3, 4, 6)                                |
| 1,2,3,4 | 10 | $h \in \mathbb{Q} \wedge 0 < h \wedge qh < (1 - q)p$                                                                                                                                                                 | $A$                                                            |
| 1,2,3,4 | 11 | $\exists m_0 \in \mathbb{N} \setminus \{0\} ($<br>$p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m_0)) \frac{h}{2} \notin A \wedge$<br>$p + \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(\text{pred}(m_0))) \frac{h}{2} \in A)$ |                                                                |
|         |    |                                                                                                                                                                                                                      | <i>Th.</i> 19.4.1.2(ii)(1, 10, 6, 8)                           |
| 1,2,3,4 | 12 | $a \in A \wedge r \notin A \wedge p \leq a < r \wedge r - a < h$                                                                                                                                                     | <i>Def.</i> 18.5.1(10, 11)                                     |
| 1,2,3,4 | 13 | $qr = qa + q(r - a)$                                                                                                                                                                                                 | <i>Th.</i> 18.3.4.9(3, 12)                                     |
| 1,2,3,4 | 14 | $qa + q(r - a) < qa + (1 - q)p$                                                                                                                                                                                      | <i>Th.</i> 18.4.2.3(10, 12)                                    |
| 1,2,3,4 | 15 | $qa + (1 - q)p \leq qa + (1 - q)a$                                                                                                                                                                                   | <i>Th.</i> 18.4.2.1(4, 12)                                     |
| 1,2,3,4 | 16 | $qa + (1 - q)a = a$                                                                                                                                                                                                  | <i>Th.</i> 18.3.4.9(3, 12)                                     |
| 1,2,3,4 | 17 | $qr < a$                                                                                                                                                                                                             | <i>Tr.</i> (13, 14, 15, 16)                                    |
| 1,2,3,4 | 18 | $q < ar^{-1}$                                                                                                                                                                                                        | <i>Th.</i> 18.4.2.3(12, 17)                                    |
| 1,2,3,4 | 19 | $\exists a \in A \exists r \in \mathbb{Q} (0 < a < r \wedge r \notin A \wedge q < ar^{-1})$                                                                                                                          |                                                                |
|         |    |                                                                                                                                                                                                                      | $\exists I(12, \exists I(12, \wedge I(12, \wedge I(12, 18))))$ |
| 1,2,3,4 | 20 | $\exists a \in A \exists r \in \mathbb{Q} (0 < a < r \wedge r \notin A \wedge q < ar^{-1}) \exists E(11, 12, 19)$                                                                                                    |                                                                |
| 1,2,3,4 | 21 | $\exists a \in A \exists r \in \mathbb{Q} (0 < a < r \wedge r \notin A \wedge q < ar^{-1}) \exists E(9, 10, 20)$                                                                                                     |                                                                |
| 1,2,3,4 | 22 | $\exists a \in A \exists r \in \mathbb{Q} (0 < a < r \wedge r \notin A \wedge q < ar^{-1}) \exists E(7, 8, 21)$                                                                                                      |                                                                |
| 1,2,3,4 | 23 | $\exists a \in A \exists r \in \mathbb{Q} (0 < a < r \wedge r \notin A \wedge q < ar^{-1}) \exists E(5, 6, 22)$                                                                                                      |                                                                |

□

### Theorem 19.4.2.3 (Gesetze der reellen Multiplikation).

**Mengen:**  $A, B, C$   
**Funktionsoperatoren:**  $\oplus, \odot, \text{inv}_{\mathbb{R}}(\cdot)$

$$\begin{aligned}
A, B, C \in \mathbb{R} &\vdash (A \odot B) \odot C = A \odot (B \odot C), \\
A \odot B &= B \odot A, \quad A \odot 1_{\mathbb{R}} = A, \\
A \odot (B \oplus C) &= (A \odot B) \oplus (A \odot C), \\
A \neq 0_{\mathbb{R}} &\rightarrow A \odot \text{inv}_{\mathbb{R}}(A) = 1_{\mathbb{R}}.
\end{aligned}$$

*Beweis.*

## Nullpunkt einer nichtnegativen Summe

Th. 19.4.2.3(i)

$$A, B \in \mathbb{R}, 0_{\mathbb{R}} \leq_{\mathbb{R}} A, 0_{\mathbb{R}} \leq_{\mathbb{R}} B \\ \vdash 0_{\mathbb{R}} \leq_{\mathbb{R}} A \oplus B \wedge (0 \in A \oplus B \leftrightarrow (0 \in A \vee 0 \in B))$$

|       |    |                                                                                |                                                                                                                                                                                         |
|-------|----|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | 1  | $A, B \in \mathbb{R}$                                                          | $A$                                                                                                                                                                                     |
| 2     | 2  | $0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B$ | $A$                                                                                                                                                                                     |
| 1,2   | 3  | $0_{\mathbb{R}} \subseteq A \wedge 0_{\mathbb{R}} \subseteq B$                 | Th. 19.3.1.1(1,2)                                                                                                                                                                       |
| 4     | 4  | $q \in 0_{\mathbb{R}}$                                                         | $A$                                                                                                                                                                                     |
| 4     | 5  | $\exists a \in 0_{\mathbb{R}} \exists b \in 0_{\mathbb{R}} (q < a + b)$        | $q < 0$ . Wähle mit der rationalen Dichte zuerst $q < a < 0$ ,<br>dann $q - a < b < 0$ ; somit $q < a + b$ .<br>Def. 19.4.1.2, Def. 19.2.2.1,<br>Th. 18.6.1.2, Th. 18.4.2.1             |
| 1,2,4 | 6  | $\exists a \in A \exists b \in B (q < a + b)$                                  | Th. 3.5.2.6(3,5)                                                                                                                                                                        |
| 1,2,4 | 7  | $q \in A \oplus B$                                                             | Def. 19.4.1.1(1,6)                                                                                                                                                                      |
| 1,2   | 8  | $0_{\mathbb{R}} \subseteq A \oplus B$                                          | Def. 3.5.1.1( $\forall I(\rightarrow I(4,7))$ )                                                                                                                                         |
| 1     | 9  | $A \oplus B \in \mathbb{R}$                                                    | Th. 19.4.1.1(1)                                                                                                                                                                         |
| 1,2   | 10 | $0_{\mathbb{R}} \leq_{\mathbb{R}} A \oplus B$                                  | Th. 19.3.1.1(8,9)                                                                                                                                                                       |
| 11    | 11 | $0 \in A \oplus B$                                                             | $A$                                                                                                                                                                                     |
| 1,11  | 12 | $\exists a \in A \exists b \in B (0 < a + b)$                                  | Def. 19.4.1.1(1,11)                                                                                                                                                                     |
| 1     | 13 | $a \in A, 0 \notin A \vdash a < 0$                                             | $a \in \mathbb{Q}$ . Aus $0 \leq a$ folgte bei $a = 0$ sofort,<br>bei $0 < a$ durch Abwärtsabschluss jeweils $0 \in A$ .<br>Def. 19.2.1.1(Th. 19.2.1.1( $\wedge E1(1)$ )), Th. 18.4.1.4 |
| 1     | 14 | $b \in B, 0 \notin B \vdash b < 0$                                             | $b \in \mathbb{Q}$ . Aus $0 \leq b$ folgte wie in 13<br>durch Gleichheit oder Abwärtsabschluss $0 \in B$ .<br>Def. 19.2.1.1(Th. 19.2.1.1( $\wedge E2(1)$ )), Th. 18.4.1.4               |
| 1,11  | 15 | $0 \notin A \wedge 0 \notin B \vdash \perp$                                    | Für die Zeugen aus 12 liefern 13 und 14 die Ungleichungen $a < 0, b < 0$ ,<br>also $a + b < 0$ , im Widerspruch zu $0 < a + b$ .<br>Def. 18.5.1(12, 13, 14)                             |
| 1,11  | 16 | $0 \in A \vee 0 \in B$                                                         | Th. 2.1.8.2(15), Th. 2.1.6.1                                                                                                                                                            |
| 17    | 17 | $0 \in A$                                                                      | $A$                                                                                                                                                                                     |
| 1,17  | 18 | $\exists a \in A (0 < a)$                                                      | Def. 19.2.1.1(Th. 19.2.1.1( $\wedge E1(1)$ )), 17)                                                                                                                                      |
| 17    | 19 | $a \in A \wedge 0 < a$                                                         | $A$                                                                                                                                                                                     |
| 17    | 20 | $\exists b \in 0_{\mathbb{R}} (-a < b \wedge 0 < a + b)$                       | $-a < 0$ . Wähle rational $b$ mit $-a < b < 0$ ;<br>Addition von $a$ ergibt $0 < a + b$ .<br>Th. 18.6.1.2( $\wedge E2(19)$ ), Th. 18.4.2.1,<br>Def. 19.4.1.2, Def. 19.2.2.1             |

|        |    |                                                                                                                                 |                                                                       |
|--------|----|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 17     | 21 | $b \in 0_{\mathbb{R}} \wedge -a < b \wedge 0 < a + b$                                                                           | $A$                                                                   |
| 1,2,17 | 22 | $b \in B$                                                                                                                       |                                                                       |
|        |    |                                                                                                                                 | <i>Th.</i> 3.5.2.6( $\wedge E2(3), \wedge E1(21)$ )                   |
| 1,2,17 | 23 | $0 \in A \oplus B$                                                                                                              |                                                                       |
|        |    |                                                                                                                                 | <i>Def.</i> 19.4.1.1( $\wedge E1(19), 22, \wedge E2(\wedge E2(21))$ ) |
| 1,2    | 24 | $0 \in A \rightarrow 0 \in A \oplus B$                                                                                          |                                                                       |
|        |    |                                                                                                                                 | $\rightarrow I(17, \exists E(18, 19, \exists E(20, 21, 23)))$         |
| 1,2    | 25 | $0 \in B \rightarrow 0 \in A \oplus B$                                                                                          |                                                                       |
|        |    |                                                                                                                                 | <i>Th.</i> 19.4.1.3(ii)(1, 24)                                        |
| 1,2    | 26 | $(0 \in A \vee 0 \in B) \rightarrow 0 \in A \oplus B$                                                                           | $\vee E(24, 25)$                                                      |
| 1,2    | 27 | $0 \in A \oplus B \leftrightarrow (0 \in A \vee 0 \in B)$                                                                       |                                                                       |
|        |    |                                                                                                                                 | $\perp I(\rightarrow I(11, 16), 26)$                                  |
| 1,2    | 28 | $0_{\mathbb{R}} \leq_{\mathbb{R}} A \oplus B \wedge (0 \in A \oplus B \leftrightarrow (0 \in A \vee 0 \in B)) \wedge I(10, 27)$ |                                                                       |

**Nullpunkt eines positiven Produkts** *Th.* 19.4.2.3(ii)

$$A, B \in \mathbb{R}, 0_{\mathbb{R}} \leq_{\mathbb{R}} A, 0_{\mathbb{R}} \leq_{\mathbb{R}} B \\ \vdash (0 \in A \odot_+ B \leftrightarrow (0 \in A \wedge 0 \in B))$$

|       |    |                                                                                                                                                                                               |                                                                  |
|-------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1     | 1  | $A, B \in \mathbb{R}$                                                                                                                                                                         | $A$                                                              |
| 2     | 2  | $0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B$                                                                                                                | $A$                                                              |
|       | 3  | $0 \notin 0_{\mathbb{R}}$                                                                                                                                                                     |                                                                  |
|       |    |                                                                                                                                                                                               | <i>Def.</i> 19.4.1.2( <i>Def.</i> 19.2.2.1, <i>Th.</i> 18.4.1.4) |
| 4     | 4  | $0 \in A \odot_+ B$                                                                                                                                                                           | $A$                                                              |
| 1,2,4 | 5  | $\exists a \in A \exists b \in B (0 < a \wedge 0 < b \wedge 0 < ab)$                                                                                                                          | <i>Def.</i> 19.4.2.1(3, 4)                                       |
| 1,5   | 6  | $0 \in A \wedge 0 \in B$                                                                                                                                                                      |                                                                  |
|       |    | Für die positiven Zeugen $a, b$ liegt $0 < a$ bzw. $0 < b$ ;<br>der Abwärtsabschluss von $A$ und $B$ liefert daher jeweils den Nullpunkt.<br><i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1(1), 5) |                                                                  |
| 1,2   | 7  | $0 \in A \odot_+ B \rightarrow (0 \in A \wedge 0 \in B)$                                                                                                                                      | $\rightarrow I(4, 6)$                                            |
| 8     | 8  | $0 \in A \wedge 0 \in B$                                                                                                                                                                      | $A$                                                              |
| 1,8   | 9  | $\exists a \in A (0 < a) \wedge \exists b \in B (0 < b)$                                                                                                                                      |                                                                  |
|       |    |                                                                                                                                                                                               | <i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1(1), 8)                 |
| 8     | 10 | $a \in A \wedge 0 < a \wedge b \in B \wedge 0 < b$                                                                                                                                            | $A$                                                              |
| 10    | 11 | $0 < ab$                                                                                                                                                                                      | <i>Def.</i> 18.5.1(10)                                           |
| 1,2,8 | 12 | $0 \in A \odot_+ B$                                                                                                                                                                           |                                                                  |
|       |    |                                                                                                                                                                                               | <i>Def.</i> 19.4.2.1(10, 11)                                     |
| 1,2   | 13 | $(0 \in A \wedge 0 \in B) \rightarrow 0 \in A \odot_+ B$                                                                                                                                      |                                                                  |
|       |    |                                                                                                                                                                                               | $\rightarrow I(8, \exists E(9, 10, 12))$                         |
| 1,2   | 14 | $0 \in A \odot_+ B \leftrightarrow (0 \in A \wedge 0 \in B)$                                                                                                                                  | $\perp I(7, 13)$                                                 |

**Positive Produktgesetze** *Th.* 19.4.2.3(iii)

$$\begin{aligned}
& A, B, C \in \mathbb{R}, \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} A, \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} B, \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} C \\
& \vdash (A \odot_+ B) \odot_+ C = A \odot_+ (B \odot_+ C) \wedge \\
& A \odot_+ B = B \odot_+ A \wedge A \odot_+ 1_{\mathbb{R}} = A
\end{aligned}$$

- 1    1  $A, B, C \in \mathbb{R}$   $A$
- 2    2  $0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} C$   $A$
- 1,2    3  $A \odot_+ B \in \mathbb{R} \wedge B \odot_+ C \in \mathbb{R} \wedge$   
 $(A \odot_+ B) \odot_+ C \in \mathbb{R} \wedge A \odot_+ (B \odot_+ C) \in \mathbb{R},$   
 $0_{\mathbb{R}} \leq_{\mathbb{R}} A \odot_+ B \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B \odot_+ C,$   
 $0_{\mathbb{R}} \subseteq (A \odot_+ B) \odot_+ C \wedge 0_{\mathbb{R}} \subseteq A \odot_+ (B \odot_+ C)$   
 $Th. 19.4.2.1(1, 2), Def. 19.4.2.1(1, 2), Th. 19.3.1.1(1)$
- 1,2    4  $q < 0 \vdash (q \in (A \odot_+ B) \odot_+ C \leftrightarrow q \in A \odot_+ (B \odot_+ C))$   
 $Def. 19.4.2.1(3), Def. 19.4.1.2$
- 1,2    5  $0 \in (A \odot_+ B) \odot_+ C$   
 $\leftrightarrow ((0 \in A \wedge 0 \in B) \wedge 0 \in C)$   
 $\leftrightarrow (0 \in A \wedge (0 \in B \wedge 0 \in C))$   
 $\leftrightarrow 0 \in A \odot_+ (B \odot_+ C)$   
 $Th. 19.4.2.3(ii)(1, 2, 3)$
- 1,2    6  $q \in (A \odot_+ B) \odot_+ C, \quad 0 < q \vdash$   
 $q \in A \odot_+ (B \odot_+ C)$   

Wähle zunächst  $u \in A \odot_+ B$ ,  $c \in C$  mit  $q < uc$ ,  
danach  $a \in A$ ,  $b \in B$  mit  $u < ab$ .  
Aus  $q < uc < abc$  liefert die rationale Dichte  $d < bc$   
mit  $q < ad$ ; damit ist  $d \in B \odot_+ C$ .  
 $Def. 19.4.2.1(1, 2), Th. 18.3.4.5,$   
 $Th. 18.4.2.3, Th. 18.6.1.2$
- 1,2    7  $q \in A \odot_+ (B \odot_+ C), \quad 0 < q \vdash$   
 $q \in (A \odot_+ B) \odot_+ C$   

Wähle zunächst  $a \in A$ ,  $d \in B \odot_+ C$  mit  $q < ad$ ,  
danach  $b \in B$ ,  $c \in C$  mit  $d < bc$ .  
Aus  $q < ad < abc$  liefert die rationale Dichte  $u < ab$   
mit  $q < uc$ ; damit ist  $u \in A \odot_+ B$ .  
 $Def. 19.4.2.1(1, 2), Th. 18.3.4.5,$   
 $Th. 18.4.2.3, Th. 18.6.1.2$
- 1,2    8  $\forall q \in \mathbb{Q} (q \in (A \odot_+ B) \odot_+ C \leftrightarrow q \in A \odot_+ (B \odot_+ C))$   
 $Th. 18.4.1.4(4, 5, 6, 7), Def. 19.4.2.1(1, 2)$
- 1,2    9  $(A \odot_+ B) \odot_+ C = A \odot_+ (B \odot_+ C)$   $Ax. 3.4.1.1(8)$
- 1,2    10  $\forall q \in \mathbb{Q} (q \in A \odot_+ B \leftrightarrow q \in B \odot_+ A)$   
 $Th. 18.3.4.6(Def. 19.4.2.1(1, 2))$
- 1,2    11  $A \odot_+ B = B \odot_+ A$   $Ax. 3.4.1.1(10)$
- 1,2    12  $1_{\mathbb{R}} \in \mathbb{R} \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} 1_{\mathbb{R}} \wedge 0 \in 1_{\mathbb{R}}$   
 $Def. 19.4.2.3, Th. 18.2.3.6, Ax. 18.5.17, Th. 19.3.1.4$
- 1,2    13  $q < 0 \vdash (q \in A \odot_+ 1_{\mathbb{R}} \leftrightarrow q \in A)$   
 $Def. 19.4.2.1(1, 2), Th. 19.3.1.1(1, 2), Def. 19.4.1.2$
- 1,2    14  $0 \in A \odot_+ 1_{\mathbb{R}} \leftrightarrow 0 \in A$   
 $Th. 19.4.2.3(ii)(1, 2, 12)$

1,2 15  $q \in A \odot_+ 1_{\mathbb{R}}, 0 < q \vdash q \in A$

Die Produktdefinition liefert  $a \in A$  und  $0 < b < 1$   
mit  $q < ab < a$ ; Abwärtsabschluss liefert  $q \in A$ .  
*Def.* 19.4.2.1(1, 2), *Def.* 19.4.2.3,  
*Th.* 18.3.4.7, *Def.* 19.2.1.1(*Th.* 19.2.1.1( $\wedge E1(1)$ )))

1,2 16  $q \in A, 0 < q \vdash q \in A \odot_+ 1_{\mathbb{R}}$

Wähle  $a \in A$  mit  $q < a$  und dann rational  $b$  mit  $qa^{-1} < b < 1$ ; daraus folgt  $q < ab$ .  
*Def.* 19.2.1.1(*Th.* 19.2.1.1( $\wedge E1(1)$ ))), *Th.* 18.6.1.2,  
*Def.* 19.4.2.3, *Th.* 18.4.2.3

1,2 17  $\forall q \in \mathbb{Q} (q \in A \odot_+ 1_{\mathbb{R}} \leftrightarrow q \in A)$

*Th.* 18.4.1.4(13, 14, 15, 16), *Def.* 19.4.2.1(1, 2)

1,2 18  $A \odot_+ 1_{\mathbb{R}} = A$

*Ax.* 3.4.1.1(17)

1,2 19  $(A \odot_+ B) \odot_+ C = A \odot_+ (B \odot_+ C) \wedge A \odot_+ B = B \odot_+ A \wedge A \odot_+ 1_{\mathbb{R}} = A$

$\wedge I(9, \wedge I(11, 18))$

### Positive Distributivität

*Th.* 19.4.2.3(iv)

$$\begin{aligned} A, B, C \in \mathbb{R}, 0_{\mathbb{R}} \leq_{\mathbb{R}} A, 0_{\mathbb{R}} \leq_{\mathbb{R}} B, 0_{\mathbb{R}} \leq_{\mathbb{R}} C \\ \vdash A \odot_+ (B \oplus C) = (A \odot_+ B) \oplus (A \odot_+ C) \end{aligned}$$

1 1  $A, B, C \in \mathbb{R}$   $A$

2 2  $0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} C$   $A$

1,2 3  $A = 0_{\mathbb{R}} \vee B = 0_{\mathbb{R}} \vee C = 0_{\mathbb{R}} \vdash A \odot_+ (B \oplus C) = (A \odot_+ B) \oplus (A \odot_+ C)$

Ist  $A = 0$ , sind beide Produkte rechts und das Produkt links  $0_{\mathbb{R}}$ .  
Ist  $B = 0$  oder  $C = 0$ , verschwinden genau die entsprechenden Faktoren;  
die übrigen Summanden werden durch das additive Nullgesetz identifiziert.  
*Def.* 19.4.2.1(1, 2), *Th.* 19.4.1.3(1)

1,2 4  $B \oplus C \in \mathbb{R} \wedge A \odot_+ B \in \mathbb{R} \wedge A \odot_+ C \in \mathbb{R} \wedge$

$A \odot_+ (B \oplus C) \in \mathbb{R} \wedge (A \odot_+ B) \oplus (A \odot_+ C) \in \mathbb{R},$

$0_{\mathbb{R}} \leq_{\mathbb{R}} B \oplus C \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} A \odot_+ B \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} A \odot_+ C \wedge$

$0_{\mathbb{R}} \leq_{\mathbb{R}} A \odot_+ (B \oplus C) \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} (A \odot_+ B) \oplus (A \odot_+ C)$

*Th.* 19.4.2.3(i)(1, 2), *Th.* 19.4.2.1(1, 2), *Def.* 19.4.2.1(1, 2), *Th.* 19.3.1.1(1)

1,2 5  $q < 0 \vdash (q \in A \odot_+ (B \oplus C) \leftrightarrow q \in (A \odot_+ B) \oplus (A \odot_+ C))$

*Th.* 19.3.1.1(4), *Def.* 19.4.1.2

1,2 6  $0 \in A \odot_+ (B \oplus C)$

$\leftrightarrow (0 \in A \wedge (0 \in B \vee 0 \in C))$

$\leftrightarrow ((0 \in A \wedge 0 \in B) \vee (0 \in A \wedge 0 \in C))$

$\leftrightarrow 0 \in (A \odot_+ B) \oplus (A \odot_+ C)$

*Th.* 19.4.2.3(ii)(1, 2, 4), *Th.* 19.4.2.3(i)(1, 2, 4)

1,2 7  $0_{\mathbb{R}} <_{\mathbb{R}} A, 0_{\mathbb{R}} <_{\mathbb{R}} B, 0_{\mathbb{R}} <_{\mathbb{R}} C,$

$q \in A \odot_+ (B \oplus C), 0 < q \vdash$

$q \in (A \odot_+ B) \oplus (A \odot_+ C)$

Aus  $q < ad$  mit  $a \in A, d \in B \oplus C$  wähle  $b \in B, c \in C$   
so, dass  $d < b + c$ . Dann ist  $q < a(b + c) = ab + ac$ .

Rationale Dichte und Offenheit liefern  $u > ab, v > ac$

mit  $u \in A \odot_+ B, v \in A \odot_+ C$  und weiterhin  $q < u + v$ .

*Def.* 19.4.2.1(1, 2), *Def.* 19.4.1.1(1), *Th.* 18.3.4.9, *Th.* 18.6.1.2

- 1,2 8  $0_{\mathbb{R}} <_{\mathbb{R}} A, 0_{\mathbb{R}} <_{\mathbb{R}} B, 0_{\mathbb{R}} <_{\mathbb{R}} C,$   
 $q \in (A \odot_+ B) \oplus (A \odot_+ C), 0 < q \vdash$   
 $q \in A \odot_+ (B \oplus C)$   
 Schreibe  $q < u + v$  mit  $u < ab$  und  $v < a'c$ . Wähle wegen der Offenheit von  $A$  ein  $a'' \in A$   
 mit  $a, a' < a''$ ; wähle ferner  $b' > b$  in  $B$ ,  $c' > c$  in  $C$  und  $d \in B \oplus C$  mit  $b' + c' < d$ .  
 Dann folgt  $q < ab + a'c < a''(b' + c') < a''d$ .  
*Def.* 19.4.2.1(1,2), *Def.* 19.4.1.1(1), *Def.* 19.2.1.1,  
*Th.* 18.3.4.9, *Th.* 18.6.1.2, *Th.* 18.4.2.3
- 1,2 9  $0_{\mathbb{R}} <_{\mathbb{R}} A, 0_{\mathbb{R}} <_{\mathbb{R}} B, 0_{\mathbb{R}} <_{\mathbb{R}} C \vdash$   
 $\forall q \in \mathbb{Q} (q \in A \odot_+ (B \oplus C) \leftrightarrow q \in (A \odot_+ B) \oplus (A \odot_+ C))$   
*Th.* 18.4.1.4(5,6,7,8), *Def.* 19.4.2.1(1,2), *Def.* 19.4.1.1(1)
- 1,2 10  $0_{\mathbb{R}} <_{\mathbb{R}} A, 0_{\mathbb{R}} <_{\mathbb{R}} B, 0_{\mathbb{R}} <_{\mathbb{R}} C \vdash$  *Ax.* 3.4.1.1(9)  
 $A \odot_+ (B \oplus C) = (A \odot_+ B) \oplus (A \odot_+ C)$
- 1,2 11  $(A = 0_{\mathbb{R}} \vee 0_{\mathbb{R}} <_{\mathbb{R}} A) \wedge (B = 0_{\mathbb{R}} \vee 0_{\mathbb{R}} <_{\mathbb{R}} B) \wedge$  *Th.* 19.3.1.3(1,2)  
 $(C = 0_{\mathbb{R}} \vee 0_{\mathbb{R}} <_{\mathbb{R}} C)$
- 1,2 12  $A \odot_+ (B \oplus C) = (A \odot_+ B) \oplus (A \odot_+ C)$   
 11 zerlegt jeden der drei nichtnegativen Faktoren in Null- und strikt positiven Fall.  
 Sobald ein Nullfall eintritt, gilt 3; der einzige verbleibende Fall ist 10.

## Vorzeichenidentitäten

*Th.* 19.4.2.3(v)

$$A, B \in \mathbb{R} \vdash (\ominus A) \odot B = \neg(A \odot B) \wedge (\ominus A) \odot (\ominus B) = A \odot B \wedge A \odot 0_{\mathbb{R}} = 0_{\mathbb{R}}$$

- 1 1  $A, B \in \mathbb{R}$  *A*
- 1 2  $\ominus A, \ominus B \in \mathbb{R}$  *Th.* 19.4.1.1(1)
- 1 3  $A = 0_{\mathbb{R}} \vee 0_{\mathbb{R}} <_{\mathbb{R}} A \vee A <_{\mathbb{R}} 0_{\mathbb{R}}$  *Th.* 19.3.1.3(1)
- 1 4  $B = 0_{\mathbb{R}} \vee 0_{\mathbb{R}} <_{\mathbb{R}} B \vee B <_{\mathbb{R}} 0_{\mathbb{R}}$  *Th.* 19.3.1.3(1)
- 1 5  $A = 0_{\mathbb{R}} \vee B = 0_{\mathbb{R}} \vdash A \odot B = 0_{\mathbb{R}},$   
 $(\ominus A) \odot B = \neg(A \odot B),$   
 $A \odot (\ominus B) = \neg(A \odot B)$   
*Def.* 19.4.2.2(1), *Th.* 19.4.1.4, *Def.* 19.4.2.1
- 1 6  $0_{\mathbb{R}} <_{\mathbb{R}} A, 0_{\mathbb{R}} <_{\mathbb{R}} B \vdash$   
 $(\ominus A) \odot B = \neg(A \odot_+ B) = \neg(A \odot B),$   
 $A \odot (\ominus B) = \neg(A \odot_+ B) = \neg(A \odot B)$   
*Def.* 19.4.2.2(1), *Th.* 19.4.1.4(1)
- 1 7  $0_{\mathbb{R}} <_{\mathbb{R}} A, B <_{\mathbb{R}} 0_{\mathbb{R}} \vdash$   
 $(\ominus A) \odot B = (A \odot_+ \ominus B) = \neg(A \odot B),$   
 $A \odot (\ominus B) = A \odot_+ (\ominus B) = \neg(A \odot B)$   
*Def.* 19.4.2.2(1), *Th.* 19.4.1.4(1)
- 1 8  $A <_{\mathbb{R}} 0_{\mathbb{R}}, 0_{\mathbb{R}} <_{\mathbb{R}} B \vdash$   
 $(\ominus A) \odot B = (\ominus A) \odot_+ B = \neg(A \odot B),$   
 $A \odot (\ominus B) = (\ominus A) \odot_+ B = \neg(A \odot B)$   
*Def.* 19.4.2.2(1), *Th.* 19.4.1.4(1), *Th.* 19.4.2.3(iii)(1)
- 1 9  $A <_{\mathbb{R}} 0_{\mathbb{R}}, B <_{\mathbb{R}} 0_{\mathbb{R}} \vdash$   
 $(\ominus A) \odot B = \neg((\ominus A) \odot_+ (\ominus B)) = \neg(A \odot B),$   
 $A \odot (\ominus B) = \neg((\ominus A) \odot_+ (\ominus B)) = \neg(A \odot B)$   
*Def.* 19.4.2.2(1), *Th.* 19.4.1.4(1)
- 1 10  $(\ominus A) \odot B = \neg(A \odot B) \wedge A \odot (\ominus B) = \neg(A \odot B) \vee$  *E*(3,4,5,6,7,8,9)

$$\begin{array}{ll}
1 \quad 11 \quad \ominus \ominus (A \odot B) = A \odot B & \text{Th. 19.4.1.4 (Th. 19.4.2.1(1))} \\
1 \quad 12 \quad (\ominus A) \odot (\ominus B) = A \odot B & = E(10, 11) \\
1 \quad 13 \quad A \odot 0_{\mathbb{R}} = 0_{\mathbb{R}} & \text{Def. 19.4.2.2(1), Def. 19.4.2.1, Th. 19.4.1.4} \\
1 \quad 14 \quad (\ominus A) \odot B = \ominus(A \odot B) \wedge (\ominus A) \odot (\ominus B) = A \odot B \wedge A \odot 0_{\mathbb{R}} = 0_{\mathbb{R}} & \wedge I(\wedge E1(10), \wedge I(12, 13))
\end{array}$$

## Positives Produktinverses

Th. 19.4.2.3(vi)

$$A \in \mathbb{R}, \quad 0_{\mathbb{R}} <_{\mathbb{R}} A \vdash A \odot_+ \text{inv}_+(A) = 1_{\mathbb{R}}$$

$$\begin{array}{ll}
1 \quad 1 \quad A \in \mathbb{R} & A \\
2 \quad 2 \quad 0_{\mathbb{R}} <_{\mathbb{R}} A & A \\
3 \quad 3 \quad q \in A \odot_+ \text{inv}_+(A) & A \\
1,2,3 \quad 4 \quad q \in 1_{\mathbb{R}} & \text{Def. 19.4.2.1 (Def. 19.4.2.4(1, 2, 3), Th. 18.4.2.3)} \\
1,2 \quad 5 \quad A \odot_+ \text{inv}_+(A) \subseteq 1_{\mathbb{R}} & \text{Def. 3.5.1.1}(\forall I(\rightarrow I(3, 4))) \\
6 \quad 6 \quad q \in 1_{\mathbb{R}} & A \\
6 \quad 7 \quad q < 1 & \text{Def. 19.4.2.3 (Def. 19.2.2.1(6))} \\
1,2,6 \quad 8 \quad q < 0 \vee q = 0 \vee (0 < q < 1) & \text{Th. 18.4.1.4(7)} \\
9 \quad 9 \quad q < 0 & A \\
1,2,9 \quad 10 \quad q \in A \odot_+ \text{inv}_+(A) & \text{Def. 19.4.2.1(1, 2), Def. 19.4.1.2(9)} \\
11 \quad 11 \quad q = 0 & A \\
1,2,11 \quad 12 \quad \exists a \in A \exists r \in \mathbb{Q} \setminus A (0 < a < r) & \text{Th. 19.4.2.1(ii)(1, 2), Def. 19.2.1.1 (Th. 19.2.1.1(1)),} \\
& \text{Th. 19.4.2.1(i)(1)} \\
12 \quad 13 \quad a \in A \wedge r \in \mathbb{Q} \setminus A \wedge 0 < a < rA & \\
12 \quad 14 \quad \exists b \in \mathbb{Q} (0 < b < r^{-1}) & \text{Th. 18.4.2.4(13), Th. 18.6.1.2(13)} \\
12 \quad 15 \quad b \in \mathbb{Q} \wedge 0 < b < r^{-1} & A \\
1,2,12 \quad 16 \quad b \in \text{inv}_+(A) \wedge 0 < ab & \text{Def. 19.4.2.4(13, 15), Def. 18.5.1(13, 15)} \\
1,2,11,12 \quad 17 \quad q \in A \odot_+ \text{inv}_+(A) & \text{Def. 19.4.2.1(11, 13, 16)} \\
18 \quad 18 \quad 0 < q < 1 & A \\
1,2,18 \quad 19 \quad \exists a \in A \exists r \notin A (0 < a < r \wedge q < ar^{-1}) & \text{Th. 19.4.2.2(1, 2, 18)} \\
1,2,18 \quad 20 \quad \exists b \in \mathbb{Q} (qa^{-1} < b \wedge b < r^{-1}) & \text{Th. 18.6.1.2(19)} \\
1,2,18 \quad 21 \quad b \in \text{inv}_+(A) \wedge q < ab & \text{Def. 19.4.2.4(19, 20)}
\end{array}$$

$$1,2,18 \quad 22 \quad q \in A \odot_+ \text{inv}_+(A)$$

*Def.* 19.4.2.1(19, 21)

$$1,2,6 \quad 23 \quad q \in A \odot_+ \text{inv}_+(A)$$

$\vee E(8, 9, 10, 11, 17, 18, 22)$

$$1,2 \quad 24 \quad 1_{\mathbb{R}} \subseteq A \odot_+ \text{inv}_+(A)$$

*Def.* 3.5.1.1( $\forall I(\rightarrow I(6, 23))$ )

$$1,2 \quad 25 \quad A \odot_+ \text{inv}_+(A) = 1_{\mathbb{R}}$$

*Th.* 3.5.3.5(5, 24)

## Allgemeine Assoziativität, Einheit und Inverses

*Th.* 19.4.2.3(vii)

$$\begin{aligned} A, B, C \in \mathbb{R} \vdash (A \odot B) \odot C &= A \odot (B \odot C) \wedge \\ A \odot B &= B \odot A \wedge A \odot 1_{\mathbb{R}} = A \wedge \\ (A \neq 0_{\mathbb{R}} \rightarrow A \odot \text{inv}_{\mathbb{R}}(A) &= 1_{\mathbb{R}}) \end{aligned}$$

$$1 \quad 1 \quad A, B, C \in \mathbb{R}$$

$$A$$

$$\begin{aligned} 1 \quad 2 \quad & (A = 0_{\mathbb{R}} \vee 0_{\mathbb{R}} <_{\mathbb{R}} A \vee A <_{\mathbb{R}} 0_{\mathbb{R}}) \wedge \\ & (B = 0_{\mathbb{R}} \vee 0_{\mathbb{R}} <_{\mathbb{R}} B \vee B <_{\mathbb{R}} 0_{\mathbb{R}}) \wedge \\ & (C = 0_{\mathbb{R}} \vee 0_{\mathbb{R}} <_{\mathbb{R}} C \vee C <_{\mathbb{R}} 0_{\mathbb{R}}) \end{aligned}$$

*Th.* 19.3.1.3(1)

$$1 \quad 3 \quad A = 0_{\mathbb{R}} \vee B = 0_{\mathbb{R}} \vee C = 0_{\mathbb{R}} \vdash (A \odot B) \odot C = A \odot (B \odot C)$$

*Th.* 19.4.2.3(v)(1)

$$1 \quad 4 \quad 0_{\mathbb{R}} <_{\mathbb{R}} A, 0_{\mathbb{R}} <_{\mathbb{R}} B, 0_{\mathbb{R}} <_{\mathbb{R}} C \vdash$$

$$(A \odot B) \odot C = (A \odot_+ B) \odot_+ C$$

$$= A \odot_+ (B \odot_+ C) = A \odot (B \odot C)$$

*Def.* 19.4.2.2(1), *Th.* 19.4.2.3(iii)(1)

$$1 \quad 5 \quad A <_{\mathbb{R}} 0_{\mathbb{R}}, 0_{\mathbb{R}} <_{\mathbb{R}} B, 0_{\mathbb{R}} <_{\mathbb{R}} C \vdash$$

$$(A \odot B) \odot C = \ominus(((\ominus A) \odot_+ B) \odot_+ C)$$

$$= \ominus((\ominus A) \odot_+ (B \odot_+ C)) = A \odot (B \odot C)$$

*Def.* 19.4.2.2(1), *Th.* 19.4.2.3(iii)(1), *Th.* 19.4.2.3(v)(1)

$$1 \quad 6 \quad B <_{\mathbb{R}} 0_{\mathbb{R}}, 0_{\mathbb{R}} <_{\mathbb{R}} A, 0_{\mathbb{R}} <_{\mathbb{R}} C \vdash$$

$$(A \odot B) \odot C = \ominus((A \odot_+ \ominus B) \odot_+ C)$$

$$= \ominus(A \odot_+ ((\ominus B) \odot_+ C)) = A \odot (B \odot C)$$

*Def.* 19.4.2.2(1), *Th.* 19.4.2.3(iii)(1), *Th.* 19.4.2.3(v)(1)

$$1 \quad 7 \quad C <_{\mathbb{R}} 0_{\mathbb{R}}, 0_{\mathbb{R}} <_{\mathbb{R}} A, 0_{\mathbb{R}} <_{\mathbb{R}} B \vdash$$

$$(A \odot B) \odot C = \ominus((A \odot_+ B) \odot_+ \ominus C)$$

$$= \ominus(A \odot_+ (B \odot_+ \ominus C)) = A \odot (B \odot C)$$

*Def.* 19.4.2.2(1), *Th.* 19.4.2.3(iii)(1), *Th.* 19.4.2.3(v)(1)

$$1 \quad 8 \quad A <_{\mathbb{R}} 0_{\mathbb{R}}, B <_{\mathbb{R}} 0_{\mathbb{R}}, 0_{\mathbb{R}} <_{\mathbb{R}} C \vdash$$

$$(A \odot B) \odot C = ((\ominus A) \odot_+ (\ominus B)) \odot_+ C$$

$$= (\ominus A) \odot_+ ((\ominus B) \odot_+ C) = A \odot (B \odot C)$$

*Def.* 19.4.2.2(1), *Th.* 19.4.2.3(iii)(1)

$$1 \quad 9 \quad A <_{\mathbb{R}} 0_{\mathbb{R}}, C <_{\mathbb{R}} 0_{\mathbb{R}}, 0_{\mathbb{R}} <_{\mathbb{R}} B \vdash$$

$$(A \odot B) \odot C = ((\ominus A) \odot_+ B) \odot_+ (\ominus C)$$

$$= (\ominus A) \odot_+ (B \odot_+ \ominus C) = A \odot (B \odot C)$$

*Def.* 19.4.2.2(1), *Th.* 19.4.2.3(iii)(1)

$$1 \quad 10 \quad B <_{\mathbb{R}} 0_{\mathbb{R}}, C <_{\mathbb{R}} 0_{\mathbb{R}}, 0_{\mathbb{R}} <_{\mathbb{R}} A \vdash$$

$$(A \odot B) \odot C = (A \odot_+ \ominus B) \odot_+ (\ominus C)$$

$$= A \odot_+ ((\ominus B) \odot_+ (\ominus C)) = A \odot (B \odot C)$$

*Def.* 19.4.2.2(1), *Th.* 19.4.2.3(iii)(1)

|         |                                                                                                                                                                                                                                                                                                                                   |                                                                                      |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1 11    | $A <_{\mathbb{R}} 0_{\mathbb{R}}, B <_{\mathbb{R}} 0_{\mathbb{R}}, C <_{\mathbb{R}} 0_{\mathbb{R}} \vdash$<br>$(A \odot B) \odot C = \ominus(((\ominus A) \odot_+ (\ominus B)) \odot_+ (\ominus C))$<br>$= \ominus((\ominus A) \odot_+ ((\ominus B) \odot_+ (\ominus C))) = A \odot (B \odot C)$                                  | <i>Def.</i> 19.4.2.2(1), <i>Th.</i> 19.4.2.3(iii)(1), <i>Th.</i> 19.4.2.3(v)(1)      |
| 1 12    | $(A \odot B) \odot C = A \odot (B \odot C)$                                                                                                                                                                                                                                                                                       | $\vee E(2, 3, 4, 5, 6, 7, 8, 9, 10, 11)$                                             |
| 1 13    | $0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B, A <_{\mathbb{R}} 0_{\mathbb{R}} \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B, Th. 19.3.1.3(1)$<br>$0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge B <_{\mathbb{R}} 0_{\mathbb{R}}, A <_{\mathbb{R}} 0_{\mathbb{R}} \wedge B <_{\mathbb{R}} 0_{\mathbb{R}}$ |                                                                                      |
| 1 14    | $A \odot B = B \odot A$                                                                                                                                                                                                                                                                                                           | <i>Def.</i> 19.4.2.2(1, 13), <i>Th.</i> 19.4.2.3(iii)(1), <i>Th.</i> 19.4.2.3(v)(1)  |
| 1 15    | $0_{\mathbb{R}} \leq_{\mathbb{R}} A \vdash A \odot 1_{\mathbb{R}} = A \odot_+ 1_{\mathbb{R}} = A,$<br>$A <_{\mathbb{R}} 0_{\mathbb{R}} \vdash A \odot 1_{\mathbb{R}} = \ominus((\ominus A) \odot_+ 1_{\mathbb{R}}) = A$                                                                                                           | <i>Def.</i> 19.4.2.2(1), <i>Th.</i> 19.4.2.3(iii)(1), <i>Th.</i> 19.4.1.4(1)         |
| 1 16    | $A \odot 1_{\mathbb{R}} = A$                                                                                                                                                                                                                                                                                                      | <i>Th.</i> 19.3.1.3(1, 15)                                                           |
| 17 17   | $A \neq 0_{\mathbb{R}}$                                                                                                                                                                                                                                                                                                           | $A$                                                                                  |
| 1,17 18 | $0_{\mathbb{R}} <_{\mathbb{R}} A \vee A <_{\mathbb{R}} 0_{\mathbb{R}}$                                                                                                                                                                                                                                                            | <i>Th.</i> 19.3.1.3(1, 17)                                                           |
| 1,17 19 | $0_{\mathbb{R}} <_{\mathbb{R}} A$                                                                                                                                                                                                                                                                                                 | $A$                                                                                  |
| 1,17 20 | $A \odot \text{inv}_{\mathbb{R}}(A) = A \odot_+ \text{inv}_+(A)$                                                                                                                                                                                                                                                                  | <i>Def.</i> 19.4.2.2(1, 19), <i>Def.</i> 19.4.2.5(1, 17, 19)                         |
| 1,17 21 | $A \odot_+ \text{inv}_+(A) = 1_{\mathbb{R}}$                                                                                                                                                                                                                                                                                      | <i>Th.</i> 19.4.2.3(vi)(1, 19)                                                       |
| 22 22   | $A <_{\mathbb{R}} 0_{\mathbb{R}}$                                                                                                                                                                                                                                                                                                 | $A$                                                                                  |
| 1,17 23 | $A \odot \text{inv}_{\mathbb{R}}(A) = (\ominus A) \odot_+ \text{inv}_+(\ominus A)$                                                                                                                                                                                                                                                | <i>Def.</i> 19.4.2.2(1, 22), <i>Def.</i> 19.4.2.5(1, 17, 22), <i>Th.</i> 19.4.1.4(1) |
| 1,17 24 | $(\ominus A) \odot_+ \text{inv}_+(\ominus A) = 1_{\mathbb{R}}$                                                                                                                                                                                                                                                                    | <i>Th.</i> 19.4.2.3(vi)(1, <i>Th.</i> 19.4.1.4(1, 22))                               |
| 1,17 25 | $A \odot \text{inv}_{\mathbb{R}}(A) = 1_{\mathbb{R}}$                                                                                                                                                                                                                                                                             | $\vee E(18, 19, 20, 21, 22, 23, 24)$                                                 |
| 1 26    | $A \neq 0_{\mathbb{R}} \rightarrow A \odot \text{inv}_{\mathbb{R}}(A) = 1_{\mathbb{R}}$                                                                                                                                                                                                                                           | $\rightarrow I(17, 25)$                                                              |
| 1 27    | $(A \odot B) \odot C = A \odot (B \odot C) \wedge$<br>$A \odot B = B \odot A \wedge$<br>$A \odot 1_{\mathbb{R}} = A \wedge$<br>$(A \neq 0_{\mathbb{R}} \rightarrow A \odot \text{inv}_{\mathbb{R}}(A) = 1_{\mathbb{R}})$                                                                                                          | $\wedge I(12, \wedge I(14, \wedge I(16, 26)))$                                       |

## Nichtnegative Differenz

*Th.* 19.4.2.3(viii)

$$X, Y \in \mathbb{R}, X \leq_{\mathbb{R}} Y \vdash 0_{\mathbb{R}} \leq_{\mathbb{R}} Y \oplus \ominus X$$

|       |                                       |                           |
|-------|---------------------------------------|---------------------------|
| 1 1   | $X, Y \in \mathbb{R}$                 | $A$                       |
| 2 2   | $X \leq_{\mathbb{R}} Y$               | $A$                       |
| 1,2 3 | $X \subseteq Y$                       | <i>Th.</i> 19.3.1.1(1, 2) |
| 1 4   | $X \oplus \ominus X = 0_{\mathbb{R}}$ | <i>Th.</i> 19.4.1.3(1)    |
| 5 5   | $q \in 0_{\mathbb{R}}$                | $A$                       |
| 1,5 6 | $q \in X \oplus \ominus X$            | $= E(4, 5)$               |

|       |                                                          |                                                        |
|-------|----------------------------------------------------------|--------------------------------------------------------|
| 1,5   | $7 \exists x \in X \exists z \in \ominus X (q < x + z)$  | <i>Def.</i> 19.4.1.1(1,6)                              |
| 1,2,5 | $8 \exists y \in Y \exists z \in \ominus X (q < y + z)$  | <i>Th.</i> 3.5.2.6(3,7)                                |
| 1,2,5 | $9 q \in Y \oplus \ominus X$                             | <i>Def.</i> 19.4.1.1(1,8)                              |
| 1,2   | $10 0_{\mathbb{R}} \subseteq Y \oplus \ominus X$         |                                                        |
|       |                                                          | <i>Def.</i> 3.5.1.1( $\forall I(\rightarrow I(5,9))$ ) |
| 1     | $11 Y \oplus \ominus X \in \mathbb{R}$                   | <i>Th.</i> 19.4.1.1(1)                                 |
| 1,2   | $12 0_{\mathbb{R}} \leq_{\mathbb{R}} Y \oplus \ominus X$ | <i>Th.</i> 19.3.1.1(10,11)                             |

## Allgemeine Distributivität

Th. 19.4.2.3(ix)

$$A, B, C \in \mathbb{R} \vdash A \odot (B \oplus C) = (A \odot B) \oplus (A \odot C)$$

|   |                                                                                                                                                                                                                                                                                                          |                                                                                                          |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 1 | $1 A, B, C \in \mathbb{R}$                                                                                                                                                                                                                                                                               | $A$                                                                                                      |
| 1 | $2 (0_{\mathbb{R}} \leq_{\mathbb{R}} A \vee A <_{\mathbb{R}} 0_{\mathbb{R}}) \wedge$<br>$(0_{\mathbb{R}} \leq_{\mathbb{R}} B \vee B <_{\mathbb{R}} 0_{\mathbb{R}}) \wedge$<br>$(0_{\mathbb{R}} \leq_{\mathbb{R}} C \vee C <_{\mathbb{R}} 0_{\mathbb{R}})$                                                | <i>Th.</i> 19.3.1.3(1)                                                                                   |
| 1 | $3 0_{\mathbb{R}} \leq_{\mathbb{R}} A, 0_{\mathbb{R}} \leq_{\mathbb{R}} B, 0_{\mathbb{R}} \leq_{\mathbb{R}} C \vdash A \odot (B \oplus C) = (A \odot B) \oplus (A \odot C)$                                                                                                                              | <i>Th.</i> 19.4.2.3(iv)(1), <i>Def.</i> 19.4.2.2(1)                                                      |
| 1 | $4 0_{\mathbb{R}} \leq_{\mathbb{R}} A, B <_{\mathbb{R}} 0_{\mathbb{R}}, C <_{\mathbb{R}} 0_{\mathbb{R}} \vdash$<br>$A \odot (B \oplus C) = \ominus(A \odot_+ ((\ominus B) \oplus (\ominus C)))$<br>$= \ominus((A \odot_+ \ominus B) \oplus (A \odot_+ \ominus C))$<br>$= (A \odot B) \oplus (A \odot C)$ | <i>Th.</i> 19.4.2.3(iv)(1), <i>Th.</i> 19.4.2.3(v)(1), <i>Th.</i> 19.4.1.3(1), <i>Th.</i> 19.4.1.4(1)    |
| 1 | $5 0_{\mathbb{R}} \leq_{\mathbb{R}} A, 0_{\mathbb{R}} \leq_{\mathbb{R}} B, C <_{\mathbb{R}} 0_{\mathbb{R}} \vdash$<br>$D := \ominus C, 0_{\mathbb{R}} <_{\mathbb{R}} D, (D \leq_{\mathbb{R}} B \vee B \leq_{\mathbb{R}} D)$                                                                              | <i>Th.</i> 19.4.1.4(1), <i>Th.</i> 19.3.1.3(1)                                                           |
| 1 | $6 D \leq_{\mathbb{R}} B, E := B \oplus C \vdash$<br>$0_{\mathbb{R}} \leq_{\mathbb{R}} E, B = E \oplus D,$<br>$A \odot B = (A \odot E) \oplus (A \odot D)$                                                                                                                                               | <i>Th.</i> 19.4.2.3(viii)(1), <i>Th.</i> 19.4.1.3(1), <i>Th.</i> 19.4.2.3(iv)(1)                         |
| 1 | $7 D \leq_{\mathbb{R}} B \vdash$<br>$A \odot (B \oplus C) = A \odot E$<br>$= (A \odot B) \oplus \ominus(A \odot D) = (A \odot B) \oplus (A \odot C)$                                                                                                                                                     | <i>Th.</i> 19.4.1.4(i)(6), <i>Th.</i> 19.4.2.3(v)(1)                                                     |
| 1 | $8 B \leq_{\mathbb{R}} D, E := \ominus(B \oplus C) \vdash$<br>$0_{\mathbb{R}} \leq_{\mathbb{R}} E, D = B \oplus E,$<br>$A \odot D = (A \odot B) \oplus (A \odot E)$                                                                                                                                      | <i>Th.</i> 19.4.2.3(viii)(1), <i>Th.</i> 19.4.1.3(1), <i>Th.</i> 19.4.1.4(1), <i>Th.</i> 19.4.2.3(iv)(1) |
| 1 | $9 B \leq_{\mathbb{R}} D \vdash$<br>$A \odot (B \oplus C) = \ominus(A \odot E)$<br>$= (A \odot B) \oplus \ominus(A \odot D) = (A \odot B) \oplus (A \odot C)$                                                                                                                                            | <i>Th.</i> 19.4.1.4(i)(8), <i>Th.</i> 19.4.2.3(v)(1)                                                     |
| 1 | $10 0_{\mathbb{R}} \leq_{\mathbb{R}} A, 0_{\mathbb{R}} \leq_{\mathbb{R}} B, C <_{\mathbb{R}} 0_{\mathbb{R}} \vdash A \odot (B \oplus C) = (A \odot B) \oplus (A \odot C)$                                                                                                                                | $\vee E(5, 6, 7, 8, 9)$                                                                                  |
| 1 | $11 0_{\mathbb{R}} \leq_{\mathbb{R}} A, B <_{\mathbb{R}} 0_{\mathbb{R}}, 0_{\mathbb{R}} \leq_{\mathbb{R}} C \vdash A \odot (B \oplus C) = (A \odot B) \oplus (A \odot C)$                                                                                                                                | <i>Th.</i> 19.4.1.3(ii)(1,10), <i>Th.</i> 19.4.2.3(vi)(1)                                                |
| 1 | $12 0_{\mathbb{R}} \leq_{\mathbb{R}} A \vdash A \odot (B \oplus C) = (A \odot B) \oplus (A \odot C)$                                                                                                                                                                                                     | $\vee E(2, 3, 4, 10, 11)$                                                                                |

$$\begin{array}{lcl}
1 & 13 & A <_{\mathbb{R}} 0_{\mathbb{R}} \vdash \\
& & A \odot (B \oplus C) = \ominus((\ominus A) \odot (B \oplus C)) \\
& & = \ominus(((\ominus A) \odot B) \oplus ((\ominus A) \odot C)) \\
& & = (A \odot B) \oplus (A \odot C) \\
& & \text{Th. 19.4.2.3(v)(1), Th. 19.4.1.4(1), 12, Th. 19.4.1.3(1)} \\
1 & 14 & A \odot (B \oplus C) = (A \odot B) \oplus (A \odot C) \quad \vee E(2, 12, 13)
\end{array}$$

Zusammenfassung:

$$\begin{array}{lcl}
1 & 1 & A, B, C \in \mathbb{R} \quad A \\
1 & 2 & (A \odot B) \odot C = A \odot (B \odot C) \wedge \\
& & A \odot B = B \odot A \wedge \\
& & A \odot 1_{\mathbb{R}} = A \wedge \\
& & (A \neq 0_{\mathbb{R}} \rightarrow A \odot \text{inv}_{\mathbb{R}}(A) = 1_{\mathbb{R}}) \\
& & \text{Th. 19.4.2.3(vii)(1)} \\
1 & 3 & A \odot (B \oplus C) = (A \odot B) \oplus (A \odot C) \quad \text{Th. 19.4.2.3(ix)(1)} \\
1 & 4 & (A \odot B) \odot C = A \odot (B \odot C) \wedge \quad \wedge I(2, 3) \\
& & A \odot B = B \odot A \wedge \\
& & A \odot 1_{\mathbb{R}} = A \wedge \\
& & A \odot (B \oplus C) = (A \odot B) \oplus (A \odot C) \wedge \\
& & (A \neq 0_{\mathbb{R}} \rightarrow A \odot \text{inv}_{\mathbb{R}}(A) = 1_{\mathbb{R}})
\end{array}$$

□

**Theorem 19.4.2.4 (Verträglichkeit von Ordnung und Arithmetik).**

**Mengen:**  $A, B, C$   
**Relationsoperatoren:**  $\leq_{\mathbb{R}}$   
**Funktionsoperatoren:**  $\oplus, \odot$

$$\begin{aligned}
A, B, C \in \mathbb{R} \vdash & (A \leq_{\mathbb{R}} B \rightarrow A \oplus C \leq_{\mathbb{R}} B \oplus C) \wedge \\
& (0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B \rightarrow 0_{\mathbb{R}} \leq_{\mathbb{R}} A \odot B).
\end{aligned}$$

*Beweis.*

**Translationsmonotonie** Th. 19.4.2.4(i)

$$A, B, C \in \mathbb{R}, A \leq_{\mathbb{R}} B \vdash A \oplus C \leq_{\mathbb{R}} B \oplus C$$

$$\begin{array}{lcl}
1 & 1 & A, B, C \in \mathbb{R} \quad A \\
2 & 2 & A \leq_{\mathbb{R}} B \quad A \\
1,2 & 3 & A \subseteq B \quad \text{Th. 19.3.1.1(1,2)} \\
4 & 4 & q \in A \oplus C \quad A \\
1,4 & 5 & \exists a \in A \exists c \in C (q < a + c) \text{Def. 19.4.1.1(1,4)} \\
1,2,4 & 6 & \exists a \in B \exists c \in C (q < a + c) \text{Th. 3.5.2.6(3,5)} \\
1,2,4 & 7 & q \in B \oplus C \quad \text{Def. 19.4.1.1(1,6)} \\
1,2 & 8 & A \oplus C \subseteq B \oplus C \quad \text{Def. 3.5.1.1}(\forall I(\rightarrow I(4, 7)))
\end{array}$$

$$1,2 \quad 9 \quad A \oplus C \leq_{\mathbb{R}} B \oplus C \quad \text{Def. 19.3.1.1(Th. 19.4.1.1(1),8)}$$

**Nichtnegative Produkte**

Th. 19.4.2.4(ii)

$$A, B \in \mathbb{R}, \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} A, \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} B \vdash 0_{\mathbb{R}} \leq_{\mathbb{R}} A \odot B$$

$$\begin{array}{ll} 1 \quad 1 \quad A, B \in \mathbb{R} & A \\ 2 \quad 2 \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} A & A \\ 3 \quad 3 \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} B & A \\ 1,2,3 \quad 4 \quad A \odot B = A \odot_+ B & \text{Def. 19.4.2.2(1,2,3)} \\ 1,2,3 \quad 5 \quad 0_{\mathbb{R}} \subseteq A \odot_+ B & \text{Def. 19.4.2.1(1,2,3)} \\ 1,2,3 \quad 6 \quad 0_{\mathbb{R}} \subseteq A \odot B & = E(4,5) \\ 1,2,3 \quad 7 \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} A \odot B & \text{Def. 19.3.1.1(Th. 19.4.2.1(1),6)} \end{array}$$

Zusammenfassung:

$$\begin{array}{ll} 1 \quad 1 \quad A, B, C \in \mathbb{R} & A \\ 1 \quad 2 \quad A \leq_{\mathbb{R}} B \rightarrow A \oplus C \leq_{\mathbb{R}} B \oplus C & \text{Th. 19.4.2.4(i)(1)} \\ 1 \quad 3 \quad (0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B) \rightarrow 0_{\mathbb{R}} \leq_{\mathbb{R}} A \odot B & \text{Th. 19.4.2.4(ii)(1)} \\ 1 \quad 4 \quad (A \leq_{\mathbb{R}} B \rightarrow A \oplus C \leq_{\mathbb{R}} B \oplus C) \wedge ((0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B) \rightarrow 0_{\mathbb{R}} \leq_{\mathbb{R}} A \odot B) \end{array}$$

$\wedge I(2,3)$

□

### 4.3 Erhaltung der rationalen Arithmetik

**Theorem 19.4.3.1** (Die rationale Einbettung erhält die Körperoperationen).

**Mengen:**  $p, q$   
**Funktionen:**  $\iota_{\mathbb{R}}$   
**Funktionsoperatoren:**  $\oplus, \odot, \ominus \cdot, \text{inv}_{\mathbb{R}}(\cdot)$

$$\begin{aligned} p, q \in \mathbb{Q} \vdash \iota_{\mathbb{R}}(p + q) &= \iota_{\mathbb{R}}(p) \oplus \iota_{\mathbb{R}}(q), \\ \iota_{\mathbb{R}}(pq) &= \iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(q), \\ \iota_{\mathbb{R}}(-p) &= \ominus \iota_{\mathbb{R}}(p), \\ p \neq 0 \rightarrow \iota_{\mathbb{R}}(p^{-1}) &= \text{inv}_{\mathbb{R}}(\iota_{\mathbb{R}}(p)). \end{aligned}$$

*Beweis.*

**Erhaltung der Addition**

Th. 19.4.3.1(i)

$$p, q \in \mathbb{Q} \vdash \iota_{\mathbb{R}}(p + q) = \iota_{\mathbb{R}}(p) \oplus \iota_{\mathbb{R}}(q)$$

|     |                                                                                               |                                                   |
|-----|-----------------------------------------------------------------------------------------------|---------------------------------------------------|
| 1   | $1 \ p, q \in \mathbb{Q}$                                                                     | $A$                                               |
| 2   | $2 \ t \in \iota_{\mathbb{R}}(p) \oplus \iota_{\mathbb{R}}(q)$                                | $A$                                               |
| 1,2 | $3 \ \exists a < p \exists b < q (t < a + b)$                                                 | Def. 19.2.2.2(Def. 19.4.1.1(1,2))                 |
| 1,2 | $4 \ t < p + q$                                                                               | Th. 18.4.2.1(3)                                   |
| 1,2 | $5 \ t \in \iota_{\mathbb{R}}(p + q)$                                                         | Def. 19.2.2.2(Def. 19.2.2.1(1,4))                 |
| 1   | $6 \ \iota_{\mathbb{R}}(p) \oplus \iota_{\mathbb{R}}(q) \subseteq \iota_{\mathbb{R}}(p + q)$  | Def. 3.5.1.1( $\forall I(\rightarrow I(2, 5))$ )  |
| 7   | $7 \ t \in \iota_{\mathbb{R}}(p + q)$                                                         | $A$                                               |
| 1,7 | $8 \ t < p + q$                                                                               | Def. 19.2.2.2(Def. 19.2.2.1(1,7))                 |
| 1,7 | $9 \ \exists a < p \exists b < q (t < a + b)$                                                 | Th. 18.6.1.2(8)                                   |
| 1,7 | $10 \ t \in \iota_{\mathbb{R}}(p) \oplus \iota_{\mathbb{R}}(q)$                               | Def. 19.2.2.2(Def. 19.4.1.1(1,9))                 |
| 1   | $11 \ \iota_{\mathbb{R}}(p + q) \subseteq \iota_{\mathbb{R}}(p) \oplus \iota_{\mathbb{R}}(q)$ | Def. 3.5.1.1( $\forall I(\rightarrow I(7, 10))$ ) |
| 1   | $12 \ \iota_{\mathbb{R}}(p + q) = \iota_{\mathbb{R}}(p) \oplus \iota_{\mathbb{R}}(q)$         | Th. 3.5.3.5(11,6)                                 |

### Erhaltung von Negation und Produkt

Th. 19.4.3.1(ii)

$$p, q \in \mathbb{Q} \vdash \iota_{\mathbb{R}}(-p) = \ominus \iota_{\mathbb{R}}(p) \wedge \\ \iota_{\mathbb{R}}(pq) = \iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(q)$$

|   |                                                                                                                                                                                                                  |                                                                          |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1 | $1 \ p, q \in \mathbb{Q}$                                                                                                                                                                                        | $A$                                                                      |
| 1 | $2 \ \forall t \in \mathbb{Q} (t \in \iota_{\mathbb{R}}(-p) \leftrightarrow t \in \ominus \iota_{\mathbb{R}}(p))$                                                                                                | $Th. 18.3.1.4(Def. 19.2.2.2(1), Def. 19.4.1.3(1))$                       |
| 1 | $3 \ \iota_{\mathbb{R}}(-p) = \ominus \iota_{\mathbb{R}}(p)$                                                                                                                                                     | Ax. 3.4.1.1(2)                                                           |
| 1 | $4 \ (p = 0 \vee 0 < p \vee p < 0) \wedge$<br>$(q = 0 \vee 0 < q \vee q < 0)$                                                                                                                                    | $Th. 18.4.1.4(1)$                                                        |
| 1 | $5 \ p = 0 \vee q = 0 \vdash \iota_{\mathbb{R}}(pq) = \iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(q)$                                                                                                         | $Th. 18.3.4.8(1), Def. 19.4.1.2, Th. 19.4.2.3(v)(Th. 19.2.2.2(1))$       |
| 1 | $6 \ \forall t \in \mathbb{Q} (0 < p \wedge 0 < q \rightarrow (t \in \iota_{\mathbb{R}}(pq) \leftrightarrow t \in \iota_{\mathbb{R}}(p) \odot_+ \iota_{\mathbb{R}}(q)))$                                         | $Def. 19.2.2.2(1), Def. 19.4.2.1(1),$<br>$Th. 18.3.4.5(1), Th. 18.6.1.2$ |
| 1 | $7 \ 0 < p, 0 < q \vdash \iota_{\mathbb{R}}(pq) = \iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(q)$                                                                                                             | $Ax. 3.4.1.1(6), Th. 19.3.1.4(1), Def. 19.4.2.2(1)$                      |
| 1 | $8 \ 0 < p, q < 0 \vdash$<br>$\iota_{\mathbb{R}}(pq) = \ominus \iota_{\mathbb{R}}(p(-q)) = \ominus(\iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(-q))$<br>$= \iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(q)$ | $Th. 18.3.1.4(1), 3, 7, Th. 19.3.1.4(1), Th. 19.4.2.3(v)(1)$             |
| 1 | $9 \ p < 0, 0 < q \vdash$<br>$\iota_{\mathbb{R}}(pq) = \ominus \iota_{\mathbb{R}}((-p)q) = \ominus(\iota_{\mathbb{R}}(-p) \odot \iota_{\mathbb{R}}(q))$<br>$= \iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(q)$ | $Th. 18.3.1.4(1), 3, 7, Th. 19.3.1.4(1), Th. 19.4.2.3(v)(1)$             |
| 1 | $10 \ p < 0, q < 0 \vdash$<br>$\iota_{\mathbb{R}}(pq) = \iota_{\mathbb{R}}((-p)(-q))$<br>$= \iota_{\mathbb{R}}(-p) \odot \iota_{\mathbb{R}}(-q) = \iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(q)$             | $Th. 18.3.1.4(1), 3, 7, Th. 19.3.1.4(1), Th. 19.4.2.3(v)(1)$             |
| 1 | $11 \ \iota_{\mathbb{R}}(pq) = \iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(q)$                                                                                                                                | $\vee E(4, 5, 7, 8, 9, 10)$                                              |
| 1 | $12 \ \iota_{\mathbb{R}}(-p) = \ominus \iota_{\mathbb{R}}(p) \wedge \iota_{\mathbb{R}}(pq) = \iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(q) \wedge I(3, 11)$                                                  |                                                                          |

## Erhaltung des Kehrwertes

Th. 19.4.3.1(iii)

$$p \in \mathbb{Q}, p \neq 0 \vdash \iota_{\mathbb{R}}(p^{-1}) = \text{inv}_{\mathbb{R}}(\iota_{\mathbb{R}}(p))$$

|     |   |                                                                                               |                                     |
|-----|---|-----------------------------------------------------------------------------------------------|-------------------------------------|
| 1   | 1 | $p \in \mathbb{Q}$                                                                            | $A$                                 |
| 2   | 2 | $p \neq 0$                                                                                    | $A$                                 |
| 1,2 | 3 | $\iota_{\mathbb{R}}(p) \neq 0_{\mathbb{R}}$                                                   | Th. 19.3.1.5(1,2,Def. 19.4.1.2)     |
| 1,2 | 4 | $\iota_{\mathbb{R}}(p) \odot \text{inv}_{\mathbb{R}}(\iota_{\mathbb{R}}(p)) = 1_{\mathbb{R}}$ | Th. 19.4.2.3(3)                     |
| 1,2 | 5 | $\iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(p^{-1}) = 1_{\mathbb{R}}$                     | Th. 19.4.3.1(ii)(Th. 18.3.5.9(1,2)) |
| 1,2 | 6 | $\iota_{\mathbb{R}}(p^{-1}) = \text{inv}_{\mathbb{R}}(\iota_{\mathbb{R}}(p))$                 | Th. 19.4.2.3(vii)(3,4,5)            |

## Zusammenfassung:

|     |   |                                                                                                                                                                                                                                                                                                                                                         |                              |
|-----|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 1   | 1 | $p, q \in \mathbb{Q}$                                                                                                                                                                                                                                                                                                                                   | $A$                          |
| 1   | 2 | $\iota_{\mathbb{R}}(p+q) = \iota_{\mathbb{R}}(p) \oplus \iota_{\mathbb{R}}(q)$                                                                                                                                                                                                                                                                          | Th. 19.4.3.1(i)(1)           |
| 1   | 3 | $\iota_{\mathbb{R}}(-p) = \ominus \iota_{\mathbb{R}}(p) \wedge$<br>$\iota_{\mathbb{R}}(pq) = \iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(q)$                                                                                                                                                                                                         | Th. 19.4.3.1(ii)(1)          |
| 4   | 4 | $p \neq 0$                                                                                                                                                                                                                                                                                                                                              | $A$                          |
| 1,4 | 5 | $\iota_{\mathbb{R}}(p^{-1}) = \text{inv}_{\mathbb{R}}(\iota_{\mathbb{R}}(p))$                                                                                                                                                                                                                                                                           |                              |
|     |   |                                                                                                                                                                                                                                                                                                                                                         | Th. 19.4.3.1(iii)(1,4)       |
| 1   | 6 | $p \neq 0 \rightarrow \iota_{\mathbb{R}}(p^{-1}) = \text{inv}_{\mathbb{R}}(\iota_{\mathbb{R}}(p))$                                                                                                                                                                                                                                                      | $\rightarrow I(4,5)$         |
| 1   | 7 | $\iota_{\mathbb{R}}(p+q) = \iota_{\mathbb{R}}(p) \oplus \iota_{\mathbb{R}}(q) \wedge$<br>$\iota_{\mathbb{R}}(pq) = \iota_{\mathbb{R}}(p) \odot \iota_{\mathbb{R}}(q) \wedge$<br>$\iota_{\mathbb{R}}(-p) = \ominus \iota_{\mathbb{R}}(p) \wedge$<br>$(p \neq 0 \rightarrow \iota_{\mathbb{R}}(p^{-1}) = \text{inv}_{\mathbb{R}}(\iota_{\mathbb{R}}(p)))$ | $\wedge I(2, \wedge I(3,6))$ |

□

Insbesondere dürfen die rationalen Zahlen von nun an über  $\iota_{\mathbb{R}}$  mit ihren Hauptschnitten identifiziert werden. Diese Identifikation ändert weder Gleichungen noch Ungleichungen.

## Kapitel 5

# Betrag, Abstand und Intervalle

### 5.1 Betrag und Abstand

**Definition 19.5.1.1 (Absolutbetrag einer reellen Zahl).**

*Mengen:*  $A$   
*Relationsoperatoren:*  $\leq_{\mathbb{R}}, <_{\mathbb{R}}$   
*Funktionsoperatoren:*  $\ominus \cdot$   
*Neues Symbol:*  
*Reeller Betrag:*  $|\cdot|_{\mathbb{R}}$

$$A \in \mathbb{R} \vdash |A|_{\mathbb{R}} := \begin{cases} A, & 0_{\mathbb{R}} \leq_{\mathbb{R}} A, \\ \ominus A, & A <_{\mathbb{R}} 0_{\mathbb{R}}. \end{cases}$$

**Definition 19.5.1.2 (Abstand zweier reeller Zahlen).**

*Mengen:*  $A, B$   
*Funktionsoperatoren:*  $\oplus, \ominus \cdot, |\cdot|_{\mathbb{R}}, d_{\mathbb{R}}(\cdot, \cdot)$   
*Neues Symbol:*  
*Reeller Abstand:*  $d_{\mathbb{R}}(\cdot, \cdot)$

$$A, B \in \mathbb{R} \vdash d_{\mathbb{R}}(A, B) := |A \oplus \ominus B|_{\mathbb{R}}$$

**Theorem 19.5.1.1 (Grundgesetze des reellen Betrags).**

*Mengen:*  $A, B$   
*Relationsoperatoren:*  $\leq_{\mathbb{R}}$   
*Funktionsoperatoren:*  $\oplus, \odot, \ominus \cdot, |\cdot|_{\mathbb{R}}$

$$\begin{aligned} A, B \in \mathbb{R} \vdash 0_{\mathbb{R}} \leq_{\mathbb{R}} |A|_{\mathbb{R}}, \quad (|A|_{\mathbb{R}} = 0_{\mathbb{R}} \leftrightarrow A = 0_{\mathbb{R}}), \\ | \ominus A |_{\mathbb{R}} = |A|_{\mathbb{R}}, \quad |A \odot B|_{\mathbb{R}} = |A|_{\mathbb{R}} \odot |B|_{\mathbb{R}}, \\ |A \oplus B|_{\mathbb{R}} \leq_{\mathbb{R}} |A|_{\mathbb{R}} \oplus |B|_{\mathbb{R}}. \end{aligned}$$

*Beweis.*

### Positivität und Nullkriterium

Th. 19.5.1.1(i)

$$A \in \mathbb{R} \vdash 0_{\mathbb{R}} \leq_{\mathbb{R}} |A|_{\mathbb{R}} \wedge (|A|_{\mathbb{R}} = 0_{\mathbb{R}} \leftrightarrow A = 0_{\mathbb{R}})$$

|      |    |                                                                                                                                                   |                                                                  |
|------|----|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1    | 1  | $A \in \mathbb{R}$                                                                                                                                | $A$                                                              |
| 1    | 2  | $0_{\mathbb{R}} \leq_{\mathbb{R}} A \vee A <_{\mathbb{R}} 0_{\mathbb{R}}$                                                                         | Th. 19.3.1.3(1)                                                  |
| 3    | 3  | $0_{\mathbb{R}} \leq_{\mathbb{R}} A$                                                                                                              | $A$                                                              |
| 1,3  | 4  | $ A _{\mathbb{R}} = A$                                                                                                                            | Def. 19.5.1.1(1,3)                                               |
| 5    | 5  | $A <_{\mathbb{R}} 0_{\mathbb{R}}$                                                                                                                 | $A$                                                              |
| 1,5  | 6  | $0_{\mathbb{R}} <_{\mathbb{R}} \ominus A$                                                                                                         | Th. 19.4.1.4(1,5)                                                |
| 1,5  | 7  | $ A _{\mathbb{R}} = \ominus A$                                                                                                                    | Def. 19.5.1.1(1,5)                                               |
| 1    | 8  | $0_{\mathbb{R}} \leq_{\mathbb{R}}  A _{\mathbb{R}}$                                                                                               | $\vee E(2, 3, 4, 5, 6, 7)$                                       |
| 9    | 9  | $ A _{\mathbb{R}} = 0_{\mathbb{R}}$                                                                                                               | $A$                                                              |
| 1,9  | 10 | $A = 0_{\mathbb{R}}$                                                                                                                              | Def. 19.5.1.1(1,2,9, Th. 19.4.1.4(1))                            |
| 11   | 11 | $A = 0_{\mathbb{R}}$                                                                                                                              | $A$                                                              |
| 1,11 | 12 | $ A _{\mathbb{R}} = 0_{\mathbb{R}}$                                                                                                               | Def. 19.5.1.1(1,11, Th. 19.4.1.4)                                |
| 1    | 13 | $ A _{\mathbb{R}} = 0_{\mathbb{R}} \leftrightarrow A = 0_{\mathbb{R}}$                                                                            | $\leftrightarrow I(\rightarrow I(9, 10), \rightarrow I(11, 12))$ |
| 1    | 14 | $0_{\mathbb{R}} \leq_{\mathbb{R}}  A _{\mathbb{R}} \wedge ( A _{\mathbb{R}} = 0_{\mathbb{R}} \leftrightarrow A = 0_{\mathbb{R}}) \wedge I(8, 13)$ |                                                                  |

### Negation und Produkt

Th. 19.5.1.1(ii)

$$A, B \in \mathbb{R} \vdash |\ominus A|_{\mathbb{R}} = |A|_{\mathbb{R}} \wedge |A \odot B|_{\mathbb{R}} = |A|_{\mathbb{R}} \odot |B|_{\mathbb{R}}$$

|   |   |                                                                                                                                        |                                         |
|---|---|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 1 | 1 | $A, B \in \mathbb{R}$                                                                                                                  | $A$                                     |
| 1 | 2 | $0_{\mathbb{R}} \leq_{\mathbb{R}} A \vee A <_{\mathbb{R}} 0_{\mathbb{R}}$                                                              | Th. 19.3.1.3(1)                         |
| 1 | 3 | $ \ominus A _{\mathbb{R}} =  A _{\mathbb{R}}$                                                                                          |                                         |
|   |   |                                                                                                                                        | Def. 19.5.1.1(1, 2, Th. 19.4.1.4(1))    |
| 1 | 4 | die vier Vorzeichenkombinationen von $A, B$ sind vollständig                                                                           |                                         |
|   |   |                                                                                                                                        | Th. 19.3.1.3(1)                         |
| 1 | 5 | $ A \odot B _{\mathbb{R}} =  A _{\mathbb{R}} \odot  B _{\mathbb{R}}$                                                                   |                                         |
|   |   |                                                                                                                                        | Def. 19.5.1.1(1, 4, Th. 19.4.2.3(v)(1)) |
| 1 | 6 | $ \ominus A _{\mathbb{R}} =  A _{\mathbb{R}} \wedge  A \odot B _{\mathbb{R}} =  A _{\mathbb{R}} \odot  B _{\mathbb{R}} \wedge I(3, 5)$ |                                         |

### Dreiecksungleichung

Th. 19.5.1.1(iii)

$$A, B \in \mathbb{R} \vdash |A \oplus B|_{\mathbb{R}} \leq_{\mathbb{R}} |A|_{\mathbb{R}} \oplus |B|_{\mathbb{R}}$$

|   |   |                                                                                    |                                    |
|---|---|------------------------------------------------------------------------------------|------------------------------------|
| 1 | 1 | $A, B \in \mathbb{R}$                                                              | $A$                                |
| 1 | 2 | $A \leq_{\mathbb{R}}  A _{\mathbb{R}} \wedge B \leq_{\mathbb{R}}  B _{\mathbb{R}}$ | Def. 19.5.1.1(1, Th. 19.3.1.3(1))  |
| 1 | 3 | $A \oplus B \leq_{\mathbb{R}}  A _{\mathbb{R}} \oplus  B _{\mathbb{R}}$            | Th. 19.4.2.4(i)(1,2)               |
| 1 | 4 | $\ominus(A \oplus B) \leq_{\mathbb{R}}  A _{\mathbb{R}} \oplus  B _{\mathbb{R}}$   | Th. 19.4.1.4(1,2, Th. 19.4.1.3(1)) |

$$1 \ 5 \ |A \oplus B|_{\mathbb{R}} \leq_{\mathbb{R}} |A|_{\mathbb{R}} \oplus |B|_{\mathbb{R}} \text{ Def. 19.5.1.1(1,3,4,Th. 19.3.1.3(1))}$$

Zusammenfassung:

$$\begin{array}{ll}
1 \ 1 \ A, B \in \mathbb{R} & A \\
1 \ 2 \ 0_{\mathbb{R}} \leq_{\mathbb{R}} |A|_{\mathbb{R}} \wedge (|A|_{\mathbb{R}} = 0_{\mathbb{R}} \leftrightarrow A = 0_{\mathbb{R}}) & Th. 19.5.1.1(i)(1) \\
1 \ 3 \ |\ominus A|_{\mathbb{R}} = |A|_{\mathbb{R}} \wedge |A \odot B|_{\mathbb{R}} = |A|_{\mathbb{R}} \odot |B|_{\mathbb{R}} & Th. 19.5.1.1(ii)(1) \\
1 \ 4 \ |A \oplus B|_{\mathbb{R}} \leq_{\mathbb{R}} |A|_{\mathbb{R}} \oplus |B|_{\mathbb{R}} & Th. 19.5.1.1(iii)(1) \\
1 \ 5 \ 0_{\mathbb{R}} \leq_{\mathbb{R}} |A|_{\mathbb{R}} \wedge & \wedge I(2, \wedge I(3, 4)) \\
\quad (|A|_{\mathbb{R}} = 0_{\mathbb{R}} \leftrightarrow A = 0_{\mathbb{R}}) \wedge & \\
\quad |\ominus A|_{\mathbb{R}} = |A|_{\mathbb{R}} \wedge & \\
\quad |A \odot B|_{\mathbb{R}} = |A|_{\mathbb{R}} \odot |B|_{\mathbb{R}} \wedge & \\
\quad |A \oplus B|_{\mathbb{R}} \leq_{\mathbb{R}} |A|_{\mathbb{R}} \oplus |B|_{\mathbb{R}} &
\end{array}$$

□

**Theorem 19.5.1.2 (Grundgesetze des reellen Abstands).**

**Mengen:**  $A, B, C$   
**Relationsoperatoren:**  $\leq_{\mathbb{R}}$   
**Funktionsoperatoren:**  $\oplus, d_{\mathbb{R}}(\cdot, \cdot)$

$$\begin{aligned}
A, B, C \in \mathbb{R} \vdash 0_{\mathbb{R}} \leq_{\mathbb{R}} d_{\mathbb{R}}(A, B), \quad (d_{\mathbb{R}}(A, B) = 0_{\mathbb{R}} \leftrightarrow A = B), \\
d_{\mathbb{R}}(A, B) = d_{\mathbb{R}}(B, A), \quad d_{\mathbb{R}}(A, C) \leq_{\mathbb{R}} d_{\mathbb{R}}(A, B) \oplus d_{\mathbb{R}}(B, C).
\end{aligned}$$

*Beweis.*

**Positivität, Definitheit und Symmetrie** Th. 19.5.1.2(i)

$$\begin{aligned}
A, B \in \mathbb{R} \vdash 0_{\mathbb{R}} \leq_{\mathbb{R}} d_{\mathbb{R}}(A, B) \wedge \\
(d_{\mathbb{R}}(A, B) = 0_{\mathbb{R}} \leftrightarrow A = B) \wedge d_{\mathbb{R}}(A, B) = d_{\mathbb{R}}(B, A)
\end{aligned}$$

$$\begin{array}{ll}
1 \ 1 \ A, B \in \mathbb{R} & A \\
1 \ 2 \ d_{\mathbb{R}}(A, B) = |A \oplus \ominus B|_{\mathbb{R}} & Def. 19.5.1.2(1) \\
1 \ 3 \ 0_{\mathbb{R}} \leq_{\mathbb{R}} d_{\mathbb{R}}(A, B) & Th. 19.5.1.1(i)(1, 2) \\
1 \ 4 \ d_{\mathbb{R}}(A, B) = 0_{\mathbb{R}} \leftrightarrow A \oplus \ominus B = 0_{\mathbb{R}} & Th. 19.5.1.1(i)(1, 2) \\
1 \ 5 \ A \oplus \ominus B = 0_{\mathbb{R}} \leftrightarrow A = B & Th. 19.4.1.4(i)(1) \\
1 \ 6 \ d_{\mathbb{R}}(A, B) = 0_{\mathbb{R}} \leftrightarrow A = B & Tr.(4, 5) \\
1 \ 7 \ B \oplus \ominus A = \ominus(A \oplus \ominus B) & Th. 19.4.1.3(1, Th. 19.4.1.4(1)) \\
1 \ 8 \ d_{\mathbb{R}}(A, B) = d_{\mathbb{R}}(B, A) & Th. 19.5.1.1(ii)(Def. 19.5.1.2(1), 7)
\end{array}$$

$$\begin{array}{l}
1 \quad 9 \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} d_{\mathbb{R}}(A, B) \wedge \quad \wedge I(3, \wedge I(6, 8)) \\
\quad (d_{\mathbb{R}}(A, B) = 0_{\mathbb{R}} \leftrightarrow A = B) \wedge \\
\quad d_{\mathbb{R}}(A, B) = d_{\mathbb{R}}(B, A)
\end{array}$$

### Dreiecksungleichung des Abstands

Th. 19.5.1.2(ii)

$$A, B, C \in \mathbb{R} \vdash d_{\mathbb{R}}(A, C) \leq_{\mathbb{R}} d_{\mathbb{R}}(A, B) \oplus d_{\mathbb{R}}(B, C)$$

$$\begin{array}{ll}
1 \quad 1 \quad A, B, C \in \mathbb{R} & A \\
1 \quad 2 \quad A \oplus \ominus C = (A \oplus \ominus B) \oplus (B \oplus \ominus C) & \text{Th. 19.4.1.3(1)} \\
1 \quad 3 \quad |A \oplus \ominus C|_{\mathbb{R}} \leq_{\mathbb{R}} |A \oplus \ominus B|_{\mathbb{R}} \oplus |B \oplus \ominus C|_{\mathbb{R}} & \text{Th. 19.5.1.1(iii)(1,2)} \\
1 \quad 4 \quad d_{\mathbb{R}}(A, C) \leq_{\mathbb{R}} d_{\mathbb{R}}(A, B) \oplus d_{\mathbb{R}}(B, C) & \text{Def. 19.5.1.2(3)}
\end{array}$$

Zusammenfassung:

$$\begin{array}{ll}
1 \quad 1 \quad A, B, C \in \mathbb{R} & A \\
1 \quad 2 \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} d_{\mathbb{R}}(A, B) \wedge & \text{Th. 19.5.1.2(i)(1)} \\
\quad (d_{\mathbb{R}}(A, B) = 0_{\mathbb{R}} \leftrightarrow A = B) \wedge & \\
\quad d_{\mathbb{R}}(A, B) = d_{\mathbb{R}}(B, A) & \\
1 \quad 3 \quad d_{\mathbb{R}}(A, C) \leq_{\mathbb{R}} d_{\mathbb{R}}(A, B) \oplus d_{\mathbb{R}}(B, C) & \text{Th. 19.5.1.2(ii)(1)} \\
1 \quad 4 \quad 0_{\mathbb{R}} \leq_{\mathbb{R}} d_{\mathbb{R}}(A, B) \wedge & \wedge I(2, 3) \\
\quad (d_{\mathbb{R}}(A, B) = 0_{\mathbb{R}} \leftrightarrow A = B) \wedge & \\
\quad d_{\mathbb{R}}(A, B) = d_{\mathbb{R}}(B, A) \wedge & \\
\quad d_{\mathbb{R}}(A, C) \leq_{\mathbb{R}} d_{\mathbb{R}}(A, B) \oplus d_{\mathbb{R}}(B, C) &
\end{array}$$

□

## 5.2 Intervalle und Umgebungen

**Definition 19.5.2.1 (Beschränkte reelle Intervalle).**

**Mengen:**  $A, B, x$

**Relationsoperatoren:**  $\leq_{\mathbb{R}}, <_{\mathbb{R}}$

**Neue Symbole:**

**Beschränkte Intervalle:**  $[A, B]_{\mathbb{R}}, (A, B)_{\mathbb{R}}, [A, B)_{\mathbb{R}}, (A, B]_{\mathbb{R}}$

$$\begin{aligned}
A, B \in \mathbb{R} \vdash \quad [A, B]_{\mathbb{R}} &:= \{x \in \mathbb{R} \mid A \leq_{\mathbb{R}} x \wedge x \leq_{\mathbb{R}} B\}, \\
(A, B)_{\mathbb{R}} &:= \{x \in \mathbb{R} \mid A <_{\mathbb{R}} x \wedge x <_{\mathbb{R}} B\}, \\
[A, B)_{\mathbb{R}} &:= \{x \in \mathbb{R} \mid A \leq_{\mathbb{R}} x \wedge x <_{\mathbb{R}} B\}, \\
(A, B]_{\mathbb{R}} &:= \{x \in \mathbb{R} \mid A <_{\mathbb{R}} x \wedge x \leq_{\mathbb{R}} B\}.
\end{aligned}$$

**Definition 19.5.2.2 (Unbeschränkte reelle Intervalle).**

**Mengen:**  $A, B, x$   
**Relationsoperatoren:**  $\leq_{\mathbb{R}}, <_{\mathbb{R}}$   
**Neue Symbole:**  
**Unbeschränkte Intervalle:**  $(-\infty, B)_{\mathbb{R}}, (-\infty, B]_{\mathbb{R}}$   
**Unbeschränkte Intervalle:**  $(A, \infty)_{\mathbb{R}}, [A, \infty)_{\mathbb{R}}$

$$\begin{aligned}
 A, B \in \mathbb{R} \vdash \quad & (-\infty, B)_{\mathbb{R}} := \{x \in \mathbb{R} \mid x <_{\mathbb{R}} B\}, \\
 & (-\infty, B]_{\mathbb{R}} := \{x \in \mathbb{R} \mid x \leq_{\mathbb{R}} B\}, \\
 & (A, \infty)_{\mathbb{R}} := \{x \in \mathbb{R} \mid A <_{\mathbb{R}} x\}, \\
 & [A, \infty)_{\mathbb{R}} := \{x \in \mathbb{R} \mid A \leq_{\mathbb{R}} x\}.
 \end{aligned}$$

**Definition 19.5.2.3 (Offene Epsilon-Umgebung).**

**Mengen:**  $x, y, \varepsilon$   
**Relationsoperatoren:**  $<_{\mathbb{R}}$   
**Funktionsoperatoren:**  $d_{\mathbb{R}}(\cdot, \cdot)$   
**Neues Symbol:**  
**Offene Umgebung:**  $B_{\varepsilon}(x)$

$$\begin{aligned}
 x, \varepsilon \in \mathbb{R}, \quad & 0_{\mathbb{R}} <_{\mathbb{R}} \varepsilon \\
 \vdash \quad & B_{\varepsilon}(x) := \{y \in \mathbb{R} \mid d_{\mathbb{R}}(y, x) <_{\mathbb{R}} \varepsilon\}.
 \end{aligned}$$

**Theorem 19.5.2.1 (Intervallform der Epsilon-Umgebung).**

**Mengen:**  $x, \varepsilon$   
**Relationsoperatoren:**  $<_{\mathbb{R}}$   
**Funktionsoperatoren:**  $\oplus, \ominus \cdot, d_{\mathbb{R}}(\cdot, \cdot)$

$$x, \varepsilon \in \mathbb{R}, \quad 0_{\mathbb{R}} <_{\mathbb{R}} \varepsilon \vdash B_{\varepsilon}(x) = (x \oplus \ominus \varepsilon, x \oplus \varepsilon)_{\mathbb{R}}$$

$$\begin{array}{ll}
 1 \quad 1 \quad x, \varepsilon \in \mathbb{R} & A \\
 2 \quad 2 \quad 0_{\mathbb{R}} <_{\mathbb{R}} \varepsilon & A \\
 1,2 \quad 3 \quad y \in B_{\varepsilon}(x) \leftrightarrow d_{\mathbb{R}}(y, x) <_{\mathbb{R}} \varepsilon & \text{Def. 19.5.2.3(1, 2)} \\
 1,2 \quad 4 \quad d_{\mathbb{R}}(y, x) <_{\mathbb{R}} \varepsilon \leftrightarrow |y \oplus \ominus x|_{\mathbb{R}} <_{\mathbb{R}} \varepsilon & \text{Def. 19.5.1.2(1)} \\
 1,2 \quad 5 \quad |y \oplus \ominus x|_{\mathbb{R}} <_{\mathbb{R}} \varepsilon \leftrightarrow & \\
 \quad \ominus \varepsilon <_{\mathbb{R}} y \oplus \ominus x \wedge y \oplus \ominus x <_{\mathbb{R}} \varepsilon & \\
 & \text{Def. 19.5.1.1(1, 2, Th. 19.3.1.3(1))} \\
 1,2 \quad 6 \quad \ominus \varepsilon <_{\mathbb{R}} y \oplus \ominus x \wedge y \oplus \ominus x <_{\mathbb{R}} \varepsilon & \\
 \quad \leftrightarrow x \oplus \ominus \varepsilon <_{\mathbb{R}} y \wedge y <_{\mathbb{R}} x \oplus \varepsilon & \\
 & \text{Th. 19.4.2.4(i)(1, Th. 19.4.1.4(1))} \\
 1,2 \quad 7 \quad x \oplus \ominus \varepsilon <_{\mathbb{R}} y \wedge y <_{\mathbb{R}} x \oplus \varepsilon & \text{Def. 19.5.2.1(1)} \\
 \quad \leftrightarrow y \in (x \oplus \ominus \varepsilon, x \oplus \varepsilon)_{\mathbb{R}} & \\
 1,2 \quad 8 \quad y \in B_{\varepsilon}(x) \leftrightarrow & \text{Tr. (3, 4, 5, 6, 7)} \\
 \quad y \in (x \oplus \ominus \varepsilon, x \oplus \varepsilon)_{\mathbb{R}} & \\
 1,2 \quad 9 \quad B_{\varepsilon}(x) = (x \oplus \ominus \varepsilon, & \text{Ax. 3.4.1.1}(\forall I(8)) \\
 \quad x \oplus \varepsilon)_{\mathbb{R}} &
 \end{array}$$

## Kapitel 6

# Archimedische Grundbegriffe

Die kanonische reelle Einbettung einer natürlichen Zahl ist die Komposition

$$n \mapsto \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n))),$$

diejenige einer ganzen Zahl ist  $z \mapsto \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(z))$ . In den folgenden Formeln werden diese Kompositionen ausgeschrieben; damit bleiben alle Zwischentypen aus den Bänden 17 und 18 sichtbar.

**Definition 19.6.1 (Archimedische Einbettung).**

*Mengen:*  $K, x, n$   
*Funktionen:*  $\eta$   
*Relationsoperatoren:*  $\preccurlyeq$   
*Neues Symbol:*  
*Archimedisches Prädikat:*  $\text{Arch}(K, \preccurlyeq, \eta)$

$$\begin{aligned} \eta: \mathbb{N} &\rightarrow K, \text{TotOrd}(K, \preccurlyeq) \\ \vdash \text{Arch}(K, \preccurlyeq, \eta) &:= \forall x \in K \exists n \in \mathbb{N} (x \preccurlyeq \eta(n) \wedge x \neq \eta(n)). \end{aligned}$$

**Theorem 19.6.1 (Archimedische Eigenschaft der reellen Zahlen).**

*Mengen:*  $x, n$   
*Funktionen:*  $\iota_{\mathbb{Z}}, \iota_{\mathbb{Q}}, \iota_{\mathbb{R}}$   
*Relationsoperatoren:*  $<_{\mathbb{R}}$

$$\forall x \in \mathbb{R} \exists n \in \mathbb{N} (x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n))))$$

|   |                                          |                  |
|---|------------------------------------------|------------------|
| 1 | $x \in \mathbb{R}$                       | $A$              |
| 1 | $2 \text{Cut}_{\mathbb{Q}}(x)$           | Th. 19.2.1.1(1)  |
| 1 | $3 \exists q \in \mathbb{Q} \setminus x$ | Def. 19.2.1.1(2) |
| 1 | $4 q \in \mathbb{Q} \wedge q \notin x$   | $A$              |

|    |                                                                                                                                                                    |                                   |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1  | $5 \exists n \in \mathbb{N} q < \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n))$                                                                                         | Th. 18.6.1.3(4)                   |
| 1  | $6 n \in \mathbb{N} \wedge q < \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n))$                                                                                          | A                                 |
| 1  | $7 x \subseteq \widehat{\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n))}$                                                                                                | Def. 19.2.1.1(2,4,6)              |
| 1  | $8 q \in \widehat{\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n))} \setminus x$                                                                                          | Def. 19.2.2.1(4,6)                |
| 1  | $9 x \subsetneq \widehat{\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n))}$                                                                                               | Def. 3.5.1.2(7,8)                 |
| 1  | $10 x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))$                                                                                | Def. 19.3.1.2(Def. 19.2.2.2(6),9) |
| 1  | $11 \exists n \in \mathbb{N} x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))$                                                       | $\exists I(6, 10)$                |
| 1  | $12 \exists n \in \mathbb{N} x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))$                                                       | $\exists E(5, 6, 11)$             |
| 1  | $13 \exists n \in \mathbb{N} x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))$                                                       | $\exists E(3, 4, 12)$             |
| 14 | $\forall x \in \mathbb{R} \exists n \in \mathbb{N} x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n))) \forall I(\rightarrow I(1, 13))$ |                                   |

**Theorem 19.6.2 (Ganzzahliger Abschnitt einer reellen Zahl).**

**Mengen:**  $x, z$   
**Funktionen:**  $\iota_{\mathbb{Q}}, \iota_{\mathbb{R}}$   
**Relationsoperatoren:**  $\leq_{\mathbb{R}}, <_{\mathbb{R}}$

$$\forall x \in \mathbb{R} \exists z \in \mathbb{Z} (\iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(z)) \leq_{\mathbb{R}} x \wedge x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(z + 1)))$$

*Beweis.*

**Ganzzahlige obere und untere Vergleichswerte** Th. 19.6.2(i)

$$x \in \mathbb{R} \vdash \exists w, u \in \mathbb{Z} (\iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(w)) \leq_{\mathbb{R}} x \wedge x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(u)))$$

|   |                                                                                                                                                                                                                 |                                                                                       |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1 | $1 x \in \mathbb{R}$                                                                                                                                                                                            | A                                                                                     |
| 1 | $2 \ominus x \in \mathbb{R}$                                                                                                                                                                                    | Th. 19.4.1.1(1)                                                                       |
| 1 | $3 \exists m, n \in \mathbb{N} (x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))) \wedge \ominus x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m))))$ | Th. 19.6.1(1, 2)                                                                      |
| 1 | $4 m, n \in \mathbb{N} \wedge x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n))) \wedge \ominus x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m))))$    | A                                                                                     |
| 1 | $5 \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(-\iota_{\mathbb{Z}}(m))) \leq_{\mathbb{R}} x$                                                                                                                          | Th. 19.4.1.4(Th. 19.4.3.1(4))                                                         |
| 1 | $6 \exists w, u \in \mathbb{Z} (\iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(w)) \leq_{\mathbb{R}} x \wedge x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(u)))$                                               | $\exists I(-\iota_{\mathbb{Z}}(m), \exists I(\iota_{\mathbb{Z}}(n), \wedge I(5, 4)))$ |
| 1 | $7 \exists w, u \in \mathbb{Z} (\iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(w)) \leq_{\mathbb{R}} x \wedge x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(u))) \exists E(3, 4, 6)$                            |                                                                                       |

Kleinster Überschreitungsindex:

|   |                                                                                                                                                                   |                    |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 1 | $1 x \in \mathbb{R}$                                                                                                                                              | A                  |
| 1 | $2 \exists w, u \in \mathbb{Z} (\iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(w)) \leq_{\mathbb{R}} x \wedge x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(u)))$ | Th. 19.6.2(i)(1)   |
| 1 | $3 w, u \in \mathbb{Z} \wedge \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(w)) \leq_{\mathbb{R}} x \wedge x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(u))$    | A                  |
| 1 | $4 M := \{n \in \mathbb{N} \mid x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(w + \iota_{\mathbb{Z}}(n)))\}$                                             | $= I$              |
| 1 | $5 M \neq \emptyset$                                                                                                                                              | Th. 17.3.2.3(3, 4) |

|                                                                                                                                                                                                                              |                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 1 6 $\exists m_0 \in M \forall n \in M (m_0 \leq n)$                                                                                                                                                                         | <i>Th.</i> 10.4.5.26(4, 5)        |
| 1 7 $m_0 \in M \wedge \forall n \in M (m_0 \leq n)$                                                                                                                                                                          | <i>A</i>                          |
| 1 8 $m_0 \neq 0$                                                                                                                                                                                                             | <i>Th.</i> 19.3.1.3(3, 4, 7)      |
| 1 9 $k := \text{pred}(m_0) \in \mathbb{N}$                                                                                                                                                                                   | <i>Th.</i> 10.4.2.17(7, 8)        |
| 1 10 $m_0 = \text{succ}(k)$                                                                                                                                                                                                  | <i>Th.</i> 10.4.2.20(7, 8, 9)     |
| 1 11 $\iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(w + \iota_{\mathbb{Z}}(k))) \leq_{\mathbb{R}} x$                                                                                                                                 | <i>Th.</i> 10.4.5.26(4, 7, 9, 10) |
| 1 12 $x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(w + \iota_{\mathbb{Z}}(k) + 1))$                                                                                                                                | <i>Th.</i> 17.3.2.1(7, 10)        |
| 1 13 $z := w + \iota_{\mathbb{Z}}(k) \in \mathbb{Z}$                                                                                                                                                                         | <i>Th.</i> 17.2.4.7(3, 9)         |
| 1 14 $\exists z \in \mathbb{Z} (\iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(z)) \leq_{\mathbb{R}} x \wedge x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(z + 1)))$                                                        | $\exists I(13, \wedge I(11, 12))$ |
| 1 15 $\exists z \in \mathbb{Z} (\iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(z)) \leq_{\mathbb{R}} x \wedge x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(z + 1)))$                                                        | $\exists E(6, 7, 14)$             |
| 1 16 $\exists z \in \mathbb{Z} (\iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(z)) \leq_{\mathbb{R}} x \wedge x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(z + 1)))$                                                        | $\exists E(2, 3, 15)$             |
| 17 $\forall x \in \mathbb{R} \exists z \in \mathbb{Z} (\iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(z)) \leq_{\mathbb{R}} x \wedge x <_{\mathbb{R}} \iota_{\mathbb{R}}(\iota_{\mathbb{Q}}(z + 1))) \forall I(\rightarrow I(1, 16))$ |                                   |

□

**Theorem 19.6.3 (Dichtheit der rationalen Zahlen in den reellen Zahlen).**

**Mengen:**  $x, y, q$   
**Funktionen:**  $\iota_{\mathbb{R}}$   
**Relationsoperatoren:**  $<_{\mathbb{R}}$

$$x, y \in \mathbb{R}, x <_{\mathbb{R}} y \vdash \exists q \in \mathbb{Q} (x <_{\mathbb{R}} \iota_{\mathbb{R}}(q) \wedge \iota_{\mathbb{R}}(q) <_{\mathbb{R}} y)$$

|                                                                                                                                              |                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1 1 $x, y \in \mathbb{R}$                                                                                                                    | <i>A</i>                            |
| 2 2 $x <_{\mathbb{R}} y$                                                                                                                     | <i>A</i>                            |
| 1,2 3 $x \subsetneq y$                                                                                                                       | Def. 19.3.1.2(1,2)                  |
| 1,2 4 $\exists b \in y \setminus x$                                                                                                          | Th. 3.5.2.4(3)                      |
| 1,2 5 $b \in y \wedge b \notin x$                                                                                                            | <i>A</i>                            |
| 1,2 6 $\exists q \in y (b < q)$                                                                                                              | Def. 19.2.1.1(Th. 19.2.1.1(1),5)    |
| 1,2 7 $q \in y \wedge b < q$                                                                                                                 | <i>A</i>                            |
| 1,2 8 $x \subseteq \widehat{q}$                                                                                                              | Def. 19.2.1.1(Th. 19.2.1.1(1),5,7)  |
| 1,2 9 $b \in \widehat{q} \setminus x$                                                                                                        | Def. 19.2.2.1(5,7)                  |
| 1,2 10 $x \subsetneq \widehat{q}$                                                                                                            | Def. 3.5.1.2(8,9)                   |
| 1,2 11 $\widehat{q} \subseteq y$                                                                                                             | Def. 19.2.1.1(Th. 19.2.1.1(1),7)    |
| 1,2 12 $q \in y \setminus \widehat{q}$                                                                                                       | Def. 19.2.2.1(7)                    |
| 1,2 13 $\widehat{q} \subsetneq y$                                                                                                            | Def. 3.5.1.2(11,12)                 |
| 1,2 14 $x <_{\mathbb{R}} \iota_{\mathbb{R}}(q) \wedge \iota_{\mathbb{R}}(q) <_{\mathbb{R}} y$                                                | Def. 19.2.2.2(Def. 19.3.1.2(10,13)) |
| 1,2 15 $\exists q \in \mathbb{Q} (x <_{\mathbb{R}} \iota_{\mathbb{R}}(q) \wedge \iota_{\mathbb{R}}(q) <_{\mathbb{R}} y) \exists I(7, 14)$    |                                     |
| 1,2 16 $\exists q \in \mathbb{Q} (x <_{\mathbb{R}} \iota_{\mathbb{R}}(q) \wedge \iota_{\mathbb{R}}(q) <_{\mathbb{R}} y) \exists E(6, 7, 15)$ |                                     |
| 1,2 17 $\exists q \in \mathbb{Q} (x <_{\mathbb{R}} \iota_{\mathbb{R}}(q) \wedge \iota_{\mathbb{R}}(q) <_{\mathbb{R}} y) \exists E(4, 5, 16)$ |                                     |

**Theorem 19.6.4 (Beliebig kleine positive rationale Hauptschnitte).**

**Mengen:**  $\varepsilon, n, q, m, M_m$

**Funktionen:**  $\iota_{\mathbb{Z}}, \iota_{\mathbb{Q}}, \iota_{\mathbb{R}}$

**Relationsoperatoren:**  $<_{\mathbb{R}}$

$$\varepsilon \in \mathbb{R}, 0_{\mathbb{R}} <_{\mathbb{R}} \varepsilon \vdash \exists n \in \mathbb{N} (0 < n \wedge \\ 0_{\mathbb{R}} <_{\mathbb{R}} \iota_{\mathbb{R}}((\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))^{-1}) \wedge \\ \iota_{\mathbb{R}}((\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))^{-1}) <_{\mathbb{R}} \varepsilon).$$

|     |    |                                                                                                                                                                                                                                                     |                                                                                                                                |
|-----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 1   | 1  | $\varepsilon \in \mathbb{R}$                                                                                                                                                                                                                        | $A$                                                                                                                            |
| 2   | 2  | $0_{\mathbb{R}} <_{\mathbb{R}} \varepsilon$                                                                                                                                                                                                         | $A$                                                                                                                            |
|     | 3  | $0_{\mathbb{R}} \in \mathbb{R}$                                                                                                                                                                                                                     | $Th. 19.4.1.1$                                                                                                                 |
| 1,2 | 4  | $\exists q \in \mathbb{Q} (0_{\mathbb{R}} <_{\mathbb{R}} \iota_{\mathbb{R}}(q) \wedge \iota_{\mathbb{R}}(q) <_{\mathbb{R}} \varepsilon)$                                                                                                            | $Th. 19.6.3(3, 1, 2)$                                                                                                          |
| 1,2 | 5  | $q \in \mathbb{Q} \wedge 0_{\mathbb{R}} <_{\mathbb{R}} \iota_{\mathbb{R}}(q) \wedge \iota_{\mathbb{R}}(q) <_{\mathbb{R}} \varepsilon$                                                                                                               | $A$                                                                                                                            |
| 1,2 | 6  | $q \in \mathbb{Q} \wedge 0 < q \wedge q \neq 0 \wedge q^{-1} \in \mathbb{Q} \wedge 0 < q^{-1}$                                                                                                                                                      | $Th. 19.3.1.4(Ax. 18.5.1, \wedge E1(5), Def. 19.4.1.2, \wedge E1(\wedge E2(5))),$<br>$Def. 18.4.1.4, Ax. 18.5.14, Ax. 18.5.22$ |
| 1,2 | 7  | $\exists m \in \mathbb{N} q^{-1} < \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m))$                                                                                                                                                                       | $Th. 18.6.1.3(6)$                                                                                                              |
| 1,2 | 8  | $m \in \mathbb{N} \wedge q^{-1} < \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m))$                                                                                                                                                                        | $A$                                                                                                                            |
| 1,2 | 9  | $M_m := \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m)) \in \mathbb{Q},$<br>$0 < m, \quad 0 < M_m, \quad q^{-1} < M_m$                                                                                                                                    | $Th. 17.2.3.1, Th. 18.2.3.1,$<br>$Th. 18.4.2.5, Th. 17.4.2.2,$<br>$Th. 17.2.3.2, Def. 17.2.3.4, Th. 18.4.1.4(6, 8)$            |
| 1,2 | 10 | $M_m^{-1} \in \mathbb{Q} \wedge 0 < M_m^{-1} \wedge M_m^{-1} < q$                                                                                                                                                                                   | $Def. 18.4.1.4, Ax. 18.5.14,$<br>$Ax. 18.5.22, Th. 18.6.1.1(9, 6),$<br>$Def. 18.5.1(\wedge E1(6), 6)$                          |
| 1,2 | 11 | $0_{\mathbb{R}} <_{\mathbb{R}} \iota_{\mathbb{R}}(M_m^{-1}) \wedge \iota_{\mathbb{R}}(M_m^{-1}) <_{\mathbb{R}} \iota_{\mathbb{R}}(q)$                                                                                                               | $Th. 19.3.1.4(Ax. 18.5.1, 10, Def. 19.4.1.2),$<br>$Th. 19.3.1.4(10, 6)$                                                        |
| 1,2 | 12 | $\iota_{\mathbb{R}}(M_m^{-1}) <_{\mathbb{R}} \varepsilon$                                                                                                                                                                                           | $Th. 19.3.1.3(11, \wedge E2(\wedge E2(5)))$                                                                                    |
| 1,2 | 13 | $\exists n \in \mathbb{N} (0 < n \wedge 0_{\mathbb{R}} <_{\mathbb{R}} \iota_{\mathbb{R}}((\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))^{-1}) \wedge \iota_{\mathbb{R}}((\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))^{-1}) <_{\mathbb{R}} \varepsilon)$ | $\exists I(m, 9, 11, 12)$                                                                                                      |
| 1,2 | 14 | $\exists n \in \mathbb{N} (0 < n \wedge 0_{\mathbb{R}} <_{\mathbb{R}} \iota_{\mathbb{R}}((\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))^{-1}) \wedge \iota_{\mathbb{R}}((\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))^{-1}) <_{\mathbb{R}} \varepsilon)$ | $\exists E(7, 8, 13)$                                                                                                          |
| 1,2 | 15 | $\exists n \in \mathbb{N} (0 < n \wedge 0_{\mathbb{R}} <_{\mathbb{R}} \iota_{\mathbb{R}}((\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))^{-1}) \wedge \iota_{\mathbb{R}}((\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n)))^{-1}) <_{\mathbb{R}} \varepsilon)$ | $\exists E(4, 5, 14)$                                                                                                          |

Die letzte Aussage ist die Form der archimedischen Eigenschaft, die in späteren Epsilon-Argumenten am häufigsten verwendet wird. Sie setzt nur die positive reelle Zahl  $\varepsilon$  voraus und liefert eine rationale Vergleichsgröße innerhalb derselben Zielmenge  $\mathbb{R}$ .

## Kapitel 7

# Axiomatische Zusammenfassung

### 7.1 Vollständig angeordnete Körper

Die folgende Definition fasst genau die Eigenschaften zusammen, welche die Schnittkonstruktion bereitstellt. Die Vollständigkeitszeile quantifiziert über Teilmengen des Trägers und ist daher in dieser Form zweitstufig zu lesen.

**Definition 19.7.1.1 (Vollständig angeordneter Körper).**

|                                                                                                   |                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mengen:</b>                                                                                    | $K, x, y, z$                                                                                                                                                                       |
| <b>Mengen:</b>                                                                                    | $S, s, u$                                                                                                                                                                          |
| <b>Relationsoperatoren:</b>                                                                       | $\preceq$                                                                                                                                                                          |
| <b>Funktionsoperatoren:</b>                                                                       | $\boxplus, \boxtimes$                                                                                                                                                              |
| <b>Funktionsoperatoren:</b>                                                                       | $\boxminus, \text{inv}$                                                                                                                                                            |
| <b>Typen</b>                                                                                      |                                                                                                                                                                                    |
| <b>Operationen:</b>                                                                               | $\boxplus, \boxtimes: K \times K \rightarrow K$                                                                                                                                    |
| <b>Negation:</b>                                                                                  | $\boxminus: K \rightarrow K$                                                                                                                                                       |
| <b>Kehrwert:</b>                                                                                  | $\text{inv}: K \setminus \{0_K\} \rightarrow K$                                                                                                                                    |
| <b>Konstanten:</b>                                                                                | $0_K, 1_K \in K, 0_K \neq 1_K$                                                                                                                                                     |
| <b>Additive abelsche Gruppe</b>                                                                   |                                                                                                                                                                                    |
| <b>Ass. Addition:</b>                                                                             | $x, y, z \in K \vdash (x \boxplus y) \boxplus z = x \boxplus (y \boxplus z)$                                                                                                       |
| <b>Komm. Addition:</b>                                                                            | $x, y \in K \vdash x \boxplus y = y \boxplus x$                                                                                                                                    |
| <b>Null und Inverses:</b>                                                                         | $x \in K \vdash x \boxplus 0_K = x, x \boxplus (\boxminus x) = 0_K$                                                                                                                |
| <b>Kommutatives multiplikatives Monoid auf <math>K</math>; Inversen auf <math>K^\times</math></b> |                                                                                                                                                                                    |
| <b>Ass. Produkt:</b>                                                                              | $x, y, z \in K \vdash (x \boxtimes y) \boxtimes z = x \boxtimes (y \boxtimes z)$                                                                                                   |
| <b>Komm. Produkt:</b>                                                                             | $x, y \in K \vdash x \boxtimes y = y \boxtimes x$                                                                                                                                  |
| <b>Eins:</b>                                                                                      | $x \in K \vdash x \boxtimes 1_K = x$                                                                                                                                               |
| <b>Produktinverses:</b>                                                                           | $x \in K \setminus \{0_K\} \vdash x \boxtimes \text{inv}(x) = 1_K$                                                                                                                 |
| <b>Feldkopplung</b>                                                                               |                                                                                                                                                                                    |
| <b>Distributivität:</b>                                                                           | $x, y, z \in K \vdash x \boxtimes (y \boxplus z) = (x \boxtimes y) \boxplus (x \boxtimes z)$                                                                                       |
| <b>Totale Ordnung</b>                                                                             |                                                                                                                                                                                    |
| <b>Totalität:</b>                                                                                 | $\text{TotOrd}(K, \preceq)$                                                                                                                                                        |
| <b>Verträglichkeit von Ordnung und Körper</b>                                                     |                                                                                                                                                                                    |
| <b>Positive Eins:</b>                                                                             | $0_K \preceq 1_K$                                                                                                                                                                  |
| <b>Translation:</b>                                                                               | $x, y, z \in K, x \preceq y \vdash x \boxplus z \preceq y \boxplus z$                                                                                                              |
| <b>Positive Produkte:</b>                                                                         | $x, y \in K, 0_K \preceq x, 0_K \preceq y \vdash 0_K \preceq x \boxtimes y$                                                                                                        |
| <b>Dedekind-Vollständigkeit</b>                                                                   |                                                                                                                                                                                    |
| <b>Vollständigkeit:</b>                                                                           | $\emptyset \neq S \subseteq K, \text{UB}_{K, \preceq}(S) \neq \emptyset \vdash$<br>$\exists s \in \text{UB}_{K, \preceq}(S) \forall u \in \text{UB}_{K, \preceq}(S) (s \preceq u)$ |
| <b>Struktur</b>                                                                                   |                                                                                                                                                                                    |
| <b>Prädikat:</b>                                                                                  | $\text{COF}$                                                                                                                                                                       |
|                                                                                                   | $\text{COF}\left(\begin{array}{c} K, \boxplus, \boxtimes, \boxminus, \\ \text{inv}, 0_K, 1_K, \preceq \end{array}\right)$                                                          |

**Theorem 19.7.1.1 (Das Schnittmodell ist ein vollständig angeordneter Körper).**

|                             |                                                                |
|-----------------------------|----------------------------------------------------------------|
| <b>Mengen:</b>              | $\mathbb{R}$                                                   |
| <b>Relationsoperatoren:</b> | $\leq_{\mathbb{R}}$                                            |
| <b>Funktionsoperatoren:</b> | $\oplus, \odot, \ominus \cdot, \text{inv}_{\mathbb{R}}(\cdot)$ |

$$\text{COF}(\mathbb{R}, \oplus, \odot, \ominus \cdot, \text{inv}_{\mathbb{R}}(\cdot), 0_{\mathbb{R}}, 1_{\mathbb{R}}, \leq_{\mathbb{R}})$$

*Beweis.*

Träger und Körpergesetze:

$$1 \ 0_{\mathbb{R}}, 1_{\mathbb{R}} \in \mathbb{R}$$

*Th.* 19.4.2.1

|                                                                                                    |                                                                                        |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 2 $0_{\mathbb{R}} \neq 1_{\mathbb{R}}$                                                             | <i>Th.</i> 19.3.1.5( <i>Def.</i> 19.4.1.2, <i>Def.</i> 19.4.2.3, <i>Def.</i> 18.4.1.4) |
| 3 $\oplus: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$                                    | <i>Th.</i> 19.4.1.1                                                                    |
| 4 $\odot: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$                                     | <i>Th.</i> 19.4.2.1                                                                    |
| 5 $\ominus \cdot: \mathbb{R} \rightarrow \mathbb{R}$                                               | <i>Th.</i> 19.4.1.1                                                                    |
| 6 $\text{inv}_{\mathbb{R}}(\cdot): \mathbb{R} \setminus \{0_{\mathbb{R}}\} \rightarrow \mathbb{R}$ | <i>Th.</i> 19.4.2.1                                                                    |
| 7 alle additiven Körperzeilen gelten                                                               | <i>Th.</i> 19.4.1.3                                                                    |
| 8 alle multiplikativen Körperzeilen gelten                                                         | <i>Th.</i> 19.4.2.3                                                                    |

Ordnung und Vollständigkeit:

|                                                                                                                                                       |                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $\text{TotOrd}(\mathbb{R}, \leq_{\mathbb{R}})$                                                                                                      | <i>Th.</i> 19.3.1.3                                                                                                                                                      |
| 2 $0_{\mathbb{R}} \leq_{\mathbb{R}} 1_{\mathbb{R}}$                                                                                                   | <i>Th.</i> 19.3.1.4,<br><i>Ax.</i> 18.5.1, <i>Ax.</i> 18.5.2,<br>$\wedge E1(\text{Def. } 18.4.1.4(\text{Ax. } 18.5.17))$ ,<br><i>Def.</i> 19.4.1.2, <i>Def.</i> 19.4.2.3 |
| 3 Translation und positive Produkte sind ordnungstreu                                                                                                 | <i>Th.</i> 19.4.2.4                                                                                                                                                      |
| 4 $4 \emptyset \neq S \subseteq \mathbb{R}$                                                                                                           | <i>A</i>                                                                                                                                                                 |
| 5 $5 \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(S) \neq \emptyset$                                                                                     | <i>A</i>                                                                                                                                                                 |
| 4,5 6 $\exists U \in \mathbb{R} \forall s \in S (s \leq_{\mathbb{R}} U)$                                                                              | <i>Th.</i> 13.2.1.1(4, 5)                                                                                                                                                |
| 4,5 7 $\exists s \in \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(S) \forall u \in \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(S) (s \leq_{\mathbb{R}} u)$ | <i>Th.</i> 19.3.3.2(4, 6)                                                                                                                                                |
| 8 $\text{COF}(\mathbb{R}, \oplus, \odot, \ominus \cdot, \text{inv}_{\mathbb{R}}(\cdot), 0_{\mathbb{R}}, 1_{\mathbb{R}}, \leq_{\mathbb{R}})$           | <i>Def.</i> 19.7.1.1(1, 2, 3, 4, 5, 6, 7, 8, 1, 2, 3, 4, 5, 6, 7)                                                                                                        |

□

**Theorem 19.7.1.2 (Kanonische natürliche Abbildung eines angeordneten Körpers).**

**Mengen:**  $K, n$   
**Funktionen:**  $\eta$   
**Funktionsoperatoren:**  $\boxplus$

$\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq)$   
 $\vdash \exists! \eta: \mathbb{N} \rightarrow K \left( \eta(0) = 0_K \wedge \forall n \in \mathbb{N} \left( \eta(\text{succ}(n)) = \eta(n) \boxplus 1_K \right) \right).$

|                                                                                            |                                             |
|--------------------------------------------------------------------------------------------|---------------------------------------------|
| 1 1 $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq)$         | <i>A</i>                                    |
| 1 2 $0_K \in K \wedge 1_K \in K$                                                           | <i>Def.</i> 19.7.1.1(1)                     |
| 1 3 $\boxplus: K \times K \rightarrow K$                                                   | <i>Def.</i> 19.7.1.1(1)                     |
| 1 4 $\forall x \in K (x \boxplus 1_K \in K)$                                               | <i>Def.</i> 19.7.1.1(1, 3, $\wedge E2(2)$ ) |
| 1 5 $\exists! s_K (s_K: K \rightarrow K \wedge \forall x \in K (s_K(x) = x \boxplus 1_K))$ | <i>Th.</i> 5.2.6.9(4)                       |

$$\begin{array}{ll}
1 \quad 6 \quad s_K: K \rightarrow K \wedge \forall x \in K (s_K(x) = x \boxplus 1_K) & A \\
1 \quad 7 \quad \exists! \eta: \mathbb{N} \rightarrow K (\eta(0) = 0_K \wedge \forall n \in \mathbb{N} (\eta(\text{succ}(n)) = s_K(\eta(n)))) & \text{Th. 10.4.3.20}(\wedge E1(2), \wedge E1(6)) \\
1 \quad 8 \quad \exists! \eta: \mathbb{N} \rightarrow K (\eta(0) = 0_K \wedge \forall n \in \mathbb{N} (\eta(\text{succ}(n)) = \eta(n) \boxplus 1_K)) & \\
& = E(\wedge E2(6), 7) \\
1 \quad 9 \quad \exists! \eta: \mathbb{N} \rightarrow K (\eta(0) = 0_K \wedge \forall n \in \mathbb{N} (\eta(\text{succ}(n)) = \eta(n) \boxplus 1_K)) & \\
& \exists! E(5, 6, 8)
\end{array}$$

Wir schreiben  $\eta_K$  für die durch die beiden Rekursionsgleichungen eindeutig bestimmte Funktion.

**Theorem 19.7.1.3 (Archimedizität vollständig angeordneter Körper).**

**Mengen:**  $K, n, s, u, x, y, z$   
**Funktionen:**  $\eta_K$   
**Relationsoperatoren:**  $\preccurlyeq$

$$\begin{array}{l}
\text{COF} \left( K, \boxplus, \boxtimes, \boxminus, \right. \\
\left. \text{inv}, 0_K, 1_K, \preccurlyeq \right), \\
\eta_K: \mathbb{N} \rightarrow K, \quad \eta_K(0) = 0_K, \\
\forall n \in \mathbb{N} (\eta_K(\text{succ}(n)) = \eta_K(n) \boxplus 1_K) \\
\vdash \text{Arch}(K, \preccurlyeq, \eta_K).
\end{array}$$

*Beweis.*

**Strikte Translation**

Th. 19.7.1.3(i)

$$\begin{array}{l}
\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preccurlyeq), \quad x, y, z \in K, \\
x \preccurlyeq y, \quad x \neq y \vdash x \boxplus z \preccurlyeq y \boxplus z \wedge x \boxplus z \neq y \boxplus z
\end{array}$$

$$\begin{array}{ll}
1 \quad 1 \quad \text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preccurlyeq) & A \\
2 \quad 2 \quad x, y, z \in K & A \\
3 \quad 3 \quad x \preccurlyeq y & A \\
4 \quad 4 \quad x \neq y & A \\
1,2,3 \quad 5 \quad x \boxplus z \preccurlyeq y \boxplus z & \text{Def. 19.7.1.1}(1,2,3) \\
1,2 \quad 6 \quad \boxminus z \in K & \text{Def. 19.7.1.1}(1,2) \\
7 \quad 7 \quad x \boxplus z = y \boxplus z & A \\
1,2,7 \quad 8 \quad (x \boxplus z) \boxplus (\boxminus z) = (y \boxplus z) \boxplus (\boxminus z) = E(7, = I) \\
1,2,7 \quad 9 \quad x = y & = E(8, \text{Def. 19.7.1.1}(1, 2, 6)) \\
1,2,4,7 \quad 10 \quad \perp & \perp I(4, 9) \\
1,2,4 \quad 11 \quad x \boxplus z \neq y \boxplus z & \neg I(7, 10) \\
1,2,3,4 \quad 12 \quad x \boxplus z \preccurlyeq y \boxplus z \wedge x \boxplus z \neq y \boxplus z & \wedge I(5, 11)
\end{array}$$

**Nichtoberschränken liefern strikte Bildzeugen**

Th. 19.7.1.3(ii)

$$\text{TotOrd}(K, \preccurlyeq), \eta_K: \mathbb{N} \rightarrow K, u \in K, u \notin \text{UB}_{K, \preccurlyeq}(\eta_K[\mathbb{N}]) \\ \vdash \exists n \in \mathbb{N} (u \preccurlyeq \eta_K(n) \wedge u \neq \eta_K(n))$$

|             |    |                                                                                                     |                                          |
|-------------|----|-----------------------------------------------------------------------------------------------------|------------------------------------------|
| 1           | 1  | $\text{TotOrd}(K, \preccurlyeq)$                                                                    | $A$                                      |
| 2           | 2  | $\eta_K: \mathbb{N} \rightarrow K$                                                                  | $A$                                      |
| 3           | 3  | $u \in K$                                                                                           | $A$                                      |
| 4           | 4  | $u \notin \text{UB}_{K, \preccurlyeq}(\eta_K[\mathbb{N}])$                                          | $A$                                      |
| 2           | 5  | $\eta_K[\mathbb{N}] \subseteq K$                                                                    | $Th. 5.2.4.18(2)$                        |
| 6           | 6  | $\forall x \in \eta_K[\mathbb{N}] (x \preccurlyeq u)$                                               | $A$                                      |
| 2,3,6       | 7  | $u \in \text{UB}_{K, \preccurlyeq}(\eta_K[\mathbb{N}])$                                             | $Th. 13.2.1.3(5, 3, 6)$                  |
| 2,3,4,6     | 8  | $\perp$                                                                                             | $\perp I(4, 7)$                          |
| 2,3,4       | 9  | $\neg \forall x \in \eta_K[\mathbb{N}] (x \preccurlyeq u)$                                          | $\neg I(6, 8)$                           |
| 2,3,4       | 10 | $\exists x \in \eta_K[\mathbb{N}] \neg (x \preccurlyeq u)$                                          | $Th. 2.10.4.4(9)$                        |
| 11          | 11 | $x \in \eta_K[\mathbb{N}] \wedge \neg (x \preccurlyeq u)$                                           | $A$                                      |
| 2,11        | 12 | $x \in K$                                                                                           |                                          |
|             |    |                                                                                                     | $Th. 3.5.2.6(5, \wedge E1(11))$          |
| 1,3,11      | 13 | $x \preccurlyeq u \vee u \preccurlyeq x$                                                            |                                          |
|             |    |                                                                                                     | $Def. 15.2.1.1(1, 3, 12)$                |
| 14          | 14 | $x \preccurlyeq u$                                                                                  | $A$                                      |
| 11,14       | 15 | $\perp$                                                                                             | $\perp I(\wedge E2(11), 14)$             |
| 11,14       | 16 | $u \preccurlyeq x$                                                                                  | $Th. 2.1.7.4(15)$                        |
| 17          | 17 | $u \preccurlyeq x$                                                                                  | $A$                                      |
| 1,3,11      | 18 | $u \preccurlyeq x$                                                                                  |                                          |
|             |    |                                                                                                     | $\vee E(13, 14, 16, 17, 17)$             |
| 19          | 19 | $u = x$                                                                                             | $A$                                      |
| 1,3         | 20 | $u \preccurlyeq u$                                                                                  | $Def. 15.2.1.1(1, 3)$                    |
| 1,3,19      | 21 | $x \preccurlyeq u$                                                                                  | $= E(19, 20)$                            |
| 1,3,11,19   | 22 | $\perp$                                                                                             | $\perp I(\wedge E2(11), 21)$             |
| 1,3,11      | 23 | $u \neq x$                                                                                          | $\neg I(19, 22)$                         |
| 2,11        | 24 | $\exists n \in \mathbb{N} x = \eta_K(n)$                                                            |                                          |
|             |    |                                                                                                     | $Th. 5.2.4.7(2, \wedge E1(11))$          |
| 25          | 25 | $n \in \mathbb{N} \wedge x = \eta_K(n)$                                                             | $A$                                      |
| 1,3,11,25   | 26 | $u \preccurlyeq \eta_K(n) \wedge u \neq \eta_K(n)$                                                  |                                          |
|             |    |                                                                                                     | $= E(\wedge E2(25), \wedge I(18, 23))$   |
| 1,2,3,11,25 | 27 | $\exists n \in \mathbb{N} (u \preccurlyeq \eta_K(n) \wedge u \neq \eta_K(n))$                       |                                          |
|             |    |                                                                                                     | $\exists I(\wedge I(\wedge E1(25), 26))$ |
| 1,2,3,11    | 28 | $\exists n \in \mathbb{N} (u \preccurlyeq \eta_K(n) \wedge u \neq \eta_K(n)) \exists E(24, 25, 27)$ |                                          |
| 1,2,3,4     | 29 | $\exists n \in \mathbb{N} (u \preccurlyeq \eta_K(n) \wedge u \neq \eta_K(n)) \exists E(10, 11, 28)$ |                                          |

**Das natürliche Bild ist unbeschränkt**

Th. 19.7.1.3(iii)

$$\begin{aligned} & \text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq), \eta_K: \mathbb{N} \rightarrow K \\ & \eta_K(0) = 0_K, \forall n \in \mathbb{N} (\eta_K(\text{succ}(n)) = \eta_K(n) \boxplus 1_K) \\ & \vdash \text{UB}_{K, \preceq}(\eta_K[\mathbb{N}]) = \emptyset \end{aligned}$$

$$\begin{aligned} & 1 \quad 1 \text{ COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq) & A \\ & 2 \quad 2 \eta_K: \mathbb{N} \rightarrow K & A \\ & 3 \quad 3 \eta_K(0) = 0_K & A \\ & 4 \quad 4 \forall n \in \mathbb{N} (\eta_K(\text{succ}(n)) = \eta_K(n) \boxplus 1_K) & A \\ & 5 \quad 5 \text{UB}_{K, \preceq}(\eta_K[\mathbb{N}]) \neq \emptyset & A \\ & 6 \quad 6 \mathbb{N} \in \mathcal{P}(\mathbb{N}) \setminus \{\emptyset\} & \\ & & \text{Th. 3.16.3.1(Ax. 10.3.1)} \\ & 2 \quad 7 \eta_K[\mathbb{N}] \in \mathcal{P}(K) \setminus \{\emptyset\} & \text{Th. 5.2.4.17(2, 6)} \\ & 2 \quad 8 \eta_K[\mathbb{N}] \subseteq K \wedge \eta_K[\mathbb{N}] \neq \emptyset & \text{Th. 3.16.3.1(7)} \\ & 1,2,5 \quad 9 \exists s \in \text{UB}_{K, \preceq}(\eta_K[\mathbb{N}]) \forall u \in \text{UB}_{K, \preceq}(\eta_K[\mathbb{N}]) (s \preceq u) & \\ & & \text{Def. 19.7.1.1(1, } \wedge I(\text{Th. 2.9.3.1}(\wedge E2(8)), \wedge E1(8)), 5) \\ & 10 \quad 10 s \in \text{UB}_{K, \preceq}(\eta_K[\mathbb{N}]) \wedge \forall u \in \text{UB}_{K, \preceq}(\eta_K[\mathbb{N}]) (s \preceq u) & A \\ & 1,2,10 \quad 11 s \in K \wedge \forall x \in \eta_K[\mathbb{N}] (x \preceq s) & \\ & & \text{Th. 13.2.1.2}(\wedge E1(8), \wedge E1(10)) \\ & 1 \quad 12 \boxminus 1_K \in K & \text{Def. 19.7.1.1(1)} \\ & 1,10 \quad 13 s \boxplus (\boxminus 1_K) \in K & \\ & & \text{Def. 19.7.1.1(1, } \wedge E1(11), 12) \\ & 1 \quad 14 0_K \preceq 1_K \wedge 0_K \neq 1_K & \text{Def. 19.7.1.1(1)} \\ & 1,10 \quad 15 s \boxplus (\boxminus 1_K) \preceq s \wedge & \\ & \quad s \boxplus (\boxminus 1_K) \neq s & \\ & = E(\text{Th. 19.7.1.3(i)}(1, \text{Def. 19.7.1.1(1), 13, } \wedge E1(14), \wedge E2(14)), \text{Def. 19.7.1.1(1, } \wedge E1(11), 12)) \\ & 16 \quad 16 s \boxplus (\boxminus 1_K) \in \text{UB}_{K, \preceq}(\eta_K[\mathbb{N}]) & A \\ & 10,16 \quad 17 s \preceq s \boxplus (\boxminus 1_K) & \\ & & \rightarrow E(\forall E(\wedge E2(10)), 16) \\ & 1,10,16 \quad 18 s = s \boxplus (\boxminus 1_K) & \\ & & \text{Def. 15.2.1.1(Def. 19.7.1.1(1), } \wedge E1(11), 13, 17, \wedge E1(15)) \\ & 1,10,16 \quad 19 \perp & \\ & & \perp I(\wedge E2(15), \text{Th. 2.9.1.1(18)}) \\ & 1,10 \quad 20 s \boxplus (\boxminus 1_K) \notin \text{UB}_{K, \preceq}(\eta_K[\mathbb{N}]) & \neg I(16, 19) \\ & 1,2,10 \quad 21 \exists n \in \mathbb{N} (s \boxplus (\boxminus 1_K) \preceq \eta_K(n) \wedge s \boxplus (\boxminus 1_K) \neq \eta_K(n)) & \\ & & \text{Th. 19.7.1.3(ii)}(\text{Def. 19.7.1.1(1), 2, 13, 20}) \\ & 22 \quad 22 n \in \mathbb{N} \wedge s \boxplus (\boxminus 1_K) \preceq \eta_K(n) \wedge s \boxplus (\boxminus 1_K) \neq \eta_K(n) & A \\ & 2,22 \quad 23 \eta_K(n) \in K & \\ & & \text{Th. 5.2.3.8(2, } \wedge E1(22)) \\ & 22 \quad 24 \text{succ}(n) \in \mathbb{N} & \\ & & \text{Ax. 10.3.4}(\wedge E1(22)) \\ & 2,22 \quad 25 \eta_K(\text{succ}(n)) \in \eta_K[\mathbb{N}] & \\ & & \text{Th. 5.2.4.5(Th. 3.5.3.1, 2, 24)} \end{aligned}$$

|              |    |                                                                                                                                                                                                                                                                                 |
|--------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1,2,22       | 26 | $(s \boxplus (\boxplus 1_K)) \boxplus 1_K \preceq \eta_K(n) \boxplus 1_K \wedge$<br>$(s \boxplus (\boxplus 1_K)) \boxplus 1_K \neq \eta_K(n) \boxplus 1_K$<br><i>Th.</i> 19.7.1.3(i)(1, 13, 23, <i>Def.</i> 19.7.1.1(1), $\wedge E1(\wedge E2(22)), \wedge E2(\wedge E2(22))$ ) |
| 1,2,4,10,22  | 27 | $s \preceq \eta_K(\text{succ}(n)) \wedge s \neq \eta_K(\text{succ}(n))$<br>$= E(26, \text{Def. } 19.7.1.1(1, \wedge E1(11), 12), \rightarrow E(\forall E(4), \wedge E1(22)))$                                                                                                   |
| 1,2,10,22    | 28 | $\eta_K(\text{succ}(n)) \preceq s$<br>$\rightarrow E(\forall E(\wedge E2(11)), 25)$                                                                                                                                                                                             |
| 1,2,10,22    | 29 | $s = \eta_K(\text{succ}(n))$<br><i>Def.</i> 15.2.1.1(1, $\wedge E1(11), 25, \wedge E1(27), 28)$                                                                                                                                                                                 |
| 1,2,4,10,22  | 30 | $\perp$<br>$\perp I(\wedge E2(27), 29)$                                                                                                                                                                                                                                         |
| 1,2,4,10     | 31 | $\perp$<br>$\exists E(21, 22, 30)$                                                                                                                                                                                                                                              |
| 1,2,3,4,5,10 | 32 | $\perp$<br>$\exists E(9, 10, 31)$                                                                                                                                                                                                                                               |
| 1,2,3,4      | 33 | $\text{UB}_{K, \preceq}(\eta_K[\mathbb{N}]) = \emptyset$<br>$\neg I(5, 32)$                                                                                                                                                                                                     |

Archimedisches Prädikat:

|           |    |                                                                                               |                                                            |
|-----------|----|-----------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 1         | 1  | $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preccurlyeq)$           | $A$                                                        |
| 2         | 2  | $\eta_K: \mathbb{N} \rightarrow K$                                                            | $A$                                                        |
| 3         | 3  | $\eta_K(0) = 0_K$                                                                             | $A$                                                        |
| 4         | 4  | $\forall n \in \mathbb{N} (\eta_K(\text{succ}(n)) = \eta_K(n) \boxplus 1_K)$                  | $A$                                                        |
| 1,2,3,4   | 5  | $\text{UB}_{K, \preccurlyeq}(\eta_K[\mathbb{N}]) = \emptyset$                                 |                                                            |
|           |    |                                                                                               | <i>Th.</i> 19.7.1.3(iii)(1, 2, 3, 4)                       |
| 6         | 6  | $x \in K$                                                                                     | $A$                                                        |
| 1,2,3,4,6 | 7  | $x \notin \text{UB}_{K, \preccurlyeq}(\eta_K[\mathbb{N}])$                                    | $= E(5, \text{Th. } 3.6.2.2)$                              |
| 1,2,3,4,6 | 8  | $\exists n \in \mathbb{N} (x \preccurlyeq \eta_K(n) \wedge x \neq \eta_K(n))$                 |                                                            |
|           |    |                                                                                               | <i>Th.</i> 19.7.1.3(ii)( <i>Def.</i> 19.7.1.1(1), 2, 6, 7) |
| 1,2,3,4   | 9  | $\forall x \in K \exists n \in \mathbb{N} (x \preccurlyeq \eta_K(n) \wedge x \neq \eta_K(n))$ | $\forall I(\rightarrow I(6, 8))$                           |
| 1,2,3,4   | 10 | $\text{Arch}(K, \preccurlyeq, \eta_K)$                                                        | <i>Def.</i> 19.6.1(9)                                      |

□

**Definition 19.7.1.2 (Gesetze der ganzzahligen Primkörpereinbettung).**

|                             |                                  |
|-----------------------------|----------------------------------|
| <b>Mengen:</b>              | $K, z, w$                        |
| <b>Funktionen:</b>          | $\theta_{\mathbb{Z}}$            |
| <b>Relationsoperatoren:</b> | $<, \preceq$                     |
| <b>Funktionsoperatoren:</b> | $\boxplus, \boxtimes, \boxminus$ |
| <b>Prädikat:</b>            | $\text{ZEmb}_K$                  |

$$\begin{aligned} \text{ZEmb}_K(\theta_{\mathbb{Z}}) &:= \theta_{\mathbb{Z}}(0) = 0_K \wedge \theta_{\mathbb{Z}}(1) = 1_K \wedge \\ &\forall z, w \in \mathbb{Z} (\theta_{\mathbb{Z}}(z + w) = \theta_{\mathbb{Z}}(z) \boxplus \theta_{\mathbb{Z}}(w) \wedge \\ &\quad \theta_{\mathbb{Z}}(zw) = \theta_{\mathbb{Z}}(z) \boxtimes \theta_{\mathbb{Z}}(w) \wedge \theta_{\mathbb{Z}}(-z) = \boxminus \theta_{\mathbb{Z}}(z) \wedge \\ &\quad (z < w \leftrightarrow (\theta_{\mathbb{Z}}(z) \preceq \theta_{\mathbb{Z}}(w) \wedge \theta_{\mathbb{Z}}(z) \neq \theta_{\mathbb{Z}}(w))))). \end{aligned}$$

**Definition 19.7.1.3 (Gesetze der rationalen Primkörpereinbettung).**

|                             |                                  |
|-----------------------------|----------------------------------|
| <b>Mengen:</b>              | $K, p, q$                        |
| <b>Funktionen:</b>          | $\theta_{\mathbb{Q}}$            |
| <b>Relationsoperatoren:</b> | $<, \preceq$                     |
| <b>Funktionsoperatoren:</b> | $\boxplus, \boxtimes, \boxminus$ |
| <b>Funktionsoperatoren:</b> | $\text{inv}$                     |
| <b>Prädikat:</b>            | $\text{QEmb}_K$                  |

$$\begin{aligned} \text{QEmb}_K(\theta_{\mathbb{Q}}) := & \theta_{\mathbb{Q}}(0) = 0_K \wedge \theta_{\mathbb{Q}}(1) = 1_K \wedge \\ & \forall p, q \in \mathbb{Q} \left( \theta_{\mathbb{Q}}(p + q) = \theta_{\mathbb{Q}}(p) \boxplus \theta_{\mathbb{Q}}(q) \wedge \right. \\ & \quad \theta_{\mathbb{Q}}(pq) = \theta_{\mathbb{Q}}(p) \boxtimes \theta_{\mathbb{Q}}(q) \wedge \\ & \quad \left. (p < q \leftrightarrow (\theta_{\mathbb{Q}}(p) \preceq \theta_{\mathbb{Q}}(q) \wedge \theta_{\mathbb{Q}}(p) \neq \theta_{\mathbb{Q}}(q))) \right) \wedge \\ & \forall p \in \mathbb{Q} \left( \theta_{\mathbb{Q}}(-p) = \boxminus \theta_{\mathbb{Q}}(p) \wedge \right. \\ & \quad \left. (p \neq 0 \rightarrow \theta_{\mathbb{Q}}(p^{-1}) = \text{inv}(\theta_{\mathbb{Q}}(p))) \right). \end{aligned}$$

**Theorem 19.7.1.4 (Eindeutigkeit des vollständig angeordneten reellen Körpers).**

|                             |                                      |
|-----------------------------|--------------------------------------|
| <b>Mengen:</b>              | $K, x, y$                            |
| <b>Funktionen:</b>          | $\Phi$                               |
| <b>Relationsoperatoren:</b> | $\leq_{\mathbb{R}}, \preceq$         |
| <b>Funktionsoperatoren:</b> | $\oplus, \odot, \boxplus, \boxtimes$ |

$$\begin{aligned} & \text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq) \\ & \vdash \exists! \Phi: \mathbb{R} \xrightarrow{\sim} K \left( \begin{array}{l} \forall x, y \in \mathbb{R} \left( \Phi(x \oplus y) = \Phi(x) \boxplus \Phi(y) \right) \wedge \\ \forall x, y \in \mathbb{R} \left( \Phi(x \odot y) = \Phi(x) \boxtimes \Phi(y) \right) \wedge \\ \forall x, y \in \mathbb{R} \left( x \leq_{\mathbb{R}} y \leftrightarrow \Phi(x) \preceq \Phi(y) \right) \wedge \\ \Phi(0_{\mathbb{R}}) = 0_K \wedge \Phi(1_{\mathbb{R}}) = 1_K \end{array} \right). \end{aligned}$$

*Beweis.*

**Arithmetik der natürlichen Einbettung**

Th. 19.7.1.4(i)

$$\begin{aligned} & \text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq), \\ & \eta_K: \mathbb{N} \rightarrow K, \quad \eta_K(0) = 0_K, \\ & \forall n \in \mathbb{N} \left( \eta_K(\text{succ}(n)) = \eta_K(n) \boxplus 1_K \right) \\ & \vdash \forall m, n \in \mathbb{N} \left( \eta_K(m + n) = \eta_K(m) \boxplus \eta_K(n) \wedge \right. \\ & \quad \eta_K(mn) = \eta_K(m) \boxtimes \eta_K(n) \wedge \\ & \quad \left. (m < n \leftrightarrow (\eta_K(m) \preceq \eta_K(n) \wedge \eta_K(m) \neq \eta_K(n))) \right) \end{aligned}$$

|              |    |                                                                                                                                                                                                                                                                                  |                                                     |
|--------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 1            | 1  | $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq)$                                                                                                                                                                                                   | $A$                                                 |
| 2            | 2  | $\eta_K: \mathbb{N} \rightarrow K$                                                                                                                                                                                                                                               | $A$                                                 |
| 3            | 3  | $\eta_K(0) = 0_K$                                                                                                                                                                                                                                                                | $A$                                                 |
| 4            | 4  | $\forall n \in \mathbb{N} \eta_K(\text{succ}(n)) = \eta_K(n) \boxplus 1_K$                                                                                                                                                                                                       | $A$                                                 |
| 1,2,3,4      | 5  | $\eta_K(1) = 1_K$                                                                                                                                                                                                                                                                | $= E(\text{Def. 10.4.4.2, 4, 3, Def. 19.7.1.1(1)})$ |
| 1,2,3,4      | 6  | $\forall m \in \mathbb{N} \eta_K(m + 0) = \eta_K(m) \boxplus \eta_K(0)$                                                                                                                                                                                                          | $\text{Th. 10.4.4.2, 3, Def. 19.7.1.1(1)}$          |
| 1,2,3,4      | 7  | $\eta_K(m + n) = \eta_K(m) \boxplus \eta_K(n) \vdash$<br>$\eta_K(m + \text{succ}(n)) = \eta_K(m) \boxplus \eta_K(\text{succ}(n))$                                                                                                                                                | $\text{Th. 10.4.4.3, 4, Def. 19.7.1.1(1)}$          |
| 1,2,3,4      | 8  | $\forall m, n \in \mathbb{N} \eta_K(m + n) = \eta_K(m) \boxplus \eta_K(n) \text{ Ax. 10.3.8(6, 7)}$                                                                                                                                                                              |                                                     |
| 1,2,3,4      | 9  | $\forall m \in \mathbb{N} \eta_K(m \cdot 0) = \eta_K(m) \boxtimes \eta_K(0)$                                                                                                                                                                                                     | $\text{Th. 10.4.7.4, 3, Def. 19.7.1.1(1)}$          |
| 1,2,3,4,8    | 10 | $\eta_K(mn) = \eta_K(m) \boxtimes \eta_K(n) \vdash$<br>$\eta_K(m \cdot \text{succ}(n)) = \eta_K(m) \boxtimes \eta_K(\text{succ}(n))$                                                                                                                                             | $\text{Th. 10.4.7.5, 4, 8, Def. 19.7.1.1(1)}$       |
| 1,2,3,4,8    | 11 | $\forall m, n \in \mathbb{N} \eta_K(mn) = \eta_K(m) \boxtimes \eta_K(n) \text{ Ax. 10.3.8(9, 10)}$                                                                                                                                                                               |                                                     |
| 1,2,4,5      | 12 | $\forall n \in \mathbb{N} (\eta_K(n) \preceq \eta_K(\text{succ}(n)) \wedge \eta_K(n) \neq \eta_K(\text{succ}(n)))$                                                                                                                                                               | $\text{Th. 19.7.1.3(i)(1), Def. 19.7.1.1(1, 5)}$    |
| 1,2,4,12     | 13 | $\forall m, n \in \mathbb{N} (m < n \rightarrow (\eta_K(m) \preceq \eta_K(n) \wedge \eta_K(m) \neq \eta_K(n)))$                                                                                                                                                                  | $\text{Th. 10.4.5.22, Ax. 10.3.8(12)}$              |
| 1,2,13       | 14 | $\forall m, n \in \mathbb{N} ((\eta_K(m) \preceq \eta_K(n) \wedge \eta_K(m) \neq \eta_K(n)) \rightarrow m < n)$                                                                                                                                                                  | $\text{Th. 10.4.5.23(13), Def. 15.2.1.1(1)}$        |
| 1,8,11,13,14 | 15 | $\forall m, n \in \mathbb{N} (\eta_K(m + n) = \eta_K(m) \boxplus \eta_K(n) \wedge$<br>$\eta_K(mn) = \eta_K(m) \boxtimes \eta_K(n) \wedge (m < n \leftrightarrow (\eta_K(m) \preceq \eta_K(n) \wedge \eta_K(m) \neq \eta_K(n))))$<br>$\wedge I(8, \wedge I(11, \perp I(13, 14)))$ |                                                     |

## Fortsetzung auf die ganzen Zahlen

Th. 19.7.1.4(ii)

$$\begin{aligned} & \text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq), \\ & \vdash \exists! \eta_{\mathbb{Z}}: \mathbb{Z} \rightarrow K \left( \eta_{\mathbb{Z}}(0) = 0_K \wedge \eta_{\mathbb{Z}}(1) = 1_K \wedge \right. \\ & \quad \forall z, w \in \mathbb{Z} (\eta_{\mathbb{Z}}(z + w) = \eta_{\mathbb{Z}}(z) \boxplus \eta_{\mathbb{Z}}(w) \wedge \\ & \quad \eta_{\mathbb{Z}}(zw) = \eta_{\mathbb{Z}}(z) \boxtimes \eta_{\mathbb{Z}}(w) \wedge \eta_{\mathbb{Z}}(-z) = \boxminus \eta_{\mathbb{Z}}(z) \wedge \\ & \quad \left. (z < w \leftrightarrow (\eta_{\mathbb{Z}}(z) \preceq \eta_{\mathbb{Z}}(w) \wedge \eta_{\mathbb{Z}}(z) \neq \eta_{\mathbb{Z}}(w)))) \right) \end{aligned}$$

|     |   |                                                                                                                                                                                                                                                                                                                             |                                                                |
|-----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| 1   | 1 | $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq) A$                                                                                                                                                                                                                                            |                                                                |
| 1   | 2 | $\eta_K: \mathbb{N} \rightarrow K \wedge \eta_K(0) = 0_K \wedge \eta_K(1) = 1_K \wedge$<br>$\forall m, n \in \mathbb{N} (\eta_K(m + n) = \eta_K(m) \boxplus \eta_K(n) \wedge$<br>$\eta_K(mn) = \eta_K(m) \boxtimes \eta_K(n) \wedge (m < n \leftrightarrow (\eta_K(m) \preceq \eta_K(n) \wedge \eta_K(m) \neq \eta_K(n))))$ | $\text{Th. 19.7.1.2(1), Th. 19.7.1.4(i)(1)}$                   |
| 1,2 | 3 | $\eta_{\mathbb{Z}}([a, b]_{\mathbb{Z}}) := \eta_K(a) \boxplus \boxminus \eta_K(b) = I$                                                                                                                                                                                                                                      |                                                                |
| 1,2 | 4 | $[a, b]_{\mathbb{Z}} = [c, d]_{\mathbb{Z}} \vdash$<br>$\eta_K(a) \boxplus \boxminus \eta_K(b) = \eta_K(c) \boxplus \boxminus \eta_K(d)$                                                                                                                                                                                     | $\text{Th. 17.2.2.2, Th. 19.7.1.4(i)(1, 2), Def. 19.7.1.1(1)}$ |

|         |       |                                                                                                                                                                        |                                                                                               |
|---------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 1,2     | 5     | $\eta_{\mathbb{Z}}: \mathbb{Z} \rightarrow K$                                                                                                                          | <i>Th.</i> 17.2.2.3(3, 4)                                                                     |
| 1,2     | 6     | $\forall z, w \in \mathbb{Z} \eta_{\mathbb{Z}}(z + w) = \eta_{\mathbb{Z}}(z) \boxplus \eta_{\mathbb{Z}}(w)$                                                            | <i>Def.</i> 17.2.4.2, 2, 3, <i>Th.</i> 17.2.2.3                                               |
| 1,2     | 7     | $\forall z, w \in \mathbb{Z} \eta_{\mathbb{Z}}(zw) = \eta_{\mathbb{Z}}(z) \boxtimes \eta_{\mathbb{Z}}(w)$                                                              | <i>Def.</i> 17.2.5.1, 2, 3, <i>Th.</i> 17.2.2.3, <i>Def.</i> 19.7.1.1(1)                      |
| 1,2     | 8     | $\eta_{\mathbb{Z}}(0) = 0_K \wedge \eta_{\mathbb{Z}}(1) = 1_K \wedge \forall z \in \mathbb{Z} \eta_{\mathbb{Z}}(-z) = \boxminus \eta_{\mathbb{Z}}(z)$                  | <i>Th.</i> 17.2.3.6, <i>Def.</i> 17.2.3.5, <i>Def.</i> 17.2.4.1, 2, 3                         |
| 1,2     | 9     | $\forall z, w \in \mathbb{Z} (z < w \leftrightarrow (\eta_{\mathbb{Z}}(z) \preccurlyeq \eta_{\mathbb{Z}}(w) \wedge \eta_{\mathbb{Z}}(z) \neq \eta_{\mathbb{Z}}(w)))$   | <i>Th.</i> 17.2.2.3, <i>Th.</i> 17.4.2.2, <i>Def.</i> 17.4.2.3, 2, 3                          |
| 1,5,9   | 10    | $\eta_{\mathbb{Z}}: \mathbb{Z} \rightarrowtail K$                                                                                                                      | <i>Th.</i> 17.4.2.3(9), <i>Def.</i> 6.2.1.1(5)                                                |
| 1,2,5,6 | 7,8,9 | 11 $\theta_{\mathbb{Z}}: \mathbb{Z} \rightarrow K, \mathbf{ZEmb}_K(\theta_{\mathbb{Z}}) \vdash \forall z \in \mathbb{Z} \theta_{\mathbb{Z}}(z) = \eta_{\mathbb{Z}}(z)$ | <i>Def.</i> 19.7.1.2, <i>Th.</i> 17.2.2.3,<br><i>Th.</i> 19.7.1.2(1), <i>Def.</i> 17.2.4.2(3) |
| 1       | 12    | $\exists! \eta_{\mathbb{Z}}: \mathbb{Z} \rightarrowtail K \mathbf{ZEmb}_K(\eta_{\mathbb{Z}})$                                                                          | <i>Def.</i> 19.7.1.2, $\exists! I(5, 6, 7, 8, 9, 10, 11)$                                     |

## Rationaler Primkörper in $K$

*Th.* 19.7.1.4(iii)

$$\begin{aligned} & \text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preccurlyeq) \\ & \vdash \exists! \eta_{\mathbb{Q}}: \mathbb{Q} \rightarrowtail K \mathbf{QEmb}_K(\eta_{\mathbb{Q}}) \end{aligned}$$

|           |   |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                        |
|-----------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1         | 1 | $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preccurlyeq) \vdash A$                                                                                                                                                                                                                                                                                        |                                                                                                                                                        |
| 1         | 2 | $\exists! \eta_{\mathbb{Z}}: \mathbb{Z} \rightarrowtail K \mathbf{ZEmb}_K(\eta_{\mathbb{Z}})$                                                                                                                                                                                                                                                                                       | <i>Th.</i> 19.7.1.4(ii)(1)                                                                                                                             |
| 2         | 3 | $\eta_{\mathbb{Z}}: \mathbb{Z} \rightarrowtail K \wedge \mathbf{ZEmb}_K(\eta_{\mathbb{Z}}) \vdash A$                                                                                                                                                                                                                                                                                | <i>A</i>                                                                                                                                               |
| 1,3       | 4 | $a \in \mathbb{Z}, b \in \mathbb{Z}_{>0} \vdash \eta_{\mathbb{Z}}(b) \neq 0_K$                                                                                                                                                                                                                                                                                                      | <i>Def.</i> 17.4.3.1, <i>Th.</i> 19.7.1.4(ii)(1, 3)                                                                                                    |
| 1,3       | 5 | $\eta_{\mathbb{Q}}([a, b]_{\mathbb{Q}}) := \eta_{\mathbb{Z}}(a) \boxtimes \text{inv}(\eta_{\mathbb{Z}}(b)) = I$                                                                                                                                                                                                                                                                     |                                                                                                                                                        |
| 1,3,4,5   | 6 | $[a, b]_{\mathbb{Q}} = [c, d]_{\mathbb{Q}} \vdash \eta_{\mathbb{Z}}(a) \boxtimes \text{inv}(\eta_{\mathbb{Z}}(b)) = \eta_{\mathbb{Z}}(c) \boxtimes \text{inv}(\eta_{\mathbb{Z}}(d))$                                                                                                                                                                                                | <i>Th.</i> 18.2.2.2, <i>Th.</i> 18.2.2.3, 3, 4, 5, <i>Def.</i> 19.7.1.1(1)                                                                             |
| 1,3,5,6   | 7 | $\eta_{\mathbb{Q}}: \mathbb{Q} \rightarrow K \text{ und } \eta_{\mathbb{Q}}(0) = 0_K, \eta_{\mathbb{Q}}(1) = 1_K$                                                                                                                                                                                                                                                                   | <i>Th.</i> 18.2.2.3(5, 6), <i>Th.</i> 18.2.3.6,<br><i>Th.</i> 19.7.1.4(ii)(1, 3), <i>Def.</i> 19.7.1.1(1)                                              |
| 1,3,5,6,7 | 8 | $\forall p, q \in \mathbb{Q} (\eta_{\mathbb{Q}}(p + q) = \eta_{\mathbb{Q}}(p) \boxplus \eta_{\mathbb{Q}}(q) \wedge \eta_{\mathbb{Q}}(pq) = \eta_{\mathbb{Q}}(p) \boxtimes \eta_{\mathbb{Q}}(q)),$<br>$\forall p \in \mathbb{Q} (\eta_{\mathbb{Q}}(-p) = \boxminus \eta_{\mathbb{Q}}(p) \wedge (p \neq 0 \rightarrow \eta_{\mathbb{Q}}(p^{-1}) = \text{inv}(\eta_{\mathbb{Q}}(p))))$ | <i>Th.</i> 18.3.2.3, <i>Th.</i> 18.3.3.3,<br><i>Th.</i> 18.3.4.4, <i>Th.</i> 18.3.5.9(5, 6),<br><i>Th.</i> 19.7.1.4(ii)(1, 3), <i>Def.</i> 19.7.1.1(1) |
| 1,3,5,6,7 | 9 | $\forall p, q \in \mathbb{Q} (p < q \leftrightarrow (\eta_{\mathbb{Q}}(p) \preccurlyeq \eta_{\mathbb{Q}}(q) \wedge \eta_{\mathbb{Q}}(p) \neq \eta_{\mathbb{Q}}(q)))$                                                                                                                                                                                                                | <i>Def.</i> 18.4.1.4, <i>Th.</i> 18.4.1.3, 3, 5, 6,<br><i>Th.</i> 19.7.1.4(ii)(1, 3), <i>Def.</i> 19.7.1.1(1)                                          |

1,7,8,9 10  $\eta_{\mathbb{Q}}: \mathbb{Q} \rightarrow K \wedge \mathbf{QEmb}_K(\eta_{\mathbb{Q}})$   
*Def.* 19.7.1.3, *Def.* 6.2.1.1(7,9),  
punktweise Konjunktion der vollständig quantifizierten Regeln aus 7,8,9  
1,3,10 11  $\theta: \mathbb{Q} \rightarrow K, \mathbf{QEmb}_K(\theta) \vdash \forall q \in \mathbb{Q} \theta(q) = \eta_{\mathbb{Q}}(q)$   
*Def.* 19.7.1.3, *Th.* 18.2.2.3, *Th.* 19.7.1.4(ii)(1,3), *Def.* 19.7.1.1(1)  
1 12  $\exists! \eta_{\mathbb{Q}}: \mathbb{Q} \rightarrow K \mathbf{QEmb}_K(\eta_{\mathbb{Q}}) \quad \exists! I(2,3,10,11)$

□

Für den im Folgenden jeweils festen vollständig angeordneten Körper schreiben wir  $\eta_{\mathbb{Z}}$  und  $\eta_{\mathbb{Q}}$  für die durch *Th.* 19.7.1.4(ii) beziehungsweise *Th.* 19.7.1.4(iii) eindeutig bestimmten Zeugen. Zusammen mit der bereits festgelegten natürlichen Einbettung  $\eta_K$  gelten damit stets deren Typ-, Rekursions-, Konstanten-, Operations-, Inversen- und Ordnungsregeln; die folgenden Teilbeweise führen die jeweils benötigten Regeln durch eine Formelreferenz ausdrücklich ein.

*Beweis.*

**Dichtheit des rationalen Primkörpers**

*Th.* 19.7.1.4(iv)

$$\begin{aligned} & \text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq), \quad x, y \in K \\ & , \quad x \preceq y, \quad x \neq y \vdash \exists q \in \mathbb{Q} ( \\ & \quad x \preceq \eta_{\mathbb{Q}}(q) \wedge x \neq \eta_{\mathbb{Q}}(q) \wedge \eta_{\mathbb{Q}}(q) \preceq y \wedge \eta_{\mathbb{Q}}(q) \neq y) \end{aligned}$$

|         |    |                                                                                                                                                                                                                                                                                                                                                            |                                                                                 |
|---------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1       | 1  | $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq)$                                                                                                                                                                                                                                                                             | <i>A</i>                                                                        |
| 2       | 2  | $x, y \in K$                                                                                                                                                                                                                                                                                                                                               | <i>A</i>                                                                        |
| 3       | 3  | $x \preceq y$                                                                                                                                                                                                                                                                                                                                              | <i>A</i>                                                                        |
| 4       | 4  | $x \neq y$                                                                                                                                                                                                                                                                                                                                                 | <i>A</i>                                                                        |
| 1       | 5  | $(\eta_K: \mathbb{N} \rightarrow K \wedge \eta_K(0) = 0_K \wedge$<br>$\forall n \in \mathbb{N} (\eta_K(\text{succ}(n)) = \eta_K(n) \boxplus 1_K)) \wedge$<br>$(\eta_{\mathbb{Z}}: \mathbb{Z} \rightarrow K \wedge \mathbf{ZEmb}_K(\eta_{\mathbb{Z}})) \wedge$<br>$(\eta_{\mathbb{Q}}: \mathbb{Q} \rightarrow K \wedge \mathbf{QEmb}_K(\eta_{\mathbb{Q}}))$ | <i>Th.</i> 19.7.1.2(1), <i>Th.</i> 19.7.1.4(ii)(1), <i>Th.</i> 19.7.1.4(iii)(1) |
| 1,2,3,4 | 6  | $0_K \preceq \delta := y \boxplus (\boxminus x) \wedge 0_K \neq \delta$                                                                                                                                                                                                                                                                                    | <i>Th.</i> 19.7.1.3(i)(1,2,3,4)                                                 |
| 1,6     | 7  | $\text{inv}(\delta) \in K \wedge 0_K \preceq \text{inv}(\delta) \wedge 0_K \neq \text{inv}(\delta)$                                                                                                                                                                                                                                                        | <i>Def.</i> 19.7.1.1(1,6)                                                       |
| 1,7     | 8  | $\exists n \in \mathbb{N} (\text{inv}(\delta) \preceq \eta_K(n) \wedge \text{inv}(\delta) \neq \eta_K(n))$                                                                                                                                                                                                                                                 | <i>Th.</i> 19.7.1.3(1,5,7)                                                      |
| 8       | 9  | $n \in \mathbb{N} \wedge \text{inv}(\delta) \preceq \eta_K(n) \wedge \text{inv}(\delta) \neq \eta_K(n)$                                                                                                                                                                                                                                                    | <i>A</i>                                                                        |
| 1,6-8   | 10 | $n \neq 0 \wedge 0_K \preceq \text{inv}(\eta_K(n)) \wedge \text{inv}(\eta_K(n)) \neq 0_K \wedge$<br>$\text{inv}(\eta_K(n)) \preceq \delta \wedge \text{inv}(\eta_K(n)) \neq \delta$                                                                                                                                                                        | <i>Def.</i> 19.7.1.1(1), <i>Th.</i> 19.7.1.4(i)(1,5)                            |

1,2,9 11  $\exists N \in \mathbb{N} (\exists \eta_K(N) \preceq x \boxtimes \eta_K(n) \wedge x \boxtimes \eta_K(n) \preceq \eta_K(N))$   
Wende *Th.* 19.7.1.3 mit den Daten aus 5  
auf  $z = x \boxtimes \eta_K(n)$  und auf  $\exists z$  an.  
Dies liefert  $N_+, N_- \in \mathbb{N}$  mit  $z < \eta_K(N_+)$   
und  $-z < \eta_K(N_-)$ . Setze  $N = N_+ + N_-$ .  
Die Positivität der natürlichen Bilder und  
 $\eta_K(N) = \eta_K(N_+) \boxplus \eta_K(N_-)$  liefern beide Schranken.  
*Th.* 19.7.1.4(i)(1, 5), *Def.* 19.7.1.1(1)

11 12  $N \in \mathbb{N} \wedge \exists \eta_K(N) \preceq x \boxtimes \eta_K(n) \wedge x \boxtimes \eta_K(n) \preceq \eta_K(N) A$

1,2,9,12 13  $M := \{k \in \mathbb{N} \mid x \boxtimes \eta_K(n) \preceq \eta_K(k) \boxplus \exists \eta_K(N) \wedge$   
 $x \boxtimes \eta_K(n) \neq \eta_K(k) \boxplus \exists \eta_K(N)\},$   
 $\text{succ}(N + N) \in M$  und damit  $M \neq \emptyset$   
 $x \boxtimes \eta_K(n) \preceq \eta_K(N) < \eta_K(N) \boxplus 1_K$   
 $= \eta_K(\text{succ}(N + N)) \boxplus \exists \eta_K(N),$   
*Ax.* 10.3.4, *Th.* 19.7.1.4(i)(1, 5), *Def.* 19.7.1.1(1, 12)

1,2,9,12 14  $\exists k_0 \in M \forall k \in M (k_0 \leq k)$  *Th.* 10.4.5.26(13)

14 15  $k_0 \in M \wedge \forall k \in M (k_0 \leq k)$  *A*

1,2,9  
12,15 16  $k_0 \neq 0$   
Für  $k_0 = 0$  wäre nach 12 und der Definition von  $M$  zugleich  
 $\exists \eta_K(N) \preceq x \boxtimes \eta_K(n)$  sowie  $x \boxtimes \eta_K(n) \preceq \exists \eta_K(N)$   
mit  $x \boxtimes \eta_K(n) \neq \exists \eta_K(N)$ , also einen Widerspruch.

1,2,9  
12,15 17  $\exists \ell \in \mathbb{N} (k_0 = \text{succ}(\ell))$   
*Ax.* 10.3.6( $\wedge E1(15), 16$ )

17 18  $\ell \in \mathbb{N} \wedge k_0 = \text{succ}(\ell)$  *A*

1,2,9  
12,15,18 19  $\ell < k_0$   
 $= E(\wedge E2(18), \text{Th. } 10.4.5.13(\wedge E1(18)))$

1,2,9  
12,15,18 20  $\ell \notin M$   
 $\ell \in M$  lieferte aus der Minimalität von  $k_0$  die Ungleichung  $k_0 \leq \ell$ ,  
im Widerspruch zu  $\ell < k_0$ .

1-2,9,12  
15,18 21  $\eta_K(\ell) \boxplus \exists \eta_K(N) \preceq x \boxtimes \eta_K(n)$   
 $\ell \in \mathbb{N} \wedge \ell \notin M$  und die Definition von  $M$ ,  
*Def.* 15.2.1.1, *Def.* 19.7.1.1(1)

1-2,9,12  
15,18 22  $m := \iota_{\mathbb{Z}}(k_0) - \iota_{\mathbb{Z}}(N) \in \mathbb{Z} \wedge$   
 $\eta_{\mathbb{Z}}(m - 1) \preceq x \boxtimes \eta_K(n) \wedge x \boxtimes \eta_K(n) \preceq \eta_{\mathbb{Z}}(m) \wedge$   
 $x \boxtimes \eta_K(n) \neq \eta_{\mathbb{Z}}(m)$   
 $\wedge E1(15), \wedge E2(15), \wedge E2(18), 19, 21,$   
*Th.* 19.7.1.4(ii)(1, 5), *Th.* 19.7.1.4(i)(1, 5)

1-4,9,12  
15,18 23  $x \preceq \eta_{\mathbb{Q}}([m, \iota_{\mathbb{Z}}(n)]_{\mathbb{Q}}) \wedge x \neq \eta_{\mathbb{Q}}([m, \iota_{\mathbb{Z}}(n)]_{\mathbb{Q}}) \wedge$   
 $\eta_{\mathbb{Q}}([m, \iota_{\mathbb{Z}}(n)]_{\mathbb{Q}}) \preceq x \boxplus \text{inv}(\eta_K(n)) \wedge$   
 $x \boxplus \text{inv}(\eta_K(n)) \preceq y \wedge x \boxplus \text{inv}(\eta_K(n)) \neq y$   
*Def.* 19.7.1.1(1), *Th.* 19.7.1.4(iii)(1), 6, 10, 22

1-4,9,12  
15 24  $x \preceq \eta_{\mathbb{Q}}([m, \iota_{\mathbb{Z}}(n)]_{\mathbb{Q}}) \wedge x \neq \eta_{\mathbb{Q}}([m, \iota_{\mathbb{Z}}(n)]_{\mathbb{Q}}) \wedge$   $\exists E(17, 18, 23)$   
 $\eta_{\mathbb{Q}}([m, \iota_{\mathbb{Z}}(n)]_{\mathbb{Q}}) \preceq y \wedge \eta_{\mathbb{Q}}([m, \iota_{\mathbb{Z}}(n)]_{\mathbb{Q}}) \neq y$

1-4 25  $\exists q \in \mathbb{Q} (x \preceq \eta_{\mathbb{Q}}(q) \wedge x \neq \eta_{\mathbb{Q}}(q) \wedge \eta_{\mathbb{Q}}(q) \preceq y \wedge \eta_{\mathbb{Q}}(q) \neq y)$   
 $\exists E(8, 9, \exists E(11, 12, \exists E(14, 15, \exists I([m, \iota_{\mathbb{Z}}(n)]_{\mathbb{Q}}, 24))))$

## Definition durch Suprema

Th. 19.7.1.4(v)

$$\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preccurlyeq), A \in \mathbb{R}$$

$$\vdash \eta_{\mathbb{Q}}[A] \neq \emptyset \wedge \text{UB}_{K, \preccurlyeq}(\eta_{\mathbb{Q}}[A]) \neq \emptyset$$

|       |    |                                                                                                                                                |                                |
|-------|----|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 1     | 1  | $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preccurlyeq) A$                                                          |                                |
| 2     | 2  | $A \in \mathbb{R}$                                                                                                                             | $A$                            |
| 1     | 3  | $\eta_{\mathbb{Q}}: \mathbb{Q} \rightarrow K$                                                                                                  | Th. 19.7.1.4(iii)(1)           |
| 2     | 4  | $\exists a \in A \wedge \exists r \in \mathbb{Q} \setminus A$                                                                                  | Def. 19.2.1.1(Th. 19.2.1.1(2)) |
| 1,2   | 5  | $\eta_{\mathbb{Q}}[A] \neq \emptyset$                                                                                                          | Th. 5.2.4.17(3,4)              |
| 4     | 6  | $a \in A \wedge r \in \mathbb{Q} \setminus A$                                                                                                  | $A$                            |
| 1,2,6 | 7  | $\forall q \in A (q < r)$                                                                                                                      |                                |
|       |    |                                                                                                                                                | Th. 19.4.2.1(i)(2,6)           |
| 1,2,6 | 8  | $\forall q \in A \eta_{\mathbb{Q}}(q) \preccurlyeq \eta_{\mathbb{Q}}(r)$                                                                       |                                |
|       |    |                                                                                                                                                | Th. 19.7.1.4(iii)(1,7)         |
| 1,2,6 | 9  | $\eta_{\mathbb{Q}}(r) \in \text{UB}_{K, \preccurlyeq}(\eta_{\mathbb{Q}}[A])$                                                                   | Th. 13.2.1.3(3,8)              |
| 1,2,6 | 10 | $\text{UB}_{K, \preccurlyeq}(\eta_{\mathbb{Q}}[A]) \neq \emptyset$                                                                             | Th. 3.6.3.5(9)                 |
| 1,2   | 11 | $\eta_{\mathbb{Q}}[A] \neq \emptyset \wedge \text{UB}_{K, \preccurlyeq}(\eta_{\mathbb{Q}}[A]) \neq \emptyset \exists E(4, 6, \wedge I(5, 10))$ |                                |

## Strikte Approximation eines Supremums

Th. 19.7.1.4(vi)

$$\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preccurlyeq), \emptyset \neq S \subseteq K, \text{UB}_{K, \preccurlyeq}(S) \neq \emptyset$$

$$, \alpha = \sup_{K, \preccurlyeq}(S), u \in K, u \preccurlyeq \alpha, u \neq \alpha$$

$$\vdash \exists s \in S (u \preccurlyeq s \wedge u \neq s)$$

|                 |    |                                                                                                                         |                              |
|-----------------|----|-------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 1               | 1  | $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preccurlyeq) A$                                   |                              |
| 2               | 2  | $\emptyset \neq S \subseteq K$                                                                                          | $A$                          |
| 3               | 3  | $\text{UB}_{K, \preccurlyeq}(S) \neq \emptyset$                                                                         | $A$                          |
| 4               | 4  | $\alpha = \sup_{K, \preccurlyeq}(S)$                                                                                    | $A$                          |
| 5               | 5  | $u \in K$                                                                                                               | $A$                          |
| 6               | 6  | $u \preccurlyeq \alpha$                                                                                                 | $A$                          |
| 7               | 7  | $u \neq \alpha$                                                                                                         | $A$                          |
| 1,2,3,4         | 8  | $\alpha \in \text{UB}_{K, \preccurlyeq}(S) \wedge \forall v \in \text{UB}_{K, \preccurlyeq}(S) (\alpha \preccurlyeq v)$ |                              |
|                 |    |                                                                                                                         | Th. 13.2.1.14(1, 2, 3, 4)    |
| 9               | 9  | $u \in \text{UB}_{K, \preccurlyeq}(S)$                                                                                  | $A$                          |
| 1,2,3,4,9       | 10 | $\alpha \preccurlyeq u$                                                                                                 | $\forall E(\wedge E2(8), 9)$ |
| 1,5,6,9         | 11 | $u = \alpha$                                                                                                            |                              |
|                 |    |                                                                                                                         | Def. 15.2.1.1(1, 5, 6, 10)   |
| 1,2,3,4,5,6,7,9 | 12 | $\perp$                                                                                                                 | $\perp I(7, 11)$             |
| 1,2,3,4,5,6,7   | 13 | $u \notin \text{UB}_{K, \preccurlyeq}(S)$                                                                               | $\neg I(9, 12)$              |
| 1,2,5,13        | 14 | $\exists s \in S \neg (s \preccurlyeq u)$                                                                               |                              |
|                 |    |                                                                                                                         | Th. 13.2.1.1(1, 2, 5, 13)    |
| 14              | 15 | $s \in S \wedge \neg (s \preccurlyeq u)$                                                                                | $A$                          |

$$1,2,5,15 \quad 16 \quad u \preceq s \wedge u \neq s$$

*Def.* 15.2.1.1(1, 2, 5, 15)

$$1,2,3,4,5,6,7 \quad 17 \quad \exists s \in S (u \preceq s \wedge u \neq s)$$

$\exists E(14, 15, \exists I(s, 16))$

## Ordnung und Bijektivität der Supremumsabbildung

Th. 19.7.1.4(vii)

$$\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq),$$

$$\Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A])$$

$$\vdash \Phi: \mathbb{R} \xrightarrow{\sim} K \wedge \forall A, B \in \mathbb{R} (A \leq_{\mathbb{R}} B \leftrightarrow \Phi(A) \preceq \Phi(B))$$

$$1 \quad 1 \quad \text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq), \quad \Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A])$$

*A*

$$2 \quad \Phi: \mathbb{R} \rightarrow K$$

*Def.* 19.7.1.1(*Th.* 19.7.1.4(v)(1))

$$3 \quad 3 \quad A, B \in \mathbb{R} \wedge A \subsetneq B$$

*A*

$$3 \quad 4 \quad \exists q \in B \setminus A \exists r \in B (q < r)$$

*Th.* 3.5.2.3(3), *Def.* 19.2.1.1(3)

$$4 \quad 5 \quad q \in B \setminus A \wedge r \in B \wedge q < r$$

*A*

$$3,5 \quad 6 \quad \forall a \in A (a < q)$$

*Th.* 19.4.2.1(i)(3, 5)

$$1,3,5 \quad 7 \quad \Phi(A) \preceq \eta_{\mathbb{Q}}(q)$$

*Th.* 13.2.1.3(6), *Th.* 13.2.1.14(1)

$$1,3,5 \quad 8 \quad \eta_{\mathbb{Q}}(q) \preceq \eta_{\mathbb{Q}}(r) \wedge \eta_{\mathbb{Q}}(q) \neq \eta_{\mathbb{Q}}(r)$$

*Th.* 19.7.1.4(iii)(1, 5)

$$1,3,5 \quad 9 \quad \eta_{\mathbb{Q}}(r) \preceq \Phi(B)$$

*Th.* 13.2.1.14(1, 5)

$$1,3 \quad 10 \quad A \subsetneq B \vdash \Phi(A) \preceq \Phi(B) \wedge \Phi(A) \neq \Phi(B)$$

$\exists E(4, 5, \text{Def. } 15.2.1.1(7, 8, 9))$

$$1 \quad 11 \quad \Phi \text{ ist ordnungserhaltend und injektiv}$$

*Th.* 19.3.1.3(2, 10)

$$12 \quad 12 \quad y \in K$$

*A*

$$1,12 \quad 13 \quad \exists p, r \in \mathbb{Q} (\eta_{\mathbb{Q}}(p) \preceq y \wedge \eta_{\mathbb{Q}}(p) \neq y \wedge y \preceq \eta_{\mathbb{Q}}(r) \wedge y \neq \eta_{\mathbb{Q}}(r))$$

Das archimedische Lemma, auf  $y$  und  $\boxminus y$  angewandt, liefert  $m, n \in \mathbb{N}$  mit

$$\boxminus \eta_K(m) \preceq y \wedge \boxminus \eta_K(m) \neq y \wedge y \preceq \eta_K(n) \wedge y \neq \eta_K(n).$$

Setze  $p = -\iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(m))$  und  $r = \iota_{\mathbb{Q}}(\iota_{\mathbb{Z}}(n))$ .

*Th.* 19.7.1.2(1), *Th.* 19.7.1.3(1, *Th.* 19.7.1.2(1), 12),

*Th.* 19.7.1.4(iii)(1), *Th.* 19.7.1.4(ii)(1)

$$13 \quad 14 \quad p, r \in \mathbb{Q} \wedge \eta_{\mathbb{Q}}(p) \preceq y \wedge \eta_{\mathbb{Q}}(p) \neq y \wedge y \preceq \eta_{\mathbb{Q}}(r) \wedge y \neq \eta_{\mathbb{Q}}(r)$$

*A*

$$12 \quad 15 \quad A_y := \{q \in \mathbb{Q} \mid \eta_{\mathbb{Q}}(q) \preceq y \wedge \eta_{\mathbb{Q}}(q) \neq y\}$$

$= I$

$$12,14 \quad 16 \quad p \in A_y \wedge r \notin A_y$$

*Th.* 19.7.1.4(iii)(1, 14, 15)

$$12,15 \quad 17 \quad q \in A_y, \quad s \in \mathbb{Q}, \quad s < q \vdash s \in A_y$$

*Th.* 19.7.1.4(iii)(1, 15), *Def.* 15.2.1.1(12)

|         |    |                                                                                                                                                             |                                                                 |
|---------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 1,12,15 | 18 | $q \in A_y \vdash \exists s \in A_y (q < s)$                                                                                                                | <i>Th.</i> 19.7.1.4(iv)(1, 12, 15), <i>Th.</i> 19.7.1.4(iii)(1) |
| 1,12,14 | 19 | $\text{Cut}_{\mathbb{Q}}(A_y)$                                                                                                                              | <i>Def.</i> 19.2.1.1(15, 16, 17, 18)                            |
| 12,15   | 20 | $y \in \text{UB}_{K, \preceq}(\eta_{\mathbb{Q}}[A_y])$                                                                                                      | <i>Th.</i> 13.2.1.3(15)                                         |
| 1,12,15 | 21 | $u \in K, u \preceq y, u \neq y \vdash u \notin \text{UB}_{K, \preceq}(\eta_{\mathbb{Q}}[A_y])$                                                             | <i>Th.</i> 19.7.1.4(iv)(1, 12, 15)                              |
| 1,12,14 | 22 | $\Phi(A_y) = y$                                                                                                                                             | <i>Th.</i> 13.2.1.15(1, 19, 20, 21)                             |
| 1       | 23 | $\Phi: \mathbb{R} \xrightarrow{\sim} K$                                                                                                                     | <i>Def.</i> 8.2.1.1(2, 11, 12, 22)                              |
| 1       | 24 | $\forall A, B \in \mathbb{R} (A \leq_{\mathbb{R}} B \leftrightarrow \Phi(A) \preceq \Phi(B))$                                                               | <i>Th.</i> 19.3.1.1(10, 11)                                     |
| 1       | 25 | $\Phi: \mathbb{R} \xrightarrow{\sim} K \wedge \forall A, B \in \mathbb{R} (A \leq_{\mathbb{R}} B \leftrightarrow \Phi(A) \preceq \Phi(B)) \wedge I(23, 24)$ |                                                                 |

### Schnittrekonstruktion aus dem Supremum

*Th.* 19.7.1.4(viii)

$$\begin{aligned} & \text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq), \\ & \Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A]) \\ & \vdash \forall A \in \mathbb{R} \forall q \in \mathbb{Q} \left( q \in A \leftrightarrow \right. \\ & \quad \left. (\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A)) \right) \end{aligned}$$

|        |    |                                                                                                                                            |                                                                                    |
|--------|----|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1      | 1  | $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq), \Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A])$         | <i>A</i>                                                                           |
| 2      | 2  | $A \in \mathbb{R} \wedge q \in \mathbb{Q}$                                                                                                 | <i>A</i>                                                                           |
| 3      | 3  | $q \in A$                                                                                                                                  | <i>A</i>                                                                           |
| 2,3    | 4  | $\exists r \in A (q < r)$                                                                                                                  | <i>Def.</i> 19.2.1.1( <i>Th.</i> 19.2.1.1( $\wedge E1(2)$ ), 3)                    |
| 4      | 5  | $r \in A \wedge q < r$                                                                                                                     | <i>A</i>                                                                           |
| 2,5    | 6  | $\eta_{\mathbb{Q}}(q) \preceq \eta_{\mathbb{Q}}(r) \wedge \eta_{\mathbb{Q}}(q) \neq \eta_{\mathbb{Q}}(r)$                                  | <i>Th.</i> 19.7.1.4(iii)(1, $\wedge E2(2)$ , $\wedge E2(5)$ )                      |
| 1,2,5  | 7  | $\eta_{\mathbb{Q}}(r) \preceq \Phi(A)$                                                                                                     | <i>Th.</i> 13.2.1.14(1, <i>Th.</i> 19.7.1.4(v)( $\wedge E1(2)$ ), $\wedge E1(5)$ ) |
| 1,2,5  | 8  | $\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A)$                                                            | <i>Def.</i> 15.2.1.1(6, 7)                                                         |
| 1,2    | 9  | $q \in A \rightarrow (\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A)) \exists E(4, 5, \rightarrow I(3, 8))$ |                                                                                    |
| 10     | 10 | $\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A)$                                                            | <i>A</i>                                                                           |
| 11     | 11 | $q \notin A$                                                                                                                               | <i>A</i>                                                                           |
| 2,11   | 12 | $\forall a \in A (a < q)$                                                                                                                  | <i>Th.</i> 19.4.2.1(i)( $\wedge E1(2)$ , 11)                                       |
| 1,2,11 | 13 | $\eta_{\mathbb{Q}}(q) \in \text{UB}_{K, \preceq}(\eta_{\mathbb{Q}}[A])$                                                                    | <i>Th.</i> 13.2.1.3( <i>Th.</i> 19.7.1.4(iii)(1, 12)                               |
| 1,2,11 | 14 | $\Phi(A) \preceq \eta_{\mathbb{Q}}(q)$                                                                                                     | <i>Th.</i> 13.2.1.14(1, 13)                                                        |

1,2,11 15  $\Phi(A) = \eta_{\mathbb{Q}}(q)$

*Def.* 15.2.1.1(10, 14)

1,2,11 16  $\perp$

$\perp I(\wedge E2(10), Th. 2.9.1.1(15))$

1,2 17  $q \in A$   $\neg I(11, 16)$

1,2 18  $(\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A)) \rightarrow q \in A \rightarrow I(10, 17)$

1,2 19  $q \in A \leftrightarrow (\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A)) \perp I(9, 18)$

1 20  $\forall A \in \mathbb{R} \forall q \in \mathbb{Q} (q \in A \leftrightarrow (\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A)))$

$\forall I(\rightarrow I(2, 19))$

**Hauptschnitte werden auf rationale Elemente abgebildet** Th. 19.7.1.4(ix)

$\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq),$

$\Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A])$

$\vdash \forall q \in \mathbb{Q} \Phi(\iota_{\mathbb{R}}(q)) = \eta_{\mathbb{Q}}(q)$

1 1  $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq), \Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A])$

$A$

2 2  $q \in \mathbb{Q}$   $A$

1,2 3  $\iota_{\mathbb{R}}(q) = \widehat{q} \in \mathbb{R}$

*Def.* 19.2.2.2(2), *Th.* 19.2.2.1(2)

4 4  $\Phi(\iota_{\mathbb{R}}(q)) \neq \eta_{\mathbb{Q}}(q)$   $A$

1,2,4 5  $(\Phi(\iota_{\mathbb{R}}(q)) \preceq \eta_{\mathbb{Q}}(q) \wedge \Phi(\iota_{\mathbb{R}}(q)) \neq \eta_{\mathbb{Q}}(q)) \vee$   
 $(\eta_{\mathbb{Q}}(q) \preceq \Phi(\iota_{\mathbb{R}}(q)) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(\iota_{\mathbb{R}}(q)))$  *Def.* 15.2.1.1(4)

6 6  $\Phi(\iota_{\mathbb{R}}(q)) \preceq \eta_{\mathbb{Q}}(q) \wedge \Phi(\iota_{\mathbb{R}}(q)) \neq \eta_{\mathbb{Q}}(q)$   $A$

1,2,6 7  $\exists r \in \mathbb{Q} (\Phi(\iota_{\mathbb{R}}(q)) \preceq \eta_{\mathbb{Q}}(r) \wedge \Phi(\iota_{\mathbb{R}}(q)) \neq \eta_{\mathbb{Q}}(r) \wedge \eta_{\mathbb{Q}}(r) \preceq \eta_{\mathbb{Q}}(q) \wedge \eta_{\mathbb{Q}}(r) \neq \eta_{\mathbb{Q}}(q))$

*Th.* 19.7.1.4(iv)(1, 6)

7 8  $r \in \mathbb{Q} \wedge \Phi(\iota_{\mathbb{R}}(q)) \preceq \eta_{\mathbb{Q}}(r) \wedge \Phi(\iota_{\mathbb{R}}(q)) \neq \eta_{\mathbb{Q}}(r) \wedge A$   
 $\eta_{\mathbb{Q}}(r) \preceq \eta_{\mathbb{Q}}(q) \wedge \eta_{\mathbb{Q}}(r) \neq \eta_{\mathbb{Q}}(q)$

1,2,8 9  $r < q \wedge r \in \iota_{\mathbb{R}}(q)$

*Th.* 19.7.1.4(iii)(1, 8), *Def.* 19.2.2.2(2)

1,2,8 10  $\eta_{\mathbb{Q}}(r) \preceq \Phi(\iota_{\mathbb{R}}(q)) \wedge \eta_{\mathbb{Q}}(r) \neq \Phi(\iota_{\mathbb{R}}(q))$

*Th.* 19.7.1.4(viii)(1, 3, 9)

1,2,8 11  $\perp$

*Def.* 15.2.1.1(8, 10)

12 12  $\eta_{\mathbb{Q}}(q) \preceq \Phi(\iota_{\mathbb{R}}(q)) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(\iota_{\mathbb{R}}(q))$   $A$

1,2,12 13  $\exists r \in \mathbb{Q} (\eta_{\mathbb{Q}}(q) \preceq \eta_{\mathbb{Q}}(r) \wedge \eta_{\mathbb{Q}}(q) \neq \eta_{\mathbb{Q}}(r) \wedge \eta_{\mathbb{Q}}(r) \preceq \Phi(\iota_{\mathbb{R}}(q)) \wedge \eta_{\mathbb{Q}}(r) \neq \Phi(\iota_{\mathbb{R}}(q)))$

*Th.* 19.7.1.4(iv)(1, 12)

13 14  $r \in \mathbb{Q} \wedge \eta_{\mathbb{Q}}(q) \preceq \eta_{\mathbb{Q}}(r) \wedge \eta_{\mathbb{Q}}(q) \neq \eta_{\mathbb{Q}}(r) \wedge$   $A$   
 $\eta_{\mathbb{Q}}(r) \preceq \Phi(\iota_{\mathbb{R}}(q)) \wedge \eta_{\mathbb{Q}}(r) \neq \Phi(\iota_{\mathbb{R}}(q))$

1,2,14 15  $q < r \wedge r < q$

*Th.* 19.7.1.4(iii)(1, 14), *Th.* 19.7.1.4(viii)(1, 3, 14), *Def.* 19.2.2.2(2)

1,2,14 16  $\perp$

*Th.* 18.4.1.4(15)

1,2,4 17  $\perp$

$\vee E(5, 6, \exists E(7, 8, 11), 12, \exists E(13, 14, 16))$

1,2 18  $\Phi(\iota_{\mathbb{R}}(q)) = \eta_{\mathbb{Q}}(q)$

$\neg I(4, 17)$

1 19  $\forall q \in \mathbb{Q} \Phi(\iota_{\mathbb{R}}(q)) = \eta_{\mathbb{Q}}(q)$

$\forall I(\rightarrow I(2, 18))$

## Erhaltung der Addition

Th. 19.7.1.4(x)

$$\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq), \quad \Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A])$$

$$\vdash \forall A, B \in \mathbb{R} \Phi(A \oplus B) = \Phi(A) \boxplus \Phi(B)$$

1 1  $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq)$   $A$

2 2  $\Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A])$   $A$

1,2 3  $\Phi: \mathbb{R} \xrightarrow{\sim} K \wedge \forall X, Y \in \mathbb{R} (X \leq_{\mathbb{R}} Y \leftrightarrow \Phi(X) \preceq \Phi(Y))$

Th. 19.7.1.4(vii)(1, 2)

1,2 4  $\forall C \in \mathbb{R} \forall q \in \mathbb{Q} (q \in C \leftrightarrow (\eta_{\mathbb{Q}}(q) \preceq \Phi(C) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(C)))$

Th. 19.7.1.4(viii)(1, 2)

5 5  $A, B \in \mathbb{R}$   $A$

6 6  $q \in \mathbb{Q}$   $A$

7 7  $q \in A \oplus B$   $A$

5,6,7 8  $\exists a \in A \exists b \in B (q < a + b)$

Def. 19.4.1.1(5, 6, 7)

8 9  $a \in A \wedge b \in B \wedge q < a + b$   $A$

1,5,6,9 10  $\eta_{\mathbb{Q}}(q) \preceq \eta_{\mathbb{Q}}(a) \boxplus \eta_{\mathbb{Q}}(b) \wedge \eta_{\mathbb{Q}}(q) \neq \eta_{\mathbb{Q}}(a) \boxplus \eta_{\mathbb{Q}}(b)$

Th. 19.7.1.4(iii)(1, 6, 9)

1,2,5,9 11  $\eta_{\mathbb{Q}}(a) \preceq \Phi(A) \wedge \eta_{\mathbb{Q}}(b) \preceq \Phi(B)$

Th. 13.2.1.14(2, Th. 19.7.1.4(v)(1, 5), 9)

1,2,5,6,9 12  $\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \boxplus \Phi(B) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A) \boxplus \Phi(B)$

Def. 19.7.1.1(1, 10, 11)

1,2,5,6 13  $q \in A \oplus B \rightarrow (\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \boxplus \Phi(B) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A) \boxplus \Phi(B))$

$\exists E(8, 9, \rightarrow I(7, 12))$

14 14  $\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \boxplus \Phi(B) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A) \boxplus \Phi(B) A$

1,2,5,6,14 15  $u := \eta_{\mathbb{Q}}(q) \boxplus \Phi(B) \in K,$

$u \preceq \Phi(A) \wedge u \neq \Phi(A)$

Th. 19.7.1.3(i)(1, 14), Def. 19.7.1.1(1)

1,2,5,6,14 16  $\exists a \in A (u \preceq \eta_{\mathbb{Q}}(a) \wedge u \neq \eta_{\mathbb{Q}}(a))$

Th. 19.7.1.4(vi)(1, 2, Th. 19.7.1.4(v)(1, 5), 15)

16 17  $a \in A \wedge u \preceq \eta_{\mathbb{Q}}(a) \wedge u \neq \eta_{\mathbb{Q}}(a)$   $A$

1,2,5,6,14,17 18  $\eta_{\mathbb{Q}}(q) \preceq \eta_{\mathbb{Q}}(a) \boxplus \Phi(B) \wedge \eta_{\mathbb{Q}}(q) \neq \eta_{\mathbb{Q}}(a) \boxplus \Phi(B)$

Th. 19.7.1.3(i)(1, 15, 17)

1,2,5,6,14,17 19  $v := \eta_{\mathbb{Q}}(q) \boxplus \eta_{\mathbb{Q}}(a) \in K,$

$v \preceq \Phi(B) \wedge v \neq \Phi(B)$

Th. 19.7.1.3(i)(1, 18), Def. 19.7.1.1(1)

1,2,5,6,14,17 20  $\exists b \in B (v \preceq \eta_{\mathbb{Q}}(b) \wedge v \neq \eta_{\mathbb{Q}}(b))$

Th. 19.7.1.4(vi)(1, 2, Th. 19.7.1.4(v)(1, 5), 19)

|                  |    |                                                                                                                                                                                 |                                   |
|------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 20               | 21 | $b \in B \wedge v \preceq \eta_{\mathbb{Q}}(b) \wedge v \neq \eta_{\mathbb{Q}}(b)$                                                                                              | $A$                               |
| 1,2,5,6,14,17,21 | 22 | $q < a + b \wedge q \in A \oplus B$                                                                                                                                             |                                   |
|                  |    | <i>Th.</i> 19.7.1.3(i)(1, 19, 21), <i>Th.</i> 19.7.1.4(iii)(1), <i>Def.</i> 19.4.1.1(17, 21)                                                                                    |                                   |
| 1,2,5,6          | 23 | $(\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \boxplus \Phi(B) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A) \boxplus \Phi(B)) \rightarrow q \in A \oplus B$                                |                                   |
|                  |    | $\exists E(16, 17, \exists E(20, 21, \rightarrow I(14, 22)))$                                                                                                                   |                                   |
| 1,2,5,6          | 24 | $q \in A \oplus B \leftrightarrow (\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \boxplus \Phi(B) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A) \boxplus \Phi(B))$                            |                                   |
|                  |    | $\perp I(13, 23)$                                                                                                                                                               |                                   |
| 1,2,5            | 25 | $\forall q \in \mathbb{Q} (q \in A \oplus B \leftrightarrow (\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \boxplus \Phi(B) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A) \boxplus \Phi(B)))$ |                                   |
|                  |    | $\forall I(\rightarrow I(6, 24))$                                                                                                                                               |                                   |
| 1,2,3,5          | 26 | $\exists C \in \mathbb{R} \Phi(C) = \Phi(A) \boxplus \Phi(B)$                                                                                                                   |                                   |
|                  |    | <i>Def.</i> 7.2.1.1(3), <i>Def.</i> 19.7.1.1(1)                                                                                                                                 |                                   |
| 26               | 27 | $C \in \mathbb{R} \wedge \Phi(C) = \Phi(A) \boxplus \Phi(B)$                                                                                                                    | $A$                               |
| 1,2,4,27         | 28 | $\forall q \in \mathbb{Q} (q \in C \leftrightarrow (\eta_{\mathbb{Q}}(q) \preceq \Phi(A) \boxplus \Phi(B) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(A) \boxplus \Phi(B)))$          |                                   |
|                  |    | $= E(\forall E(4, \wedge E1(27)), \wedge E2(27))$                                                                                                                               |                                   |
| 1,2,5,27         | 29 | $C = A \oplus B$                                                                                                                                                                | <i>Ax.</i> 3.4.1.1(25, 28)        |
| 1,2,5,27         | 30 | $\Phi(A \oplus B) = \Phi(A) \boxplus \Phi(B)$                                                                                                                                   | $= E(29, \wedge E2(27))$          |
| 1,2,5            | 31 | $\Phi(A \oplus B) = \Phi(A) \boxplus \Phi(B)$                                                                                                                                   | $\exists E(26, 27, 30)$           |
| 1,2              | 32 | $\forall A, B \in \mathbb{R} \Phi(A \oplus B) = \Phi(A) \boxplus \Phi(B)$                                                                                                       | $\forall I(\rightarrow I(5, 31))$ |

## Erhaltung des positiven Produkts

*Th.* 19.7.1.4(xi)

$$\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq), \quad \Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A])$$

$$\vdash \forall A, B \in \mathbb{R} ((0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B) \rightarrow \\ \Phi(A \odot_+ B) = \Phi(A) \boxtimes \Phi(B))$$

|         |    |                                                                                                                                                                                                                                              |     |
|---------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1       | 1  | $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq)$                                                                                                                                                               | $A$ |
| 2       | 2  | $\Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A])$                                                                                                                                                                                         | $A$ |
| 1,2     | 3  | $\Phi: \mathbb{R} \xrightarrow{\sim} K \quad \text{ordnungserhaltend}$                                                                                                                                                                       |     |
|         |    | <i>Th.</i> 19.7.1.4(vii)(1, 2)                                                                                                                                                                                                               |     |
| 1,2     | 4  | $\Phi(0_{\mathbb{R}}) = 0_K \quad \text{und} \quad \forall C \in \mathbb{R} \forall q \in \mathbb{Q} (q \in C \leftrightarrow (\eta_{\mathbb{Q}}(q) \preceq \Phi(C) \wedge \eta_{\mathbb{Q}}(q) \neq \Phi(C)))$                              |     |
|         |    | <i>Th.</i> 19.7.1.4(ix)(1, 2), <i>Def.</i> 19.4.1.2, <i>Th.</i> 19.7.1.4(viii)(1, 2)                                                                                                                                                         |     |
| 5       | 5  | $A, B \in \mathbb{R}$                                                                                                                                                                                                                        | $A$ |
| 6       | 6  | $0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B$                                                                                                                                                               | $A$ |
| 1-6     | 7  | $\alpha := \Phi(A), \beta := \Phi(B),$<br>$0_K \preceq \alpha \wedge 0_K \preceq \beta$                                                                                                                                                      |     |
|         |    | Aus der Ordnungsäquivalenz in 3 und 6 folgen $\Phi(0_{\mathbb{R}}) \preceq \Phi(A)$ und $\Phi(0_{\mathbb{R}}) \preceq \Phi(B)$ ; mit 4 und den Definitionen von $\alpha, \beta$ werden daraus $0_K \preceq \alpha$ und $0_K \preceq \beta$ . |     |
| 8       | 8  | $q \in \mathbb{Q}$                                                                                                                                                                                                                           | $A$ |
| 9       | 9  | $q \in A \odot_+ B$                                                                                                                                                                                                                          | $A$ |
| 5,6,8,9 | 10 | $q < 0 \vee \exists a \in A \exists b \in B (0 < a \wedge 0 < b \wedge q < ab)$                                                                                                                                                              |     |
|         |    | <i>Def.</i> 19.4.2.1(5, 6, 8, 9), <i>Def.</i> 19.4.1.2, <i>Def.</i> 19.2.2.1                                                                                                                                                                 |     |
| 11      | 11 | $q < 0$                                                                                                                                                                                                                                      | $A$ |

|           |    |                                                                                                                                                                                                                                                        |                                                                                                                                               |
|-----------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1,7,8,11  | 12 | $\eta_{\mathbb{Q}}(q) \preceq \alpha \boxtimes \beta \wedge \eta_{\mathbb{Q}}(q) \neq \alpha \boxtimes \beta$                                                                                                                                          |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Th.</i> 19.7.1.4(iii)(1, 8, 11), <i>Def.</i> 19.7.1.1(1, 7)                                                                                |
| 11        | 13 | $\exists a \in A \exists b \in B (0 < a \wedge 0 < b \wedge q < ab)$                                                                                                                                                                                   | $A$                                                                                                                                           |
| 13        | 14 | $a \in A \wedge b \in B \wedge 0 < a \wedge 0 < b \wedge q < ab$                                                                                                                                                                                       | $A$                                                                                                                                           |
| 1,7,8,14  | 15 | $\eta_{\mathbb{Q}}(q) \preceq \eta_{\mathbb{Q}}(a) \boxtimes \eta_{\mathbb{Q}}(b) \wedge \eta_{\mathbb{Q}}(q) \neq \eta_{\mathbb{Q}}(a) \boxtimes \eta_{\mathbb{Q}}(b),$<br>$0_K \preceq \eta_{\mathbb{Q}}(a) \wedge 0_K \preceq \eta_{\mathbb{Q}}(b)$ |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Th.</i> 19.7.1.4(iii)(1, 14)                                                                                                               |
| 1,2,5     |    |                                                                                                                                                                                                                                                        |                                                                                                                                               |
| 7,14      | 16 | $\eta_{\mathbb{Q}}(a) \preceq \alpha \wedge \eta_{\mathbb{Q}}(b) \preceq \beta$                                                                                                                                                                        |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Th.</i> 13.2.1.14(2, <i>Th.</i> 19.7.1.4(v)(1, 5), 14)                                                                                     |
| 1,7,14    | 17 | $\eta_{\mathbb{Q}}(a) \boxtimes \eta_{\mathbb{Q}}(b) \preceq \alpha \boxtimes \beta$                                                                                                                                                                   |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Def.</i> 19.7.1.1(1, 7, 15, 16)                                                                                                            |
| 1-2,5     |    |                                                                                                                                                                                                                                                        |                                                                                                                                               |
| 7-8,14    | 18 | $\eta_{\mathbb{Q}}(q) \preceq \alpha \boxtimes \beta \wedge \eta_{\mathbb{Q}}(q) \neq \alpha \boxtimes \beta$                                                                                                                                          |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Def.</i> 15.2.1.1(15, 17)                                                                                                                  |
| 1-2,5-8   | 19 | $q \in A \odot_+ B \rightarrow (\eta_{\mathbb{Q}}(q) \preceq \alpha \boxtimes \beta \wedge \eta_{\mathbb{Q}}(q) \neq \alpha \boxtimes \beta)$                                                                                                          |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | $\rightarrow I(9, \vee E(10, 11, 12, 13, \exists E(13, 14, 18)))$                                                                             |
| 20        | 20 | $\eta_{\mathbb{Q}}(q) \preceq \alpha \boxtimes \beta \wedge \eta_{\mathbb{Q}}(q) \neq \alpha \boxtimes \beta$                                                                                                                                          | $A$                                                                                                                                           |
| 8         | 21 | $q < 0 \vee 0 \leq q$                                                                                                                                                                                                                                  | <i>Th.</i> 18.4.1.4(8)                                                                                                                        |
| 22        | 22 | $q < 0$                                                                                                                                                                                                                                                | $A$                                                                                                                                           |
| 5,6,8,22  | 23 | $q \in A \odot_+ B$                                                                                                                                                                                                                                    |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Def.</i> 19.4.2.1(5, 6, 8), <i>Def.</i> 19.4.1.2, <i>Def.</i> 19.2.2.1(22)                                                                 |
| 24        | 24 | $0 \leq q$                                                                                                                                                                                                                                             | $A$                                                                                                                                           |
| 1,7,8     |    |                                                                                                                                                                                                                                                        |                                                                                                                                               |
| 20,24     | 25 | $\alpha \neq 0_K \wedge \beta \neq 0_K$                                                                                                                                                                                                                |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | Aus $\alpha = 0_K$ oder $\beta = 0_K$ folgte $\alpha \boxtimes \beta = 0_K$ ,<br>im Widerspruch zu $0_K \preceq \eta_{\mathbb{Q}}(q)$ und 20. |
| 1,7,8     |    |                                                                                                                                                                                                                                                        |                                                                                                                                               |
| 20,24     | 26 | $u := \eta_{\mathbb{Q}}(q) \boxtimes \text{inv}(\beta) \in K,$<br>$0_K \preceq u \wedge u \preceq \alpha \wedge u \neq \alpha$                                                                                                                         |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Def.</i> 19.7.1.1(1, 7, 20, 24, 25)                                                                                                        |
| 1-2,5,7-8 |    |                                                                                                                                                                                                                                                        |                                                                                                                                               |
| 20,24     | 27 | $\exists a \in A (u \preceq \eta_{\mathbb{Q}}(a) \wedge u \neq \eta_{\mathbb{Q}}(a))$                                                                                                                                                                  |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Th.</i> 19.7.1.4(vi)(1, 2, <i>Th.</i> 19.7.1.4(v)(1, 5), 26)                                                                               |
| 27        | 28 | $a \in A \wedge u \preceq \eta_{\mathbb{Q}}(a) \wedge u \neq \eta_{\mathbb{Q}}(a)$                                                                                                                                                                     | $A$                                                                                                                                           |
| 1,7-8,20  |    |                                                                                                                                                                                                                                                        |                                                                                                                                               |
| 24,28     | 29 | $0 < a \wedge \eta_{\mathbb{Q}}(a) \neq 0_K$                                                                                                                                                                                                           |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Th.</i> 19.7.1.4(iii)(1, 8, 24, 26, 28)                                                                                                    |
| 1,7-8,20  |    |                                                                                                                                                                                                                                                        |                                                                                                                                               |
| 24,28     | 30 | $v := \eta_{\mathbb{Q}}(q) \boxtimes \text{inv}(\eta_{\mathbb{Q}}(a)) \in K,$<br>$0_K \preceq v \wedge v \preceq \beta \wedge v \neq \beta$                                                                                                            |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Def.</i> 19.7.1.1(1, 20, 26, 28, 29)                                                                                                       |
| 1-2,5,7-8 |    |                                                                                                                                                                                                                                                        |                                                                                                                                               |
| 20,24,28  | 31 | $\exists b \in B (v \preceq \eta_{\mathbb{Q}}(b) \wedge v \neq \eta_{\mathbb{Q}}(b))$                                                                                                                                                                  |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Th.</i> 19.7.1.4(vi)(1, 2, <i>Th.</i> 19.7.1.4(v)(1, 5), 30)                                                                               |
| 31        | 32 | $b \in B \wedge v \preceq \eta_{\mathbb{Q}}(b) \wedge v \neq \eta_{\mathbb{Q}}(b)$                                                                                                                                                                     | $A$                                                                                                                                           |
| 1,7-8,20  |    |                                                                                                                                                                                                                                                        |                                                                                                                                               |
| 24,28,32  | 33 | $0 < b \wedge q < ab$                                                                                                                                                                                                                                  |                                                                                                                                               |
|           |    |                                                                                                                                                                                                                                                        | <i>Def.</i> 19.7.1.1(1, 20, 29, 30, 32), <i>Th.</i> 19.7.1.4(iii)(1)                                                                          |

|                      |    |                                                                                                                                                                              |                                                                   |
|----------------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 5-6,8,20<br>24,28,32 | 34 | $q \in A \odot_+ B$                                                                                                                                                          | <i>Def.</i> 19.4.2.1(5, 6, 8, 28, 32, 33)                         |
| 1-2,5-8<br>20,24     | 35 | $q \in A \odot_+ B$                                                                                                                                                          | $\exists E(27, 28, \exists E(31, 32, 34))$                        |
| 1-2,5-8              | 36 | $(\eta_{\mathbb{Q}}(q) \preceq \alpha \boxtimes \beta \wedge \eta_{\mathbb{Q}}(q) \neq \alpha \boxtimes \beta) \rightarrow q \in A \odot_+ B$                                | $\rightarrow I(20, \vee E(21, 22, 23, 24, 35))$                   |
| 1-2,5-8              | 37 | $q \in A \odot_+ B \leftrightarrow (\eta_{\mathbb{Q}}(q) \preceq \alpha \boxtimes \beta \wedge \eta_{\mathbb{Q}}(q) \neq \alpha \boxtimes \beta)$                            | $\perp I(19, 36)$                                                 |
| 1-2,5-7              | 38 | $\forall q \in \mathbb{Q} (q \in A \odot_+ B \leftrightarrow (\eta_{\mathbb{Q}}(q) \preceq \alpha \boxtimes \beta \wedge \eta_{\mathbb{Q}}(q) \neq \alpha \boxtimes \beta))$ | $\forall I(\rightarrow I(8, 37))$                                 |
| 1-3,5,7              | 39 | $\exists C \in \mathbb{R} \Phi(C) = \alpha \boxtimes \beta$                                                                                                                  | <i>Def.</i> 7.2.1.1( $\wedge E1(3)$ ), <i>Def.</i> 19.7.1.1(1, 7) |
| 39                   | 40 | $C \in \mathbb{R} \wedge \Phi(C) = \alpha \boxtimes \beta$                                                                                                                   | $A$                                                               |
| 1,2,4,40             | 41 | $\forall q \in \mathbb{Q} (q \in C \leftrightarrow (\eta_{\mathbb{Q}}(q) \preceq \alpha \boxtimes \beta \wedge \eta_{\mathbb{Q}}(q) \neq \alpha \boxtimes \beta))$           | $= E(\forall E(4, \wedge E1(40)), \wedge E2(40))$                 |
| 1-2,5-7<br>40        | 42 | $C = A \odot_+ B$                                                                                                                                                            | <i>Ax.</i> 3.4.1.1(38, 41)                                        |
| 1-2,5-7<br>40        | 43 | $\Phi(A \odot_+ B) = \Phi(A) \boxtimes \Phi(B)$                                                                                                                              | $= E(7, 42, \wedge E2(40))$                                       |
| 1-2,5-7              | 44 | $\Phi(A \odot_+ B) = \Phi(A) \boxtimes \Phi(B)$                                                                                                                              | $\exists E(39, 40, 43)$                                           |
| 1,2,5                | 45 | $(0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B) \rightarrow \Phi(A \odot_+ B) = \Phi(A) \boxtimes \Phi(B)$                                   | $\rightarrow I(6, 44)$                                            |
| 1,2                  | 46 | $\forall A, B \in \mathbb{R} ((0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B) \rightarrow \Phi(A \odot_+ B) = \Phi(A) \boxtimes \Phi(B))$     | $\forall I(\rightarrow I(5, 45))$                                 |

## Suprema und Schnittoperationen

Th. 19.7.1.4(xii)

$$\begin{aligned} & \text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq), \Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A]) \vdash \\ & \forall A, B \in \mathbb{R} (\Phi(A \oplus B) = \Phi(A) \boxplus \Phi(B) \wedge \\ & \Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B)) \wedge \Phi(0_{\mathbb{R}}) = 0_K \wedge \Phi(1_{\mathbb{R}}) = 1_K \end{aligned}$$

|           |   |                                                                                |                                                                                                 |
|-----------|---|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1         | 1 | $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preceq)$ | $A$                                                                                             |
| 2         | 2 | $\Phi(A) := \sup_{K, \preceq}(\eta_{\mathbb{Q}}[A])$                           | $A$                                                                                             |
| 1,2       | 3 | $\forall A, B \in \mathbb{R} \Phi(A \oplus B) = \Phi(A) \boxplus \Phi(B)$      | <i>Th.</i> 19.7.1.4(x)(1, 2)                                                                    |
| 1,2       | 4 | $\forall q \in \mathbb{Q} \Phi(\iota_{\mathbb{R}}(q)) = \eta_{\mathbb{Q}}(q)$  | <i>Th.</i> 19.7.1.4(ix)(1, 2)                                                                   |
| 1,2       | 5 | $\Phi(0_{\mathbb{R}}) = 0_K \wedge \Phi(1_{\mathbb{R}}) = 1_K$                 | <i>Def.</i> 19.4.1.2, <i>Def.</i> 19.4.2.3, <i>Th.</i> 18.2.3.6, <i>Th.</i> 19.7.1.4(iii)(1), 4 |
| 6         | 6 | $A \in \mathbb{R}$                                                             | $A$                                                                                             |
| 1,2,3,5,6 | 7 | $\Phi(A) \boxplus \Phi(\ominus A) = 0_K$                                       | $= E(\forall E(3, A, \ominus A), \text{Th. } 19.4.1.3(iv)(6), 5)$                               |
| 1,2,6     | 8 | $\Phi(\ominus A) = \boxminus \Phi(A)$                                          | <i>Def.</i> 19.7.1.1(1, 7)                                                                      |
| 1,2       | 9 | $\forall A \in \mathbb{R} \Phi(\ominus A) = \boxminus \Phi(A)$                 | $\forall I(\rightarrow I(6, 8))$                                                                |

|              |    |                                                                                                                                                                                                         |                                                                                              |
|--------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 1,2          | 10 | $\forall A, B \in \mathbb{R} ((0_{\mathbb{R}} \leq_{\mathbb{R}} A \wedge 0_{\mathbb{R}} \leq_{\mathbb{R}} B) \rightarrow \Phi(A \odot_+ B) = \Phi(A) \boxtimes \Phi(B))$                                | <i>Th.</i> 19.7.1.4(xi)(1, 2)                                                                |
| 11           | 11 | $A, B \in \mathbb{R}$                                                                                                                                                                                   | $A$                                                                                          |
| 1,11         | 12 | $0_{\mathbb{R}} \leq_{\mathbb{R}} A \vee A <_{\mathbb{R}} 0_{\mathbb{R}}$                                                                                                                               | <i>Th.</i> 19.3.1.3(11)                                                                      |
| 1,11         | 13 | $0_{\mathbb{R}} \leq_{\mathbb{R}} B \vee B <_{\mathbb{R}} 0_{\mathbb{R}}$                                                                                                                               | <i>Th.</i> 19.3.1.3(11)                                                                      |
| 14           | 14 | $0_{\mathbb{R}} \leq_{\mathbb{R}} A$                                                                                                                                                                    | $A$                                                                                          |
| 15           | 15 | $0_{\mathbb{R}} \leq_{\mathbb{R}} B$                                                                                                                                                                    | $A$                                                                                          |
| 1,2,11,14,15 | 16 | $\Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B)$                                                                                                                                                           | <i>Def.</i> 19.4.2.2(14, 15), $\forall E(10, 11, \wedge I(14, 15))$                          |
| 17           | 17 | $B <_{\mathbb{R}} 0_{\mathbb{R}}$                                                                                                                                                                       | $A$                                                                                          |
| 1,2,11,14,17 | 18 | $\Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B)$                                                                                                                                                           | <i>Def.</i> 19.4.2.2(14, 17), <i>Th.</i> 19.4.1.4(17),<br>8, 10, <i>Def.</i> 19.7.1.1(1)     |
| 1,2,11,14    | 19 | $\Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B)$                                                                                                                                                           | $\vee E(13, 15, 16, 17, 18)$                                                                 |
| 20           | 20 | $A <_{\mathbb{R}} 0_{\mathbb{R}}$                                                                                                                                                                       | $A$                                                                                          |
| 21           | 21 | $0_{\mathbb{R}} \leq_{\mathbb{R}} B$                                                                                                                                                                    | $A$                                                                                          |
| 1,2,11,20,21 | 22 | $\Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B)$                                                                                                                                                           | <i>Def.</i> 19.4.2.2(20, 21), <i>Th.</i> 19.4.1.4(20),<br>8, 10, <i>Def.</i> 19.7.1.1(1)     |
| 23           | 23 | $B <_{\mathbb{R}} 0_{\mathbb{R}}$                                                                                                                                                                       | $A$                                                                                          |
| 1,2,11,20,23 | 24 | $\Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B)$                                                                                                                                                           | <i>Def.</i> 19.4.2.2(20, 23), <i>Th.</i> 19.4.1.4(20, 23),<br>8, 10, <i>Def.</i> 19.7.1.1(1) |
| 1,2,11,20    | 25 | $\Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B)$                                                                                                                                                           | $\vee E(13, 21, 22, 23, 24)$                                                                 |
| 1,2,11       | 26 | $\Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B)$                                                                                                                                                           | $\vee E(12, 14, 19, 20, 25)$                                                                 |
| 1,2          | 27 | $\forall A, B \in \mathbb{R} \Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B) \forall I(\rightarrow I(11, 26))$                                                                                              |                                                                                              |
| 1,2          | 28 | $\forall A, B \in \mathbb{R} (\Phi(A \oplus B) = \Phi(A) \boxplus \Phi(B) \wedge$<br>$\Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B)) \wedge \Phi(0_{\mathbb{R}}) = 0_K \wedge \Phi(1_{\mathbb{R}}) = 1_K$ | $\wedge I(3, \wedge I(27, 5))$                                                               |

## Supremumsdichte der Hauptschnitte

*Th.* 19.7.1.4(xiii)

$$A \in \mathbb{R} \vdash \emptyset \neq \iota_{\mathbb{R}}[A] \subseteq \mathbb{R} \wedge \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\iota_{\mathbb{R}}[A]) \neq \emptyset \wedge$$

$$A = \sup_{\mathbb{R}, \leq_{\mathbb{R}}}(\iota_{\mathbb{R}}[A])$$

$$1 \quad 1 \quad A \in \mathbb{R} \quad A$$

$$1 \quad 2 \quad \exists q \in A$$

*Def.* 19.2.1.1(*Th.* 19.2.1.1(1))

$$1 \quad 3 \quad \emptyset \neq \iota_{\mathbb{R}}[A] \subseteq \mathbb{R}$$

*Th.* 5.2.4.17(2), *Th.* 19.2.2.2

|         |    |                                                                                                                                                                                                                           |                       |                                                                              |
|---------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------|
| 1       | 4  | $\forall q \in A (\iota_{\mathbb{R}}(q) \leq_{\mathbb{R}} A)$                                                                                                                                                             |                       | <i>Th.</i> 19.3.1.1( <i>Def.</i> 19.2.2.1, <i>Def.</i> 19.2.1.1(1))          |
| 1       | 5  | $A \in \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\iota_{\mathbb{R}}[A]) \wedge \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\iota_{\mathbb{R}}[A]) \neq \emptyset$                                                           |                       | <i>Th.</i> 13.2.1.3(1, 3, 4), <i>Th.</i> 3.6.3.5                             |
| 6       | 6  | $x \in \bigcup \iota_{\mathbb{R}}[A]$                                                                                                                                                                                     | $A$                   |                                                                              |
| 6       | 7  | $\exists q \in A (x \in \iota_{\mathbb{R}}(q))$                                                                                                                                                                           | $A$                   | <i>Th.</i> 3.13.2.1(6)                                                       |
| 7       | 8  | $q \in A \wedge x \in \iota_{\mathbb{R}}(q)$                                                                                                                                                                              | $A$                   |                                                                              |
| 1,6,8   | 9  | $x \in A$                                                                                                                                                                                                                 |                       | <i>Def.</i> 19.2.2.1(8), <i>Def.</i> 19.2.1.1(1, 8)                          |
| 1       | 10 | $\bigcup \iota_{\mathbb{R}}[A] \subseteq A$                                                                                                                                                                               |                       | <i>Def.</i> 3.5.1.1( $\forall I(\rightarrow I(6, \exists E(7, 8, 9)))$ )     |
| 11      | 11 | $x \in A$                                                                                                                                                                                                                 | $A$                   |                                                                              |
| 1,11    | 12 | $\exists q \in A (x < q)$                                                                                                                                                                                                 |                       | <i>Def.</i> 19.2.1.1(1, 11)                                                  |
| 12      | 13 | $q \in A \wedge x < q$                                                                                                                                                                                                    | $A$                   |                                                                              |
| 1,11,13 | 14 | $x \in \iota_{\mathbb{R}}(q) \wedge \iota_{\mathbb{R}}(q) \in \iota_{\mathbb{R}}[A]$                                                                                                                                      |                       | <i>Def.</i> 19.2.2.2(13), <i>Th.</i> 5.2.4.8(13)                             |
| 1,11,13 | 15 | $x \in \bigcup \iota_{\mathbb{R}}[A]$                                                                                                                                                                                     | $A$                   | <i>Th.</i> 3.13.2.1(14)                                                      |
| 1       | 16 | $A \subseteq \bigcup \iota_{\mathbb{R}}[A]$                                                                                                                                                                               |                       | <i>Def.</i> 3.5.1.1( $\forall I(\rightarrow I(11, \exists E(12, 13, 15)))$ ) |
| 1       | 17 | $\bigcup \iota_{\mathbb{R}}[A] = A$                                                                                                                                                                                       | $Ax.$ 3.4.1.1(10, 16) |                                                                              |
| 1       | 18 | $\sup_{\mathbb{R}, \leq_{\mathbb{R}}}(\iota_{\mathbb{R}}[A]) = \bigcup \iota_{\mathbb{R}}[A]$                                                                                                                             | $A$                   | <i>Th.</i> 19.3.3.3(3, 1, 4)                                                 |
| 1       | 19 | $\emptyset \neq \iota_{\mathbb{R}}[A] \subseteq \mathbb{R} \wedge \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\iota_{\mathbb{R}}[A]) \neq \emptyset \wedge A = \sup_{\mathbb{R}, \leq_{\mathbb{R}}}(\iota_{\mathbb{R}}[A])$ |                       | $\wedge I(3, \wedge I(\wedge E2(5), = E(17, 18)))$                           |

### Ordnungsisomorphismen erhalten vorhandene Suprema *Th.* 19.7.1.4(xiv)

$$\begin{aligned} \Psi: \mathbb{R} &\xrightarrow{\sim} K, \forall X, Y \in \mathbb{R} (X \leq_{\mathbb{R}} Y \leftrightarrow \Psi(X) \preceq \Psi(Y)) \\ &, \emptyset \neq S \subseteq \mathbb{R}, \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(S) \neq \emptyset \\ &\vdash \sup_{\mathbb{R}, \leq_{\mathbb{R}}} S = \sup_{K, \preceq} (\Psi[S]) \end{aligned}$$

- 1 1  $\Psi: \mathbb{R} \xrightarrow{\sim} K$   $A$   
2 2  $\forall X, Y \in \mathbb{R} (X \leq_{\mathbb{R}} Y \leftrightarrow \Psi(X) \preceq \Psi(Y))$   $A$   
1,2 3  $\text{PartOrd}(K, \preceq)$

Surjektivitat aus 1 schreibt jedes  $x, y, z \in K$  als  $\Psi(X), \Psi(Y), \Psi(Z)$ .

Mit der aquivalenz 2 ubertragen sich Reflexivitat, Antisymmetrie und Transitivitat von  $\leq_{\mathbb{R}}$ .

*Th.* 19.3.1.3(i), *Def.* 8.2.1.1(1)

- 4 4  $\emptyset \neq S \subseteq \mathbb{R}$   $A$   
5 5  $\text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(S) \neq \emptyset$   $A$   
4,5 6  $s := \sup_{\mathbb{R}, \leq_{\mathbb{R}}} S \in \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(S),$  *Th.* 13.2.1.14(4, 5)  
 $\forall U \in \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(S) (s \leq_{\mathbb{R}} U)$   
1,2,4,5,6 7  $\Psi[S] \subseteq K \wedge \Psi(s) \in \text{UB}_{K, \preceq}(\Psi[S])$   
*Th.* 5.2.4.16(4, 1), *Th.* 13.2.1.3(1, 2, 4, 6)

|                |    |                                                                                  |                                                                       |
|----------------|----|----------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 8              | 8  | $v \in \text{UB}_{K, \preccurlyeq}(\Psi[S])$                                     | $A$                                                                   |
| 1,7,8          | 9  | $\exists V \in \mathbb{R} (\Psi(V) = v)$                                         | $\text{Def. 7.2.1.1(1), Th. 13.2.1.2}(\wedge E1(7), 8)$               |
|                | 9  | $10 V \in \mathbb{R} \wedge \Psi(V) = v$                                         | $A$                                                                   |
| 1,2,4,7,8,10   | 11 | $V \in \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(S)$                             | $\text{Th. 13.2.1.2}(\wedge E1(7), 8), \text{Th. 13.2.1.3}(2, 4, 10)$ |
| 1,2,4,5,6,8,10 | 12 | $s \leq_{\mathbb{R}} V$                                                          | $\forall E(\wedge E2(6), 11)$                                         |
| 1,2,4,5,6,8,10 | 13 | $\Psi(s) \preccurlyeq v$                                                         | $= E(\forall E(2, s, V), 12, \wedge E2(10))$                          |
| 1,2,4,5,6,8    | 14 | $\Psi(s) \preccurlyeq v$                                                         | $\exists E(9, 10, 13)$                                                |
| 1,2,4,5,6      | 15 | $\forall v \in \text{UB}_{K, \preccurlyeq}(\Psi[S]) (\Psi(s) \preccurlyeq v)$    | $\forall I(\rightarrow I(8, 14))$                                     |
| 1,2,3,4,5,6    | 16 | $\Psi(s) = \sup_{K, \preccurlyeq}(\Psi[S])$                                      | $\text{Th. 13.2.1.15}(3, \wedge E1(7), \wedge E2(7), 15)$             |
| 1,2,4,5        | 17 | $\Psi(\sup_{\mathbb{R}, \leq_{\mathbb{R}}} S) = \sup_{K, \preccurlyeq}(\Psi[S])$ | $= E(6, 16)$                                                          |

Operationen und Eindeutigkeit:

|     |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                              |
|-----|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1   | 1  | $\text{COF}(K, \boxplus, \boxtimes, \boxminus, \text{inv}, 0_K, 1_K, \preccurlyeq)A$                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                              |
| 1   | 2  | $\Phi(A) = \sup_{K, \preccurlyeq}(\eta_{\mathbb{Q}}[A])$                                                                                                                                                                                                                                                                                                                                                                                                                                                | $= I$                                                        |
| 1   | 3  | $\Phi: \mathbb{R} \xrightarrow{\sim} K \wedge$<br>$\forall A, B \in \mathbb{R}$<br>$(A \leq_{\mathbb{R}} B \leftrightarrow \Phi(A) \preccurlyeq \Phi(B))$                                                                                                                                                                                                                                                                                                                                               | $\text{Th. 19.7.1.4(vii)}(1, 2)$                             |
| 1   | 4  | $\forall A, B \in \mathbb{R} (\Phi(A \oplus B) = \Phi(A) \boxplus \Phi(B) \wedge$<br>$\Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B)) \wedge \Phi(0_{\mathbb{R}}) = 0_K \wedge \Phi(1_{\mathbb{R}}) = 1_K$                                                                                                                                                                                                                                                                                                 | $\text{Th. 19.7.1.4(xii)}(1, 2)$                             |
| 1   | 5  | $\forall A, B \in \mathbb{R}$<br>$\Phi(A \oplus B) = \Phi(A) \boxplus \Phi(B)$                                                                                                                                                                                                                                                                                                                                                                                                                          | $\wedge E1(4)$                                               |
| 1   | 6  | $\forall A, B \in \mathbb{R}$<br>$\Phi(A \odot B) = \Phi(A) \boxtimes \Phi(B)$                                                                                                                                                                                                                                                                                                                                                                                                                          | $\wedge E1(\wedge E2(4))$                                    |
| 1   | 7  | $\Phi(0_{\mathbb{R}}) = 0_K \wedge \Phi(1_{\mathbb{R}}) = 1_K$                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\wedge I(\wedge E1(\wedge E2(4)), \wedge E2(\wedge E2(4)))$ |
| 8   | 8  | $\Psi: \mathbb{R} \xrightarrow{\sim} K, \forall X, Y \in \mathbb{R} ($<br>$\Psi(X \oplus Y) = \Psi(X) \boxplus \Psi(Y) \wedge \Psi(X \odot Y) = \Psi(X) \boxtimes \Psi(Y),$<br>$\forall X, Y \in \mathbb{R} (X \leq_{\mathbb{R}} Y \leftrightarrow \Psi(X) \preccurlyeq \Psi(Y)),$<br>$\Psi(0_{\mathbb{R}}) = 0_K, \Psi(1_{\mathbb{R}}) = 1_K$                                                                                                                                                          | $A$                                                          |
| 1,8 | 9  | $(\Psi \circ \iota_{\mathbb{R}}): \mathbb{Q} \rightarrow K \wedge \text{QEmb}_K(\Psi \circ \iota_{\mathbb{R}})$<br>$\text{Th. 19.2.2.2, Th. 19.3.1.5,}$<br>$\text{Th. 19.4.3.1, Th. 19.3.1.4,}$<br>$\text{Def. 19.7.1.3,}$<br>die Additions- und Nullregeln aus 8 geben $\Psi(\ominus X) = \boxminus \Psi(X)$ ,<br>die Produkt- und Einsregeln aus 8 geben für $X \neq 0$ ebenso $\Psi(\text{inv}_{\mathbb{R}}(X)) = \text{inv}(\Psi(X))$ ;<br>$\text{Th. 19.4.1.3, Th. 19.4.2.3, Def. 19.7.1.1}(1, 8)$ |                                                              |
| 1,8 | 10 | $\forall q \in \mathbb{Q} \Psi(\iota_{\mathbb{R}}(q)) = \eta_{\mathbb{Q}}(q)$                                                                                                                                                                                                                                                                                                                                                                                                                           | $\text{Th. 19.7.1.4(iii)}(1, 9)$                             |

$$\begin{array}{ll}
1 \quad 11 \quad \forall A \in \mathbb{R} \left( \emptyset \neq \iota_{\mathbb{R}}[A] \subseteq \mathbb{R} \wedge \right. & Th. \ 19.7.1.4(xiii) \\
\quad \quad \quad \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(\iota_{\mathbb{R}}[A]) \neq \emptyset \wedge \\
\quad \quad \quad \left. A = \sup_{\mathbb{R}, \leq_{\mathbb{R}}}(\iota_{\mathbb{R}}[A]) \right) & \\
1,8 \quad 12 \quad \emptyset \neq S \subseteq \mathbb{R}, \text{UB}_{\mathbb{R}, \leq_{\mathbb{R}}}(S) \neq \emptyset \vdash & \\
\quad \quad \quad \Psi\left(\sup_{\mathbb{R}, \leq_{\mathbb{R}}} S\right) = \sup_{K, \preceq}(\Psi[S]) & \\
1,8 \quad 13 \quad \forall A \in \mathbb{R} \Psi[\iota_{\mathbb{R}}[A]] = \eta_{\mathbb{Q}}[A] & Th. \ 19.7.1.4(xiv)(8) \\
1,2,8 \quad 14 \quad \forall A \in \mathbb{R} \Psi(A) = \Phi(A) & Th. \ 5.2.4.6(10), Ax. \ 3.4.1.1 \\
& = E(10, 11, 12, 13, 2) \\
1 \quad 15 \quad \exists! \Phi: \mathbb{R} \xrightarrow{\sim} K \left( \right. & \\
\quad \quad \quad \forall x, y \in \mathbb{R} & \\
\quad \quad \quad \Phi(x \oplus y) = \Phi(x) \boxplus \Phi(y) \wedge & \\
\quad \quad \quad \forall x, y \in \mathbb{R} & \\
\quad \quad \quad \Phi(x \odot y) = \Phi(x) \boxtimes \Phi(y) \wedge & \\
\quad \quad \quad \forall x, y \in \mathbb{R} & \\
\quad \quad \quad (x \leq_{\mathbb{R}} y \leftrightarrow \Phi(x) \preceq \Phi(y)) \wedge & \\
\quad \quad \quad \Phi(0_{\mathbb{R}}) = 0_K \wedge \Phi(1_{\mathbb{R}}) = 1_K & \\
& \exists! I(3, 5, 6, 7, \rightarrow I(8, 14))
\end{array}$$

□

Die Eindeutigkeit ist als Eindeutigkeit des ordnungs- und körperstruktur-erhaltenden Isomorphismus zu verstehen. Damit kann in späteren Bänden entweder mit dem konkreten Schnittmodell oder ausschließlich mit den registrierten Axiomen eines vollständig angeordneten Körpers gearbeitet werden. Für Grenzwertaussagen stehen insbesondere die reelle Ordnung, der Abstand und positive Epsilon-Umgebungen bereits zur Verfügung.